Mundarija

Expressions……

Linear Equations

Linear System of Equations

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Research organizing(Margin of Error; Outliers)

Expressions -> Ifodalar

1. [June 2023]

Which expression is equivalent to $(3x^3 - x^2 + 4)(5x^2 + 8x)$?

A) $15x^5 + 19x^4 - 8x^3 + 20x + 32$

B) $15x^5 + 19x^4 - 8x^3 - 20x^2 + 32$

C) $15x^5 + 19x^4 - 8x^3 + 20x^2 + 32x$

D) $15x^5 + 29x^4 - 8x^3 - 20x^2 + 32x$

Which = qaysi

Expression = ifoda

Is – tarjima qilinmaydi tarjima qilsangiz: bu deb tarjima qilasiz bazida esa teng deb tarjima qilamiz

2. [June 2023]

If $\frac{6}{7}p + 12 = 54$ what is the value of $7p$?

If = agar

What = nima

Value = qiymati

Of = ...ning

what is the value of $7p$? – 7p ning qiymati nimaga teng

3. [June 2023]

$$b^2 + 5c = 9d$$

The given equation relates the real numbers b, c, and d, where $d > \frac{5}{9}c$. Which equation correctly expresses b in terms of c and d ?

A) $b = \frac{9d+5c}{2}$

B) $b = \frac{9d-5c}{2}$

C) $b = \pm\sqrt{9d + 5c}$

D) $b = \pm\sqrt{9d - 5c}$

Given = berilgan

Equation = tenglama

Relates = aloqador, bog'liq

Real number = haqiqiy son

Correctly = to'g'ri

Expresses b = b ni ifodalaydi

In terms of c and d = c va d orqali

4. [August US 2023]

If $x^3 + y^3 = 12$ and $x^3 - y^3 = 6$, what is the value of $x^6 - y^6$?

5. [August US 2023]

Which expression is equivalent to $2(xy^2 + x^2y) - 8(xy^2 - 4x^2y)$?
A) $-6xy^2 + 34x^2y$
B) $-6xy^2 + 33x^2y$
C) $-6xy^2 - 30x^2y$
D) $-6xy^2 - 31x^2y$

To .. = ..ga

Equivalent= ekvivalent , teng degani

Where = bu yerda

6. [November 2023]

If $2x + 3 = 9$, what is the value of $4x - 4$?

7. [November 2023]

Which expression is equivalent to $(131y)^{\frac{1}{2}}$, where $y > 1$?
A) $131\sqrt{y}$
B) $\sqrt{131}y$
C) $\sqrt{131y}$
D) $\sqrt{(131y)^2}$

8. [November 2023]

Which expression is equivalent to $(5a^2 - 3ab + 7b^2) - (4a^2 + ab - 2b^2)$
A) $a^2 - 4ab + 9b^2$
B) $a^2 - 2ab + 5b^2$
C) $a^4 - 4a^2b^2 + 9b^4$
D) $a^4 - 2a^2b^2 + 5b^4$

9. [November 2023]

$$7x + 16 = \frac{4}{y}$$

The given equation relates the distinct positive numbers x and y. Which equation correctly expresses y in terms of x ?

Given equation = berilgan tenglama

Distinct = turli

positive numbers x= musbat x soni

A) $y = 7x + 12$
B) $y = \frac{4}{7x+16}$
C) $y = \frac{7x+16}{4}$
D) $y = 4(7x + 16)$

10. [December 2023]

$$v = -\frac{w}{130x}$$

The given equation relates the distinct positive numbers v, w, and x. Which equation correctly expresses w in terms of v and x ?
A) $w = -130vx$
B) $w = -\frac{130v}{x}$
C) $w = -\frac{x}{130v}$
D) $w = v + 130x$

11. [December 2023]
Which expression is equivalent to

$$15(x + 10)?$$

A) $15x + 10$
C) $15x + 25$
B) $15x + 150$
D) $15x + 5$

12. [March 2024]
Which expression is equivalent to

$$(9nr^2 + 3nr) + (6n^2r + 5nr)?$$

A) $15n^2r^2 + 8nr$
B) $15n^3r^3 + 8n^2r^2$
C) $54n^3r^3 + 15n^2r^2$
D) $9nr^2 + 6n^2r + 8nr$

13. [December 2023]

If $9(x+3)=81$, what is the value of $x+3$?

A) 6
C) 69
B) 9
D) 72

14. If $3x=4y=6z$, which of the following expresses the average of x and y in terms of z ?

A) $\frac{7z}{4}$
B) $\frac{5z}{3}$
C) $\frac{10}{4z}$
D) $\frac{13z}{9}$

15. Two quantities x and y are related such that $y=7$ when $x=1$. When the value of x increases by 1 , the value of y is multiplied by 2
Which of the following represents this relationship?

A) $y=7x^2$
B) $y=7(x-1)^2$
C) $y=7(2)^x$
D) $y=7(4)^{x-1}$

16. [March 2024]

Which expression is equivalent to $16(x+5)$?

A) $16x+5$
B) $16x+80$
C) $16x+21$
D) $16x+11$

If = agar
which of the following = quyidagilarning qaysi biri

average = o'rtachasi(o'rta arifmetik)
two = ikki
quantities = miqdor
related = bog'liq

that – qo'shimcha tarif uchun ishlatiladi faqat o'zidan avval kelgan narsani tariflab beradi, alohida mano bildirmaydi
when = qachonki
value of x = x ning qiymati
increases by 1 = 1 ga oshadi $(+1)$
multiplied by 2 = 2 ga ko'payadi $(\cdot 2)$
represents = ko'rsatadi
this = bu
relationship = bo'g'lanish

17. [March 2024]

Which expression is equivalent to

$$\frac{y+2}{x-9}+\frac{y(x-9)}{x^2y-9xy}?$$

A) $\frac{xy+y-7}{x^3y-18x^2y+81xy}$
B) $\frac{xy+10y+2}{x^2y-9xy+x-9}$
C) $\frac{xy^2+3xy-9y}{x^2y-9xy}$
D) $\frac{xy^2+3xy-9y}{x^3y-18x^2y+81xy}$

18. [June 2024]

$$\frac{2}{c}-\frac{7}{d}=5$$

The given equation relates the variables c and d, where c and d are greater than 1. Which equation correctly expresses d in terms of c ?

A) $d=\frac{c}{7(2-5c)}$
B) $d=\frac{7c}{2-5c}$
C) $d=\frac{7(2-5c)}{c}$
D) $d=\frac{2-5c}{7c}$

Greater than .. -- .. dan katta (>)
Where = bu yerda
c and d are greater than 1 = c va d lar 1 dan katta

19. [March 2024]

$$x=2a(b+5)$$

The given equation relates the positive numbers a, b, and x. Which equation correctly expresses a in terms of b and x ?

A) $a=\frac{x}{2}-(b+5)$
B) $a=\frac{x}{2(b+5)}$
C) $a=\frac{2(b+5)}{x}$
D) $a=2x(b+5)$

20. [March 2024]

$$x = 2a(b + 3)$$

The given equation relates positive numbers x, a, and b. Which equation correctly expresses a in terms of b and x ?

A) $a = \frac{x}{2} - 4b + 3$

B) $a = \frac{x}{2(b+3)}$

C) $a = \frac{2(b+3)}{x}$

D) $a = 2x(b + 3)$

21. [March 2024]

If $\frac{x}{y} = 63$ and $\frac{cx}{3y} = 63$, what is the value of c ?

22. [March 2024]

The function f gives the product of a number, x, and a number that is 66 more than x. Which equation defines f ?

A) $f(x) = x^2 + x + 66$

B) $f(x) = x^2 + 66$

C) $f(x) = x^2 + 66x$

D) $f(x) = x^2 + 66x + 66$

function f = f funksiya
give = bemoq
product = ko'paytmasi
more than = .. dan ko'proq
66 more than x – x dan 66 taga ko'proq
Define = ifodalamoq

23. [March 2024]

If

$$8 + \frac{9(6 - x)}{4} = \frac{7(6 - x)}{3}$$

what is the value of $6 - x$?

24. [May 2024]

$$(5x^3 - 2) - (-6x^3 + 3)$$

The given expression is equivalent to $bx^3 - 5$ where b is a constant. What is the value of b ?

25. [June 2024]

Which expression is equivalent to

$(x^3 + 6x^2 - 8x) + 8(x^2 + 4)$

A) $x^3 + 4x^2 - 8x + 32$
B) $x^3 + 14x^2 - 8x + 32$
C) $x^3 + 8x^2 - 8x + 32$
D) $x^3 + 6x^2 - 8x + 32$

26. [May US 2024]

Which expression is equivalent to

$$(4x^2 + 6) - (3x^2)?$$

A) 7
B) $7x^2 + 6$
C) $x^2 - 6$
D) $x^2 + 6$

27. [March 2024]

If $\frac{7}{8}p + 35 = 56$

what is the value of $8p$?

28. [March 2024]

Which expression is equivalent to

$$\frac{y+4}{x-7}+\frac{y(x-7)}{x^2y-7xy}$$

A) $\frac{xy+y-3}{x^3y-14x^2y+49xy}$

B) $\frac{xy+8y+4}{x^2y-7x^2y+x-7}$

C) $\frac{xy^2+5xy-7y}{x^2y-7xy}$

D) $\frac{xy^2+5xy-7y}{x^3y-14x^2y+49xy}$

29. [August 2025]

$$65=\frac{2K}{v^2}$$

For an object with a mass of 65 kilograms, the given equation relates the kinetic energy K, in joules, of the object to the object's speed v, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v=\frac{2\ K}{65^2}$

B) $v=\frac{65^2}{2K}$

C) $v=\sqrt{\frac{65}{2k}}$

D) $v=\sqrt{\frac{2K}{65}}$

For = uchun
In = da , to .. = .. ga,
An object = bir obyekt
Speed = tezlik

30. Which of the following expressions is equivalent to $\frac{2a}{3b}=\frac{5}{a+1}$?

A) $3a+1=15b$

B) $2a^2+1=15b$

C) $2a^2+2a=15b$

D) $2a^2+2a=8b$

Linear Equations –> Chiziqli tenglamalar

1. [August 2025]

If $\frac{x-17}{18} = \frac{x-17}{6}$, what is the value of $x + 17$?

A) 0
B) 3
C) 17
D) 34

2. [August 2025]

If $4 + \frac{5(7-x)}{2} = \frac{8(7-x)}{3}$, what is the value of $7 - x$?

3. [August 2025]

$$c(x - 6) = -7(x + k)$$

In the given equation, c and k are constants. The equation has exactly one solution. Which of the following statements must be true?

A) The value of c cannot be -7 .
B) The value of c cannot be $-\frac{7}{6}$
C) The value of k cannot be $\frac{6}{7}$
D) The value of k cannot be -6

Has = ega bo'lmoq, bor bo'lmoq
Exactly = aniq
One = bir
Solution = yechim
Statetement = tasdiq
Must be = bo'lishi kerak
Cannot be = bo'la olmaydi

4. [August 2025]

If $|16 - 2x| = 58$, which of the following is a value of $8 - x$?

A) 0
B) 8
C) 29
D) 42

5. [August 2025]

If $10 + \frac{8(7-x)}{5} = \frac{5(7-x)}{3}$, what is the value of $7 - x$?

6. [August 2025]

$$|a + 10| = 21$$

What is a possible solution to the given equation?
A) -21
B) -10
C) 0
D) 11

Possible = mumkin bo'lgan
x-intercept = x ni kesish nuqtasi
in the xy-plane = xy koordinalar tekislikda

7. [September 2025]

What is the x-intercept of the graph of $4x + y = 24$ in the xy-plane?
A) $(6,0)$
B) $(0,0)$
C) $(28,0)$
D) $(4,0)$

8. . [September 2025]

$$8(t + 4) - 10(t + 4) = 34$$

What value of t is the solution to the given equation?

9. [September 2025]

If $4(2x - 0.6) + x = 0.45 + 0.3x$, what is the value of $870x$?

10. [September 2025]

If $\frac{x+5}{3} = \frac{x+5}{13}$, the value of $x + 5$ is between which of the following pairs of values?

A) 7 and 13
B) 4 and 6
C) -4 and 4
D) -6 and -3

Between = o'rtasida
pairs of values = qiymatlar juftligi
has = ega bo'lmoq, bor bo'lmoq
infinitely = cheksiz
many = ko'p

11. [October 2025]

$$9x + 17 = 9x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k ?

12. [October US 2024]

$$\frac{49}{p} = \frac{49}{q} - \frac{49}{r} - \frac{49}{s}$$

The given equation relates the positive variables p, q, r and s. Which of the following is equivalent to q ?

A) $p + r + s$
B) $(49p + r + s)$
C) $\frac{prs}{pr+ps+rs}$
D) $\frac{prs}{49p+49r+49s}$

13. [October US 2024]

If $\frac{j}{k} = 62$ and $\frac{96j}{12xk} = 62$ what is the value of x?

14. [March US 2023]

If $(x + 9) - 5 = 25$, what is the value of $x + 9$?
A) 30
B) 21
C) 20
D) 11

15. [March US 2023]

$$|15 - x| - 9 = 0$$

What are all positive solutions to the given equation?
A) -6 and -24
B) -6 and 24
C) 6 and -24
D) 6 and 24

16. [March US 2023]

$$ax - y = bxy$$

The given equation relates the positive numbers a, b, x, and y. Which equation correctly expresses a in terms of b, x, and y ?
A) $a = \frac{bxy+y}{x}$
B) $a = \frac{bxy-y}{x}$
C) $a = \frac{x}{bxy+y}$
D) $a = \frac{x}{bxy-y}$

17. [March US 2023]

Which equation has no solution?
A) $x = x + 1$
B) $x = 2x + 2$
C) $x + 1 = x + 1$
D) $x + 1 = 2x + 2$

18. If $5(x + 1) = 25$, what is the value of $x + 1$?
A) 4 B) 5
C) 19 D) 20

19. [March US 2023]

What value of x is the solution to the equation $x + 10 = 3$
A) -13
B) -7
C) 7
D) 13

20.

$$9x + x = 2x + x + 5$$

Which of the following equations has the same solution as the given equation?
A) $7x = 5$
B) $7x = -5$
C) $11x = 5$
D) $11x = -5$

Same = bir xil
Same solution bir xik yechim
As = kabi
as the given equation? = berilgan tenglama kabi
sum = yig'indisi
smallest = eng kichik

zero = nol, 0
one = bir
two = ikki
Exactly = aniq
Infinitely many = cheksiz ko'p

21. [August US 2023]

$$|3x - 12| = 18$$

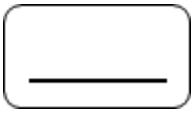
What is the sum of the solutions to the given equation?
A) -2
B) 8
C) 10
D) 12

22. How many solutions does the equation $10x = 10$ have?
A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

23. [November 2023]

$$|6x - 3| = |5x|$$

What is the smallest solution to the given equation?



24. [August US 2023]

$$\frac{1}{3}(6y - 9) = 13$$

What value of y satisfies the given equation?

Satisfies = qanoatlantiradi

25. [March 2024]

If $-9(15 - 3x) + 5 = -10(15 - 3x) + 21$, what is the value of $15 - 3x$?

Medium level-O'rtacha daraja

26. [December 2023]

$$-5x + 15rx = 135$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

A) $\frac{1}{9}$
B) $\frac{1}{3}$
C) 3
D) 9

27. [March 2024]

If $4 + \frac{5(3-x)}{2} = \frac{8(3-x)}{3}$, what is the value of $3 - x$?

28. [May 2024]

$$-5x + 15rx = 135$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

A) $\frac{1}{9}$
B) $\frac{1}{3}$
C) 3
D) 9

29. [May 2024]

If $\frac{x-4}{5} = \frac{x-4}{7}$, the value of $x - 4$ is between which of the following pairs of values?
A) -7 and -5
B) -3 and 3
C) 3.5 and 4.5
D) 4.75 and 7.25

30. [May US 2024]

If

$$12 + \frac{7(3-x)}{6} = \frac{5(3-x)}{3}$$

which equation must also be true?
A) $3 - x = 207$
B) $3 - x = -24$
C) $x - 3 = 207$
D) $x - 3 = -24$

31. [May US 2024]

$$|x - 9| + 81 = 99$$

What is the sum of the solutions to the given equation?

32. [June 2024]

Which equation has no solution?
A) $-19x - \frac{1}{7} = -19x - \frac{1}{7}$
B) $-19x - \frac{1}{7} = -19x + \frac{1}{7}$
C) $-19x - \frac{1}{7} = 19x - \frac{1}{7}$
D) $-19x - \frac{1}{7} = 19x + \frac{1}{7}$

33. [June 2024]

$$\frac{8(cx+3)}{5} = 4x - 3$$

In the given equation, c is a constant. If the equation has no solution, what is the value of c ?

A) $-\frac{5}{2}$

B) $-\frac{2}{5}$

C) $\frac{2}{5}$

D) $\frac{5}{2}$

34. [June 2024]

The equation $4|x-3| = k$, where k is a constant, has exactly one solution. Which of the following could be the value of $\frac{k}{4}$?

A) -3 only

B) -3 or 3

C) 0 only

D) 3 only

35. [August 2024]

How many solutions does the equation $6x + 4 = 6x + 8$ have?

A) Infinitely many

B) Exactly two

C) Exactly one

D) Zero

36. [August US 2024]

$$|x - 6| = 13$$

What is the sum of the solutions to the given equation?

A) 0 B) 12 C) 19 D) 26

37. [August 2024]

$$-3x + 6rx = 42$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

A) $\frac{1}{7}$
B) $\frac{1}{2}$
C) 2
D) 7

38. [October US 2024]

Which equation has no solution?

A) $3(x-1) = -3(-x+1)$
B) $3(x-1) = -3(-x-1)$
C) $3(x-1) = 4(x-1)$
D) $3(x-1) = 4(-x+1)$

39. [August US 2024]

How many solutions does the equation $5x + 3 = 5x + 7$ have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

40. [October 2024]

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?

41. [October US 2024]

$$-3x + 6rx = 54$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

A) $\frac{1}{9}$
B) $\frac{1}{2}$
C) 2
D) 9

42. [December 2024]

$$a(7x - 14) + 5a = 7(ax - 2a) - 35$$

In the given equation, a is a constant. The equation has infinitely many solutions. What are all possible values of a ?

A) 0 only
B) -7 only
C) Any real number
C) No real number

All = barcha
Possible = mumkin bo'lgan
Only = faqat
Any = biror bir, istalgan
No = hech qanday
CANNOT be = bo'la olmaydi

43. [December 2024]

$$-\frac{3}{19}rx + \frac{s}{8} = 10 - \frac{5}{57}x$$

In the given equation, r and s are constants. The equation has no solution. What is the value of r? $(s \neq 80)$

44. [March US 2025]

$$x(r - 7) + 3 = 19x + 25$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

A) 3 B) 7 C) 22 D) 26

Linear System of Equations –> Chiziqli tenglamalar sistemasi

1. [March US 2023]

$$5x + 4y = 44$$
$$3x - 7y = 17$$

The solution to the given system of equations is (x, y). What is the value of $8x - 3y$?

2. [March US 2023]

$$2x - 2y = 2$$

One of the two linear equation in a system is given.The system has exactly one solution. Which equation could be the second equation in this system?

A) $-8x + 8y = 3$
B) $3x - 3y = 8$
C) $-10x + 8y = 5$
D) $-x + y = -1$

One = 1
Of = ning
Two = 2
Linear = chiziqli
Equation = tenglama
In = da
Has = ega
Exactly = aniq
One solution = bir yechim
Could be = bo'la oladi
Second = ikkinchi

3. [March US 2023]

$$y = 18x + 25$$
$$y = -14x - 7$$

What is the solution (x, y) to the given system of equations?

A) $(-7,25)$
B) $(-1,7)$
C) $(7, -1)$
E) $(25, -7)$

4. [June 2023]

$$3x + 5y = 8$$
$$9x + 15y = 24$$

For each real number, r which of the following points lies on the graph of each equation in the xy-plane for the given system?

A) $\left(r, -\frac{5r}{3} + \frac{8}{3}\right)$
B) $\left(r, \frac{3r}{5} + \frac{8}{5}\right)$

For = uchun
Each = har bir
Real number = haqiqiy son
Points = nuqtalar
Lies = yotadi
On = da
graph of each equation = har bir tenglamaning grafigi

C) $\left(-\frac{5r}{3}+\frac{8}{3}, r\right)$
D) $\left(\frac{r}{3}+8, -\frac{r}{3}+24\right)$

5. [June 2023]

$$mx + ky = -83$$
$$2x + ky = 22$$

In the given system of equations, m and k are constants. The graph of these equations in the xy-plane intersect at the point $(5, y)$. What is the value of m ?

These = bu
in the xy-plane = xy korrdinatalar tekisligida
intersect = kesishmoq
at the point = nuqtada

6. [August US 2023]

$$2x = -4$$
$$2x + y = -2$$

The solution to the given system of equations is (x, y). What is the value of $4x + y$?
A) -6
B) -2
C) 2
D) 8

7.

$$x = y^2 + 8y$$
$$x + 2y = 24$$

The system of equations above is graphed in the xy-plane. Which of the following is the y-coordinate of an intersection point of the graphs of the two equations?
A) 8
B) 4
C) -2
D) -12

8.

$$14x + \frac{1}{2}y = -28$$
$$2x^2 = y - 40$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy-plane. What is a possible value of x ?

Possible= mumkin bo'lgan
could be = bo'la oladi
x-coordinate of the solution= yechimning x koordinatasi

9. $(x-1)^2 + (y-2)^2 = 35$
$x = 2y + 2$

Which of the following could be the x-coordinate of the solution to the system of equations above?

A) $\sqrt{6}$
B) $\frac{\sqrt{30}}{2}$
C) $2\sqrt{6} + 2$
D) $\sqrt{30} + 2$

10.

$$4ax = by + 12$$
$$2ax - 5by = 15$$

The solution to the given system of equations is $(2,3)$. What is the value of $a + b$?

11.

$$3x + 4y = 6$$
$$-5x + y = a$$

In the system of equations below, a is a constant. What is the y-value of the solution to the system in terms of a ?

A) $\frac{30+3a}{23}$ B) $-\frac{30+3a}{23}$
C) $\frac{6+4a}{23}$ D) $-\frac{6+4a}{23}$

12.

$$5\left(9x + 7 + \frac{c}{5}\right) = 45x$$

In the given equation, c is a constant. The equation has infinitely many solutions. What is the value of c ?

13.

$$8x + 32y = 38$$
$$12x + 48y = 57$$

At how many points do the graphs of the given equations intersect in the xy-plane?

A) Exactly one
B) Infinitely many
C) Exactly two
D) Zero

At how many points = nechta nuqtada

Intersect = kesib o'tmoq

14.

$$(x + k) + \frac{9}{2}(y + t) + 19 = 0$$
$$(x + k) - \frac{9}{2}(y + t) - 19 = 0$$

In the given system of equations, k and t are constants. The solution to this system is (x, y). What is the value of $18(y + t)$?

A) -76 B) 0 C) 38 D) 9

15.

$$ax - by = 52$$
$$2ax - 8y = 48$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy-plane intersect at the point $(2,4)$. What is the value of b ?

A) 3 B) 4 C) -8 D) -3

16.

$$x^2 + y^2 = 36$$
$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

A) $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B) $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C) $-20m + 4\sqrt{11}$

D) $20m - 4\sqrt{11}$

17. Which of the following systems of equations has infinite solutions?

A) $7ax + 2by = 4c$
$7ax + 2by = -4c$

B) $-8by + 3ax = -12c$
$24by + 9ax = 36c$

C) $ax + 3by = 4c$
$18by + 6ax = 24c$

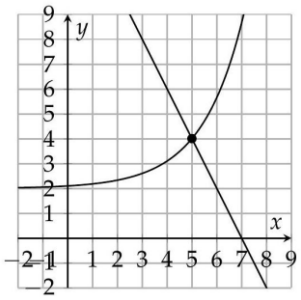
D) $15ax + 30by = 10c$
$6by - 10x = 3ax$

18.

$$4x - 6y = 10y + 2$$
$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

19.



linear equation = to'g'ri chiziqli tenglama

nonlinear equation = to'g'ri chiziqli bo'lmagan tenglama

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

A) $(4,0)$
B) $(5,4)$
C) $(0,0)$
D) $(0,5)$

20. [October 2025]

In the xy-plane, line k passes through the points $(3,0)$ and $(0,5)$. Which of the following is an equation of line k ?

A) $3x - 5y = 15$
B) $5x + 3y = 15$
C) $3x + 5y = 15$
D) $5x - 3y = 15$

21. [October 2025]

$$7j = k + 13m$$

The given equation relates the distinct positive numbers j, k, and m. Which equation correctly expresses j in terms of k and m ?

A) $j = 7(k + 13m)$
B) $j = \frac{k}{7} + 13m$
C) $j = k + \frac{13m}{7}$
D) $j = \frac{k+13m}{7}$

22. [August 2025]

$$x^2 + y + 3 = 3$$
$$8x + 16 - y = 0$$

The solution to the given system of equations is (x, y). What is the value of x ?

A) -16
B) -4
C) 2
D) 8

23. [August 2025]

$$y = (x - 6)(x + 5)$$
$$y = 11x - 66$$

The solution to the given system of equations is (x, y). What is the value of $2y$?
A) 0
B) -5
C) -6
D) 6

24. [August 2025]

$$y = 3x + 11{,}000$$
$$y = -3x + 13{,}000$$

The solution to the given system of equations is (x, y). What is the value of $2y$?
A) 48000
B) 24,000
C) 12,000
D) 2,000

25. [September 2025]

$$y = -15x + 19$$
$$y = -20x + 24$$

What is the solution (x, y) to the given system of equations?
A) (24,19)
B) (19,24)
C) (4,1)
D) (1,4)

26. [September 2025]

$$8x + 32y = 30$$
$$12x + 48y = 45$$

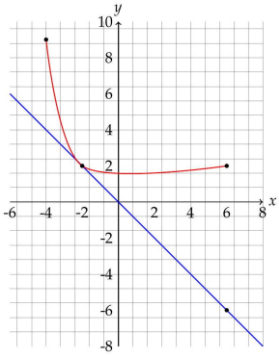
At how many points do the graphs of the given equations intersect in the xy-plane?
A) zero
B) exactly one
C) infinitely many
D) exactly two

27. [October 2025]

$$0.5x + 1.2y = 90$$
$$1.5x + 3.6y = 27$$

How many solutions does the given system of equations have??
A) Infinitely many
B) Exactly two
C) Exactly one
D) Zero

28. [October 2025]



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?
A) $(0,0)$
B) $(0,-2)$
C) $(-2,2)$
D) $(3,0)$

29. . [November 2023]

$$4x - 16y = 8$$

One of the two equations in the system of linear equations is given. The system has no solution. Which of the following be the second equation in this system?

A) $x - 8y = 0$ B) $\frac{1}{2}x - 2y = 1$
C) $\frac{1}{2}x - 2y = 0$ D) $x - 4y = 2$

30. [March 2024]

$$x + 4y = -16$$
$$5x + 20y = 0$$
$$3x + 12y = 34$$

The given equations are graphed in the xy plane. How many solutions (x, y) does the system of three equations have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

31. [March 2024]

$$x + 5y = -18$$
$$6x + 30y = 0$$
$$4x + 20y = 28$$

The given equations are graphed in the xy plane. How many solutions (x, y) does the system of equations have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

32. [March 2024]

$$y = 6x^2 - 36x + 56$$
$$y + 3 = 0$$

How many solutions are there to the given system of equations?

A) There is exactly 1 solution.
B) There are exactly 2 solutions.
C) There are exactly 3 solutions.
D) There are no solutions.

33. [March 2024]

$$2(9x) + 5(8y) = 284$$
$$2(9x) - 5(8y) = -356$$

The solution to the given system of equations is (x, y). What is the value of $9x + 8y$?

A) -82
B) -18
C) 46
D) 64

34. [May 2024]

$$31(x - n) = 31y + 31n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

A) $3x - 3y = 6n$
B) $3x + 3y = 3n$
C) $3x + 3y = 6n$
D) $3x - 3y = 3n$

35. [May 2024]

$$2x + 5y = 3$$
$$8x + 20y = 12$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A) $\left(r, -\frac{5r}{2} + \frac{3}{2}\right)$
B) $\left(r, \frac{2r}{5} + \frac{3}{5}\right)$
C) $\left(\frac{r}{4} + 3, -\frac{r}{4} + 12\right)$
D) $\left(-\frac{5r}{2} + \frac{3}{2}, r\right)$

36. [August US 2024]

Which system of linear equations has infinitely many solutions?

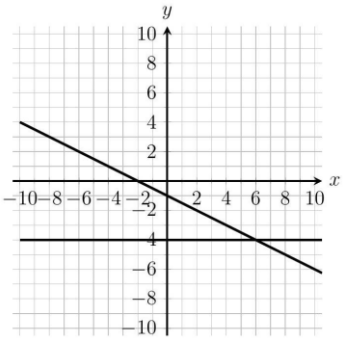
A) $4x + 2y = 12$
$4x + 8y = 48$

B) $x + 2y = 12$
$x + 8y = 48$

C) $x + 2y = 12$
$4x + 8y = 48$

D) $4x + 2y = 12$
$x + 8y = 48$

37.



The graph of a system of two linear equations is shown. If a third equation, $x + 2y =$ -8, is added to this system, how many solutions will the resulting system of three equations have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

Shown = ko'rsatilgan
Third equation = uchinchi t-ma
Added = qo'shildi
Resulting = natijaviy
resulting system =natijaviy sistema
three = 3
have = ega bo'lmoq

38. [August US 2024]

$$5x + 7y = 8$$
$$30x + 42y = 48$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A) $\left(r, -\frac{5r}{7} + \frac{8}{7}\right)$
B) $\left(-\frac{5r}{7} + \frac{8}{7}, r\right)$
C) $\left(-\frac{5r}{7} + 8, \frac{5r}{7} + 48\right)$
D) $\left(\frac{r}{6} + 8, -\frac{r}{6} + 48\right)$

39. [October US 2024]

$$2x = 20y - 8$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

A) $x = 5y$
B) $\frac{1}{2}x = 5y$
C) $x = 10y - 4$
D) $\frac{1}{2}x = 10y - 4$

40. . [October US 2024]

$$6x + 3y = 5$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

I. $9x + 4.5y = 7.5$
II. $9x - 4.5y = 7.5$

A) I only
B) II only
C) I and II
D) Neither I and II

given equation = berilgan t-ma
is = bu
one equation = bir t-ma
If = agar
at least = eng kamida
other equation= boshqa tenglama

41. [November 2024]

$$45x + 33y - 200 = -359$$
$$2wx + 11y = -53$$

In the given system of equations, w is a constant. The system has infinitely many solutions. What is the value of w ?

42. [November US 2024]

$$22x - 21y = 43$$
$$5x + 14y = 59$$

The solution to the given system of equations is $(x, y) = \left(\frac{p}{59}, \frac{w}{826}\right)$, where p and w are integers. What is the value of p ?

43. [December 2024]

$$13(x - n) = 13y + 13n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

A) $2x - 2y = 4n$
B) $2x + 2y = 2n$
C) $2x + 2y = 4n$
D) $2x - 2y = 2n$

44. [June 2025]

$$\frac{1}{4}(s + 4t) = 544$$
$$7s - 8t = 1{,}192$$

The solution to the given system of equations is (s, t). What is the value of $s - t$?

Linear function --> Chiziqli funksiya

1. [August 2025]

The function f is defined by $f(x) = 25x + 70$. What is the value of $f(x)$ when $x = 2$
A) 50
B) 97
C) 120
D) 190

2. [August 2025]

The table below gives the number of hours, h, of labor and a plumber's total charge $f(h)$, in dollars, for two different jobs.

Hours (h)	Total Charge $f(h)$
1	120
3	220

There is a linear relationship between h and $f(h)$.
Which equation represents this relationship?
A) $f(h) = 20h + 100$
B) $f(h) = 100h + 20$
C) $f(h) = 50h + 70$
D) $f(h) = 70h + 50$

3. [August 2025]

What is the slope of the graph of $60x - 10y = -48$ in the xy-plane?
A) -6
B) $-\frac{5}{4}$
C) $\frac{5}{4}$
D) 6

4. August 2025]

In the relationship between variables x and y, each increase of 8 in the value of x decreases the value of y by 3 . When the value of x is 16 , the value of y is 10 . Which equation represents this relationship?
A) $y = -\frac{3}{8}x + 10$ B) $y = -\frac{3}{8}x + 16$
C) $y = -\frac{8}{3}x + 10$ D) $y = -\frac{8}{3}x + \frac{158}{3}$

5. [August 2025]

Line k contains the points $(-3,-48)$, $(v,0)$, and $(5,56)$. What is the value of v ?

6. [August 2025]

x	y
0	n
5	$n+24$
10	$n+48$

There is a linear relationship between x and y. The table shows three values of x and their corresponding values of y in terms of a constant n. What is the slope of the line that represents this relationship in the xy-plane?

7. [August 2025]

The function m is defined by $m(x)=3x+6$ and the function p is defined by $p(x)=6-x$. What is the value of $2m(6)-p(6)$?

8. . [August 2025]

What is the slope of the graph of $48x-6y=-32$ in the $xy-$ plane?

A) -8 B) $-\frac{3}{2}$ C) $\frac{3}{2}$ D) 8

9. [August 2025]

The table gives the number of hours, h, of labor and a plumber's total charge $f(h)$, in dollars, for two different jobs.

h	$f(h)$
1	125
3	225

There is a linear relationship between h and $f(h)$. Which equation represents this relationship?
A) $f(h) = 25h + 100$
B) $f(h) = 100h + 25$
C) $f(h) = 50h + 75$
D) $f(h) = 75h + 50$

10. [August 2025]

The function f is defined by $f(x) = 7x - 65$. If the point $(k, 2k)$ lies on the graph of $y = f(x)$ in the xy-plane, where k is a constant, what is the value of k ?
A) 7
B) 14
C) 13
D) 165

11. [August 2025]

In the relationship between variables x and y, each increase of 7 in the value of x decreases the value of y by 3 . When the value of x is 14 , the value of y is 9 . Which equation represents this relationship?
A) $y = \frac{-3}{7}x + 9$
B) $y = \frac{-3}{7}x + 15$
C) $y = \frac{-7}{3}x + 9$
D) $y = \frac{-7}{3}x + \frac{125}{3}$

12. [August 2025]

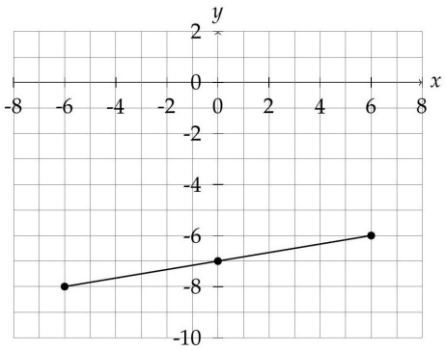
x	g(x)
1	44
2	38
3	32
4	26

For the linear function g, the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

13. [September 2025]

For the linear function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(0,6)$ and $(7,7)$. What is the slope of $y = f(x)$?

14. [October 2025]



The graph of the linear function f is shown, where $y = f(x)$. What is the slope of the graph of f ?

A) $\frac{1}{6}$ B) -7 C) $-\frac{1}{6}$ D) 7

15. [October 2025]

In the xy-plane, what is the slope of the line that passes through the points $(0,0)$ and $(13,14)$?

A) 13

B) $\frac{14}{13}$

C) $\frac{13}{14}$

D) 14

16. [October 2025]

x	y
$-2s$	17
$-s$	14
s	8

The table shows three values of a and their corresponding values of y, where s is a constant. There is a linear relationship between a and y. Which of the following equations represents this relationship?

A) $sx + 3y = 11s$

B) $3x + sy = 11s$

C) 5

D) 10

17. [September 2025]

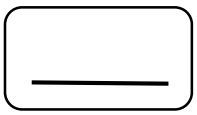
Number of cars	Maximum number of passengers and crew
2	102
6	294
10	486

The table shows the linear relationship between the number of cars, c , on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A) $48p - c = -6$
B) $48c - p = 6$
C) $48p - c = 6$
D) $48c - p = -6$

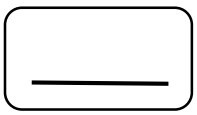
18. [March US 2023]

Line k in the xy-plane has slope $-\frac{2p}{5}$ and y-intercept $(0, p)$, where p is a positive constant. What is the x-coordinate of the x intercept of line k ?



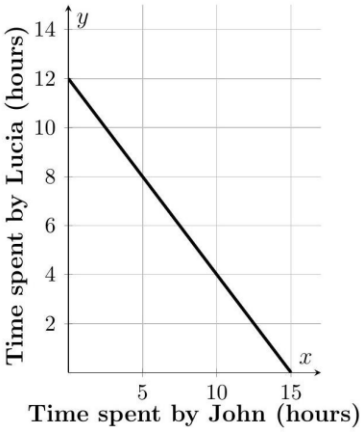
19. [March US 2023]

Line k passes through the points $(1,1)$ and $(2,6)$ in the xy-plane. If the equation for the line k is written in the form $y = mx + b$, where m and b are constants, what is the value of m ?



20. [March US 2023]

Lucia and John will work together to make 60 paper flowers for a school party. The line shown represents the possible combinations of time, in hours, spent by Lucia and John to fulfill this task.



According to the graph, on average, how many paper flowers will Lucia make per hour.

A) 4
B) 5
C) 12
D) 15

21. [June 2023]

x	y
0	n
4	$n + 19$
8	$n + 38$

There is a linear relationship between x and y. The table shows three values of x and their corresponding values of y in terms of a constant n. What is the slope of the line that represents this relationship in the xy plane?

22. [June 2023]

What is the y-intercept of the graph of $3x + 2y = 96$ in the xy-plane?
A) $(0,5)$
B) $(0,6)$
C) $(0,32)$
D) $(0,48)$

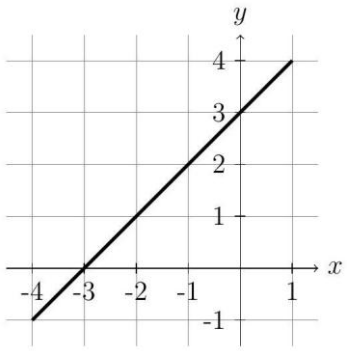
23. June 2023]

For the linear function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(0,2)$ and $(3,3)$. What is the slope of $y = f(x)$?

24. [August US 2023]}

A line in the xy-plane passes through the points $(0,12)$ and $(1,14)$. The equation of the line is $y = mx + b$, where m and b are constants. What is the value of b ?

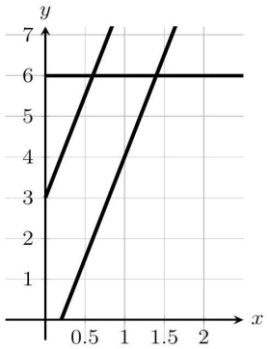
25. [August US 2023]



Which is an equation of the graph shown?
A) $y = x + 1$
B) $y = x - 1$
C) $y = x + 3$

D) $y = x - 3$

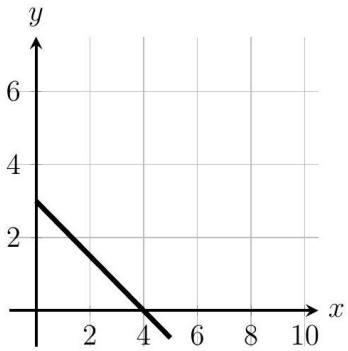
26. [August US 2023]



How many solutions (x, y) does the systems of three linear equations represented in the graph have?

A) 0
B) 1
C) 2
D) 3

27. [November 2023]



The graph of the linear equation

$$15x + By = 60$$

is shown, where B is a constant. What is the value of B ?

A) $\frac{3}{4}$
B) 3
C) 4
D) 20

28. [November 2023]

An equation of the line in the xy-plane that contains the points $(0,2)$ and $(8,0)$ is $ax + by = 1$, where a and b are constants. What is the value of a ?

29. . [December 2023]

The function g is defined by $g(x) = \frac{4}{5}x - 32$. What is the x coordinate of x intercept of the graph of $y = g(x)$ in the xy-plane?

30. [December 2023]

When the linear function h is graphed in the xy-plane, where $y = h(x)$, its slope is -2 . If $h(5) = 3$, which equation defines h ?

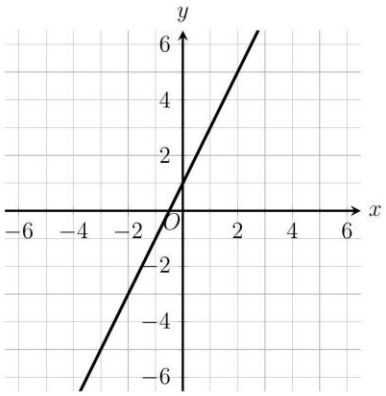
A) $h(x) = -2x + 5$
B) $h(x) = -2x + 13$
C) $h(x) = -2x + 3$
D) $h(x) = -2x - 7$

31. [March 2024]

x	y
-11	-25
9	55

The table shows two values of x and their corresponding values of y. The graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{3}, a\right)$. What is the value of a ?

32. [March 2024]



The graph of $y = f(x)$ is shown. What is the y-intercept of the graph of $y = f(x) + 13$?

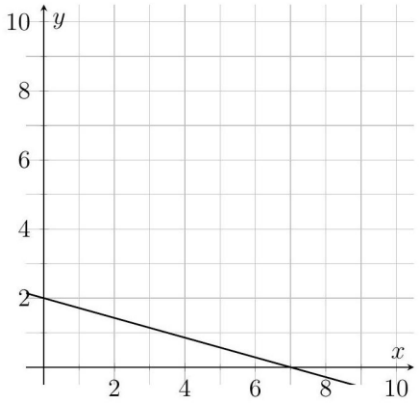
A) $(0, -12)$

B) $(0,14)$

C) $(1,13)$

D) $(0,13)$

33.



The graph of the linear equation

$$8x + By = 56$$

is shown, where B is a constant. What is the value of B ?

34. . [March 2024]

$y = -2x + 70$ Which table gives the values of x and the corresponding values of y for the given equation?

A)

x	y
0	0
1	2
2	4

C)

x	y
0	0
1	72
2	144

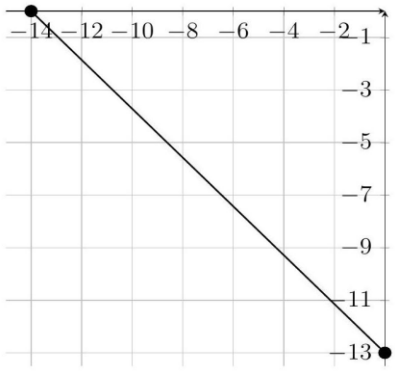
B)

x	y
0	0
1	72
2	74

D)

x	y
0	70
1	68
2	66

35.



The point ($v, -7$) lies on the line shown. What is the value of v ?

A) $-\frac{84}{13}$ B) $\frac{13}{14}$ C) $\frac{83}{14}$ D) $\frac{280}{13}$

36. For the linear function f, the graph of $y = f(x)$ in the xy-plane has a slope of $\frac{4}{5}$ and passes through the point $(20,19)$. Which equation defines f ?

A) $f(x) = \frac{4}{5}x + \frac{19}{20}$
B) $f(x) = \frac{4}{5}x + \frac{24}{5}$
C) $f(x) = \frac{4}{5}x + 19$
D) $f(x) = \frac{4}{5}x + 3$

37. [March 2024]

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to x kelvins. If a temperature is increased by 11.70 kelvins, by how much did the temperature increase in degrees Fahrenheit?

38. [May US 2024]

$$g(x) = 3(16x - 17)$$

What is the y-coordinate of the y-intercept of the graph of $y = g(x) - 5$ in the xy-plane?
A) -56
B) -51
C) -22
D) -17

39. [May 2024]

$$f(x) = 5x + b$$

For the linear function f, b is a constant and $f(7) = 35$. What is the value of b ?

A) 0 B) 1 C) 5 D) 7

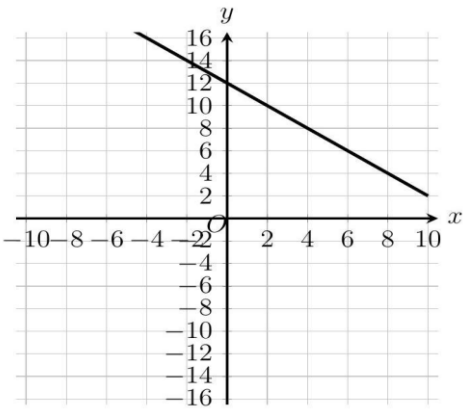
40. [May US 2024]

Number of cars	Maximum number of passengers and crew
4	164
6	242
8	320

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p ?

A) $39c - p = -8$
B) $39c - p = 8$
C) $39p - c = -8$
D) $39p - c = 8$

41.



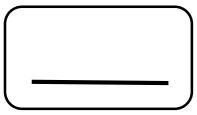
What is an equation of the graph shown?

A) $y = x + 12$
B) $y = -x + 12$
C) $y = 2x + 12$
D) $y = -2x + 12$

42. [May US 2024]

x	y
-11	-33
6	52

The table shows two values of x and their corresponding values of y. The graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{4}, a\right)$. What is the value of a ?



43. . [June 2024]

$$5x - 4y = -20$$

For the given equation, which table gives three values of x and their corresponding values of y ?

A)

x	0	4	8
y	5	10	15

B)

x	0	4	8
y	15	10	5

C)

x	5	10	15
y	0	4	8

D)

x	5	10	15
y	8	4	0

44. [June 2024]

$$7x - 5y = -70$$

For the given equation, which table gives three values of x and their corresponding values of y ?

A)

x	0	5	10
y	14	21	28

B)

x	0	5	10
y	28	21	14

C)

x	14	21	28
y	0	5	10

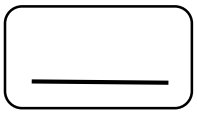
D)

x	14	21	28
y	10	5	0

Linear functions 2-vazifa

1. . [October 2025]

Charles saves $\frac{3}{5}$ of the $225 he earns each week from his summer job. If Charles continues to save at this rate, how much money, in dollars, will Charles save in 3 weeks?



Saves = yig'moq, saqlamoq
He = u
Earns = pul ishlab topmoq
each = har
week = hafta
from ....=... dan
job = ish
at this rate = shu ko'rsatgichda
how much money = qancha pul
in 3 weeks = 3 haftada

charges = haq oladi
for = uchun
first hour = dastlabki soat, birinchi soat
each additional hour = har bir qo'shimcha soat uchun
give =beradi, bermoq
charge = haq, to'lov
for h hours of work = isghning h soati uchun
integer = butun son

2. [June 2023]

A business consultant charges $408 for the first hour and $204 for each additional hour of work. Which of the following functions gives the charge $C(h)$, in dollars, for h hours of work, where h is a positive integer?
A) $C(h) = 204h + 204$
B) $C(h) = 204h + 408$
C) $C(h) = 408h + 204$
D) $C(h) = 408h + 612$

3. [December 2024]

The function f is defined by $f(x) = \frac{x+11}{3}$, and $f(a) = 16$, where a is a constant. What is the value of a ?
A) 9
B) 16
C) 59
D) 37

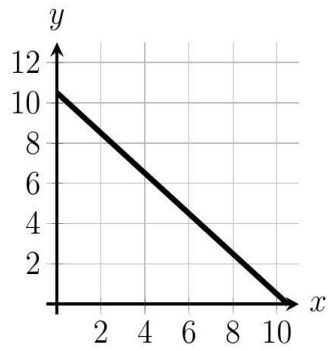
4. [August US 2023]

The function p is defined by

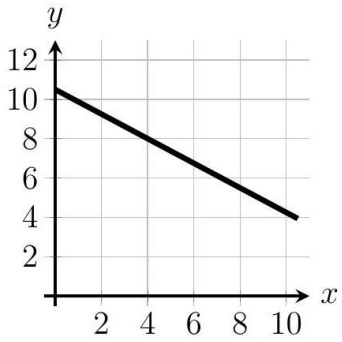
$$p(x) = -\frac{5}{8}x + \frac{21}{2}$$

What is the graph of $y = p(x)$?

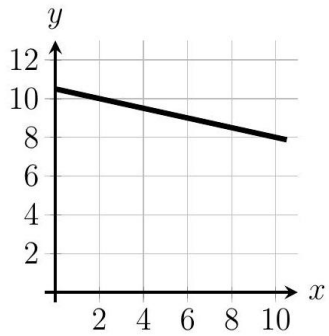
A)



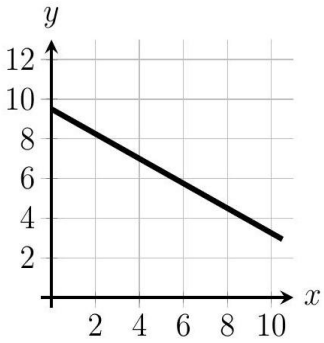
B)



C)



D)



5. A line in the xy-plane passes through the points $(0,12)$ and $(1,14)$. The equation of the line is $y = mx + b$, where m and b are constants. What is the value of b ?

A line = bir chiziq

in the xy-plane = xy koordinatalar tekisligida

passes through =orqali o'tmoq

6. [November 2023]

A business consultant charges $\$444$ for the first hour and $\$222$ for each additional hour of work. Which of the following functions gives the charges $C(h)$, in dollars, for h hours of work, where h is a positive integer?

A) $C(h) = 222h + 222$

B) $C(h) = 222h + 444$

C) $C(h) = 444h + 222$
D) $C(h) = 444h + 666$

7. [December 2023]

The cost of renting a piece of constructing equipment for up to 5 days is $350 for the first day and $175 for each additional day. Which of the following equations gives the cost y, in dollars, of renting the equipment for x days, where x is a positive integer and $x \leq 5$?

A) $y = 175x + 175$
B) $y = 350x - 175$
C) $y = 175x + 350$
D) $y = 350x + 175$

Cost =ijara
Of ...= ..ning
Renting = ijara
The cost of renting = ijaraning narxi
Equipment = jihoz
a piece of = ...ning bir bo'lagi
for = uchun
up to 5 days = 5kungacha

8. [March 2024]

The cost to rent a bus from Company X is $750 for the first 2 hours and an additional $50 per hour for each hour after the first 2 hours. If the total cost to rent the bus from Company X for t hours, where $t > 2$, is $1,050, which equation represents this situation?

A) $750(t - 2) + 50t = 1{,}050$
B) $750(2t) + 50t = 1{,}050$
C) $750 + 50(t - 2) = 1{,}050$
D) $750 + 50(2t) = 1{,}050$

The cost to rent = ijaraga olishga narx
a bus = bir avtobus
from = dan
after = dan keyingi
after the first 2 hours = dastlabki 2 soatdan keyingi
for each hour = har bir soat uchun
total cost = jami narx
represents = ifodalaydi
this = bu
situation = holat

9. [March 2024]

$$f(x) = 5x - 9$$

Function f is defined by the given equation. The function g is defined by $g(x) = f(x)(3x - 2)$ What is the x-coordinate of the x-intercept of the graph of $y = g(x)$ in the xy-plane?

A) $\frac{11}{8}$ B) $\frac{9}{5}$

C) $\frac{7}{2}$ D) $\frac{11}{2}$

10. [March 2024]

x	$g(x)$
-25	4
-9	0
15	6

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+5}$ and f is a linear function. What is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

11. [August 2025]

A book-of-the-month club charges $16 for the first book purchased during the month and an additional $11 for each book after the first. Which equation describes this situation, where y is the total cost, in dollars, for books purchased during a month, x is

Describes = tasvirlamoq
Month = oy
Purchased= xarid qilingan
During = davomida

the number of books purchased during the month,

and $x > 0$?

A) $y = x + 16$

B) $y = 11x + 16$

C) $y = 11(x - 1) + 16$

D) $y = (x - 1) + 16$

12. [March 2024]

The function f is defined by

$$f(x) = -12x + 28$$

What is the y-coordinate of the y intercept of the graph $y = f(x)$ in the xy-plane?

13. [May 2024]

The cost of renting a piece of equipment is $\$64n$ for the first day and $\$32n$ for each additional day, where n is a positive integer. Which of the following functions gives the cost $C(x)$ in dollars, of renting this equipment for x days, where x is a positive integer?

A) $C(x) = 32nx + 32n$

B) $C(x) = 32nx + 64n$

C) $C(x) = 64nx + 32n$

D) $C(x) = 64nx - 32n$

14. [May 2024]

Distance (miles)	Average time (minutes)
0.06	3
0.28	14
0.34	17

average time = o'rtacha vaqt
to travel = sayohat qilishga
certain = malum bir
distance = masofa

The table gives the average time t in minutes, it takes Adriana to travel a certain distance d in miles. Which equation could represent this linear relationship?

A) $t = \frac{1}{50}d$

B) $t = \frac{1}{2}d$

C) $t = 2d$

D) $t = 50d$

15. [May 2024]

x	y
$-2s$	20
$-s$	16
s	8

The table shows three values of x and their corresponding values of y, where s is a constant. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A) $sx + 4y = 12s$

B) $4x + sy = 12s$

C) $4x + sy = 12$

D) $sx + 4y = 12$

16. [May 2024]

A computer repair specialist charges $160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 4 hours of repair is $240. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A) $f(x) = 40x + 80$
B) $f(x) = 40x + 160$
C) $f(x) = 60x$
D) $f(x) = 60x + 160$

Repair = tamirlash

for the first two hours of repair = tamirlashning dastlabki 2 soati uchun

plus = pilus, qo'shimchasiga

an hourly fee = soatlik haq

17. [June 2024]

A carpenter charges a flat rate of $234 for the first 3 hours of work and $65 for each additional hours of work. Which equation gives the total amount y, in dollars, that the carpenter charges for x hours of work, where $x > 3$?

A) $y = 65x + 39$
B) $y = 234x + 65$
C) $y = 65x + 234$
D) $y = 234x + 429$

18. [August 2024]

For the linear function $p, p(c) = -6$, where c is a constant, $p(3) = 26$, and the slope of the graph of $y = p(x)$ in the xy-plane is 8. For the linear function $t, t(c) = -7$ and $t(4) = 38$. What is the slope of the graph of $y = t(x)$ in the xy-plane?

A) -1 B) 2 C) 8 D) 9

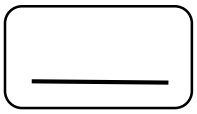
19. [August 2024]

In the xy-plane, line l has a slope of 4 and an x-intercept of $\left(\frac{4}{3}, 0\right)$. What is the y-intercept of line l ?

A) $\left(\frac{4}{3}, 0\right)$
B) $\left(-\frac{16}{3}, 0\right)$
C) $\left(0, \frac{4}{3}\right)$
D) $\left(0, -\frac{16}{3}\right)$

20. [August 2024]

If $f(x) = x + 3$ and $g(x) = 3x$, what is the value of $5f(2) - g(2)$?



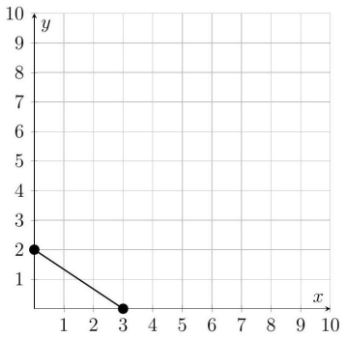
21. [August US 2024]

x	y
-12	94
-6	70
6	22
12	-2

The table shows four values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A) $24x + 6y = 46$
B) $24x + 6y = 276$
C) $6x + 24y = 46$
D) $6x + 24y = 276$

22.



The graph of the linear equation $4x + BY = 12$ is shown, Where B is a constant. What is the value of B ?

A) $\frac{2}{3}$

B) 3

C) 2

D) 6

23. [October US 2024]

x	y
0	12
1	19
2	26

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. What is the y-coordinate of the y-intercept of the line in the xy-plane that represents this relationship?

24. [October US 2024]

The cost to rent a bus from Company X is 850 for the first 2 hours and an additional 30 per hour for each hour after the first 2 hours. If the total cost to rent the bus from Company X for t hours, where $t > 2$, is 1,030 , which equation represents this situation?

A) $850(t-2)+30t=1{,}030$
B) $850(2t)+30t=1{,}030$
C) $850+30(t-2)=1{,}030$
D) $850+30(2t)=1{,}030$

25. [October US 2024]

The value of y is 49 more than twice the value of x. Which equation represents y in terms of x ?

A) $y=2x+49$
B) $y=49x+2$
C) $y=51x-2$
D) $y=51x+49$

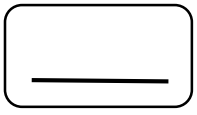
26. [October US 2024]

The linear function g is defined by $g(x)=15x-b$ where b is a constant. If $g(c+7)=c$, where c is a constant, which of the following expressions represents the value of b ?

(A) $\frac{15c}{4}$
(B) $\frac{19c}{4}+7$
(C) $\frac{61c}{4}+105$
(D) $14c+105$

27. [November 2024]

In the xy-plane, line p has a slope of $\frac{3}{4}$ and an x-intercept of $(19,0)$. What is the y-coordinate of the y-intercept of line p ?



28. [November 2024]

In the xy-plane, line p has a slope of $\frac{3}{4}$ and an x-intercept of $(19,0)$. What is the y-coordinate of the y-intercept of line p ?

29. [November 2024]

The function f and g are defined as

$$f(x) = \frac{1}{5}x - 9 \text{ and } g(x) = \frac{4}{5}x + 27$$

If the function h is defined as $h(x) = f(x) + g(x)$ what is the x-intercept of the graph of the $y = h(x)$ in the xy-plane?

A) $\left(-\frac{135}{4}, 0\right)$

B) $(-18,0)$

C) $\left(\frac{45}{4}, 0\right)$

D) $(18,0)$

30. [November US 2024]

x	y
2	$-\frac{183}{10}$
4	$-\frac{171}{10}$
7	$-\frac{153}{10}$

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. What is the x-coordinate of the x-intercept of the graph that represents this linear relationship in the xy-plane?

1. [December 2024]

$$y = \frac{2}{5}x + 2$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

A) -2

B) $-\frac{1}{2}$

C) $\frac{1}{2}$

D) 2

2. [May 2024]

Line k is defined by $y = 6x + \frac{1}{9}$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

A) -9

B) $-\frac{1}{6}$

C) $\frac{1}{9}$

D) 6

3. [May US 2024]

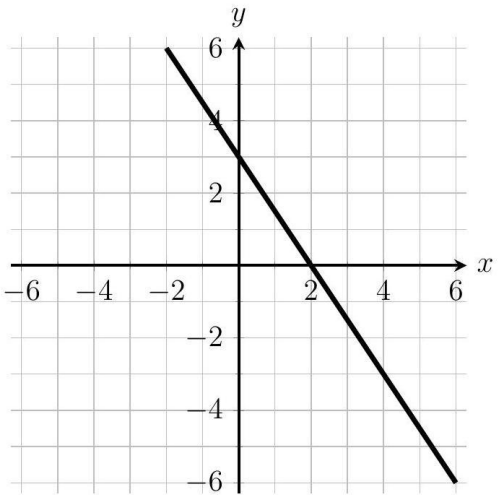
Line r is defined by the equation

$$4x - 7y = 3$$

Line s is parallel to line r in the xy-plane. What is the slope of line s ?

A) $\frac{7}{4}$ B) $\frac{4}{7}$ C) -4 D) -7

4.



Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k and passes through the point $(-12, -19)$. Which equation defines line j ?

A) $y = \frac{2}{3}x - 11$

B) $y = \frac{2}{3}x - 27$

C) $y = \frac{3}{2}x - 11$

D) $y = \frac{3}{2}x - 27$

5. [June 2024]

Line k is defined by $y = 6x + 5$. Line j is parallel to line k in the xy-plane and passes through the point $(0,11)$. Which equation defines line j ?

A) $y = 6x + 11$

B) $y = 11x + 5$

C) $y = -6x + 11$

D) $y = -11x + 5$

6. [August 2024]

Line k is defined by $y = 4x + 1$. Line j is parallel to line k in the xy-plane and passes through the point $(0,5)$. Which equation defines line j ?

A) $y = -5x + 1$

B) $y = 5x + 1$

C) $y = -4x + 5$

D) $y = 4x + 5$

7. [August 2024]

In the xy-plane, line s passes through the point (0,0) and is parallel to the line represented by the equation $y = 25x + 5$. If line s also passes through the point $(2, d)$, what is the value of d ?

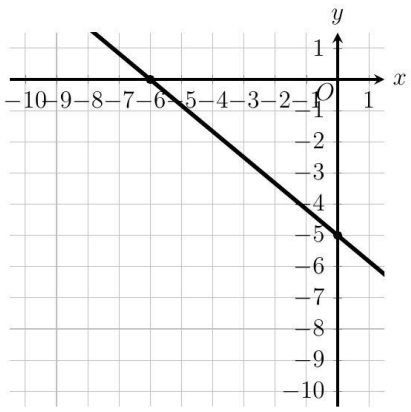
A) 5
B) 25
C) 50
D) 55

8. [October US 2024]

Line r is defined by equation $4x - 3y = 8$. Line s is parallel to line r in the xy-plane. What is the slope of line s ?

A) -4
B) -3
C) $\frac{3}{4}$
D) $\frac{4}{3}$

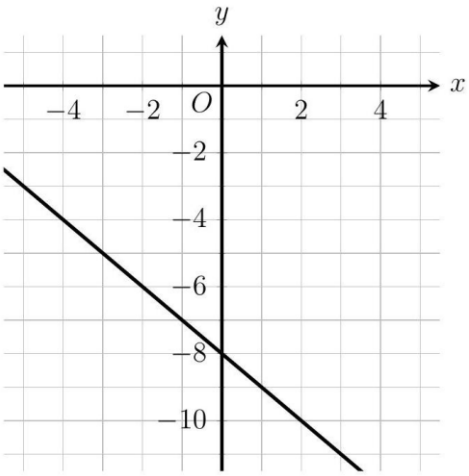
9.



Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k. What is the slope of line j ?

A) $-\frac{6}{5}$
B) $\frac{6}{5}$
C) 6
D) -5

10.



The graph of the linear function $y = f(x) - 19$ is shown. If c and d are positive constants, which equation could define f ?

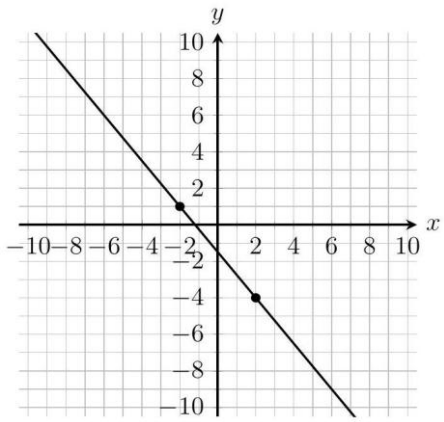
A) $f(x) = d - cx$

B) $f(x) = -d + cx$

C) $f(x) = d + cx$

D) $f(x) = -d - cx$

11.



Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k and passes through the point $(7,15)$. What is the y coordinate of the y-intercept of line j ?

A) $-\frac{47}{4}$

B) $-\frac{3}{2}$

C) $\frac{4}{5}$

D) $\frac{47}{5}$

12. In the xy-plane, lines l and k are parallel. Line l passes through the points $(-1,-7)$ and $(1,3)$. If the point (a,b) lies on line k, which of the following is another point on line k ?

A) $(a-1,b-5)$
B) $(a-1,b+5)$
C) $(a+5,b+1)$
D) $(a+5,b-1)$

13. . In the given pair of equations below, a and b are constants. The graph of this pair of equations in the xy-plane is a pair of perpendicular lines. Which of the following pairs of equations represents a pair of parallel lines?

$$7x+11y=100$$
$$ax+by=10b$$

A) $7x-11y=10$
$ax+by=10b$

B) $7x+33y=100$
$2ax-3by=50b$

C) $33x-7y=100$
$3ax+by=\mathrm{b}$

D) $11x+7y=100$
$ax+by=10b$

14. equations in the xy-plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

$$3x+8y=10$$
$$ax+by=10$$

A) $3x-8y=10$
$ax+by=10$

B) $6x-16y=10$
$2ax+2by=5$

C) $3x+16y=10$
$2ax+by=10$

D) $3x+16y=10$
$ax+2by=10$

15. In the xy-plane, lines l and k are parallel. Line l passes through the points $(-1,-7)$ and $(1,3)$. If the point (a,b) lies on line k, which of the following is another point on line k ?

A) $(a-1,b-5)$
B) $(a-1,b+5)$
C) $(a+5,b+1)$
D) $(a+5,b-1)$

16. In the xy-plane, a square has vertices at points A(1, 2), B(4, 6), C(0, 9), and D(-3, 5). Line M is perpendicular to the diagonal AC of the square and passes through the midpoint of side BD. Which of the following equations represents line M?

A) $7x + y = 6$
B) $7x - y = -5$
C) $x - 7y = -41$
D) $x + 7y = 41$

Linear Inequalities → Chiziqli tengsizliklar

1. [June 2023]

In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to $7x + 4y > 12$?
A) The region where $x > 0$ and $y > 0$
B) The region where $x < 0$ and $y > 0$
C) The region where $x < 0$ and $y < 0$
D) The region where $x > 0$ and $y < 0$

2. [June 2023]

In December 2017, the lowest temperature recorded in a certain city was 40 degrees Fahrenheit ($F°$) and the highest temperature recorded was $90°F$. Which inequality is true for all values of t, where t represents any temperature in $F°$, recorded in the city in December?
A) $40 \leq t \leq 90$
B) $t \leq 40$
C) $t \leq 50$
D) $t \geq 90$

Lowest = eng past, quyi
Recorded = qayt etildi
in a certain city = malum bir shaharda
highest =eng baland
for all values of t= t ning barcha qiymatlari uchun
any = biror bir

3. [August US 2023]

$$y \leq \frac{2}{3}x + 5$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
-1.5	6
1.5	5
3	-4

B)

x	y
-1.5	5
1.5	7
3	8

C)

x	y
-1.5	6
1.5	5
3	6

x	y
-1.5	6
1.5	5
3	-4

D)

x	y
-1.5	3
1.5	5
3	6

x	y
-1.5	6
1.5	5
3	-4

4. [December 2023]

A model estimates that a gray whale travels 73 to 78 miles each day during its migration. Based on this model, which inequality represents the estimated total number of miles, x, a gray whale could travel in 31 days of its migration?

A) $(73)(31) \leq x \leq (78)(31)$
B) $73 + 31 \leq x \leq 78 + 31$
C) $73 \leq 31x \leq 78$
D) $73 \leq 31 + x \leq 78$

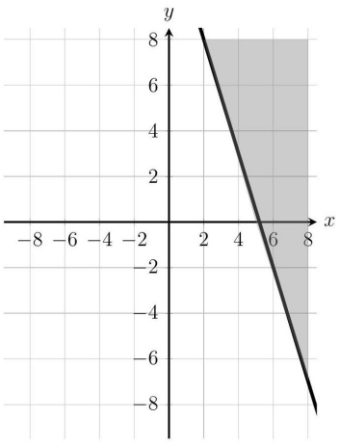
Estimates = tahmin qiladi
Travels = sayohat qiladi
each day = har kuni
during = davomida
its = uning
Based on....= ...ga asoslanib
this model = bu model
Based on this model = bu modelga asoslanib
Estimated = tahminiy
total number = jami soni

shaded= bo'yalgan
region= hudud
shown = ko'rsatilgan

5. [December 2023]

The shaded region shown represents solutions to inequality. Which ordered pair (x, y) is a solution to this inequality?

A) $(-7,0)$
C) $(0,7)$
B) $(0,-7)$
D) $(7,0)$



6. [March 2024]

The minimum value of x is 11 less than 8 times another number n. Which inequality shows the possible values of x ?

A) $x \leq 8n - 11$
B) $x \geq 8n - 11$
C) $x \leq 11 - 8n$
D) $x \geq 11 - 8n$

11 less than 8 times..= ..ning 8 martasidan ko'ra 11 taga kamroq

Possible = mumkin bo'lgan

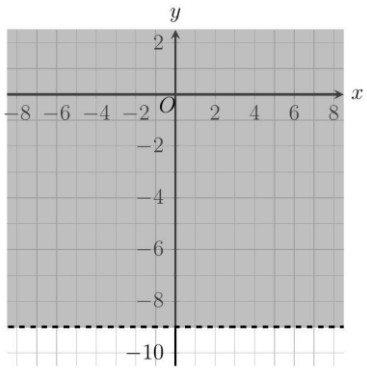
7. [November 2023]

$$y \leq x + 9$$
$$y \geq -2x - 9$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A) $(0,-10)$
C) $(-7,0)$
B) $(0,10)$
D) $(7,0)$

8.



The shaded region shown represents the solutions to the inequality $-18y < c$, where c is a constant. What is the value of c ?

A) 162
B) 9
C) -9
D) -162

9. [March 2024]

$$y \leq x + 2$$
$$y \geq -3x - 4$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A) $(0, -8)$
B) $(0,8)$
C) $(-8,0)$
D) $(8,0)$

10. [March 2024]

$$y \leq x + 2$$
$$y \geq -4x - 6$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A) $(0, -12)$
B) $(0,12)$
C) $(-12,0)$
D) $(12,0)$

11. [May US 2024]

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x. The product of 18 and the fourth odd integer is at most 22 less than the sum of the first and third odd integers. Which inequality represents this situation?

A) $18(x+6) \leq x + (x+4) - 22$
B) $18(x+6) \geq 22 - (x + (x+4))$
C) $18(x+4) \leq x + (x+3) - 22$
D) $18(x+4) \geq 22 - (x + (x+3))$

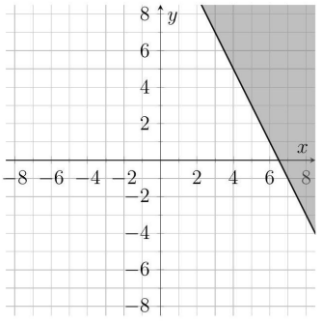
In a set = to'plamda,
in = da , set = to'plam
consecutive= ketma-ket kelgan
odd integers= toq butun son
even = juft
where = bu yerda
ordered = tartiblangan
from least = kichigidan
to greatest = kattasiga
product =ko'ppaytmasi
four = 4 fourth = to'rtinchi
one = bir first = birinchi
at most= ko'pida ($\leq$)

12. [June 2024]

During a portion of a flight, a small airplane's cruising speed varied between 125 miles per hour and 135 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

A) $s \leq 10$
B) $s \leq 125$
C) $s \leq 135$
D) $125 \leq s \leq 135$

13.



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

A) $(-8,0)$
B) $(0,8)$
C) $(0,-8)$
D) $(8,0)$

14. [August 2024]

A newspaper stand sells magazines for $5 each and newspapers for $2 each. The stand needs to sell m magazines and n newspapers for a combined total value of at least $150 each day to cover operation costs. Which inequality represents this situation?

A) $5m + 2n \leq 150$

B) $5m + 2n \geq 150$

C) $7mn \leq 150$

D) $7mn \geq 150$

Sells = sotadi
Needs = kerak
needs to sell = sotishi kerak
combined total value =birgalikdagi jami qiymati

at least $150 = kamida 150 dollar

15. [August 2024]

$$y < -6x - 22$$
$$y > -4x - 7$$

For what of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	-9	-10	-12
y	27	51	-4

B)

x	-9	-10	-12
y	32	36	-4

C)

x	-9	-10	-12
y	28	36	48

D)

x	-9	-10	-12
y	30	37	48

16. A model estimates that a gray whale travels 72 to 77 miles each day during its migration. Based on this model, which inequality represents the estimated total number of miles, x, a gray whale could travel in 14 days of its migration?

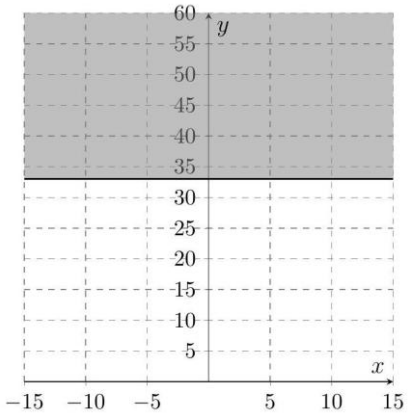
A) $(72)(14) \leq x \leq (77)(14)$

B) $72 + 14 \leq x \leq 77 + 14$

C) $72 \leq 14x \leq 77$

D) $72 \leq 14 + x \leq 77$

17.



The shaded region shown in the graph represents all the solutions to which inequality?

A) $x < 33$

B) $x \geq 33$

C) $y \leq 33$

D) $y \geq 33$

18. [October US 2024]

A botanist estimates the number of oak trees in a forest to be at least 2600 and no more than 3200 . Which inequality best represents the number oak trees, x, in the forest?

A) $x \leq 600$

B) $x \leq 2600$

C) $2600 \leq x \leq 3200$

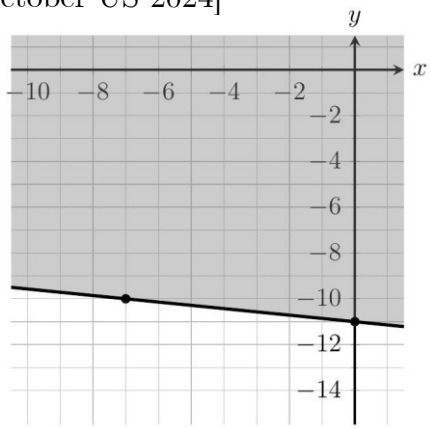
D) $x \leq 2600$ or $x \geq 3200$

Estimates = tahmin qiladi
to be = bo'lishi
no more than = dan ortiq emas

19. [October US 2024]

If $x \geq -6$ represent all solutions to the inequality $ax - 27 \leq 15$, where a is a constant, what is the greatest possible value of a ?

20. [October US 2024]



The shaded region shown represents solution to $rx + ty \geq -77$, where r and t are constants. What is the value of $r + t$?

A) 8
B) 6
C) -6
D) -7

21. [December US 2024]

$$y > 17$$
$$4x + y < 21$$

The point $(x, 54)$ is a solution to the given system of inequalities in the xy-plane. Which of the following could be the value of x ?

A) 9
B) 5
C) -5
D) -9

22. [December 2024]

Sabrina has a goal to save at least 240 USD. Sabrina has already saved 180 USD. If x represents the additional amount of money, in dollars, Sabrina needs to save to meet her goal, which inequality represents this situation?

A) $\frac{x}{180} \geq 240$
B) $180x \geq 240$
C) $180 + x \geq 240$
D) $180 - x \geq 240$

23. [December 2024]

$$y < 42 - 7x$$
$$\frac{y}{7} > 10$$

Which inequality represents the x values for all solutions (x, y) that satisfy the given system of inequalities in the xy-plane?

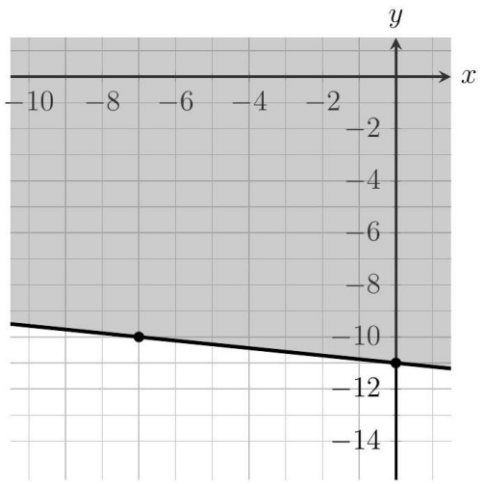
A) $x > 6$

B) $x < 6$

C) $x > -4$

D) $x < -4$

24.



The shaded region shown represents the solutions to $rx + ty \geq -77$, where r and t are constants. What is the value of $r + t$?

A) 8

B) 6

C) -6

D) -7

25. March US 2025]

$$y \leq 5x - 21$$

Which point (x, y) is a solution to the given inequality in the xy-plane?

A) $(-5, -21)$

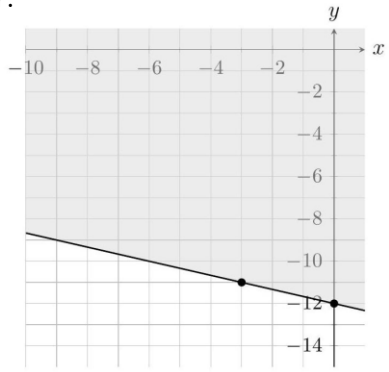
B) $(0,21)$

C) $(5, -21)$

D) $(6,21)$

26. In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x. The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

27.



The shaded region shown represents the solutions to $rx + ty \geq -36$, where r and t are constants. What is the value of $r + t$?

A) 4

B) 2

C) -2

C) -4

28. [May 2025]

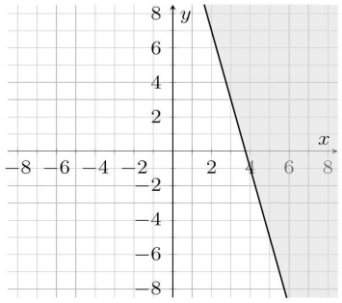
$$y \leq x + 3$$
$$y \geq -3x - 7$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A) $(0, -14)$ B) $(0,14)$

C) $(-14,0)$ D) $(14,0)$

29.



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

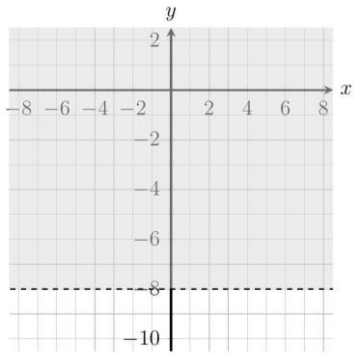
A) $(-5,0)$
B) $(0,5)$
C) $(0,-5)$
D) $(5,0)$

30. [June US 2025]

For a certain classification, the positive mass m, in grams, of an object must be less than 510 grams. Which inequality represents this situation?

A) $m > 510$
B) $m < 0$
C) $-510 < m < 0$
D) $0 < m < 510$

31.



The shaded region shown represents the solutions to the inequality $ry < 128$, where r is a constant. What is the value of r ?

A) 16
B) 8
C) -8
D) -16

32. A car travels at a speed of at least 25 miles per hour but no more than 50 miles per hour for a certain part of a trip. Which inequality represents this situation, where x is the speed of the car, in miles per hour, on this part of the trip?

A) $x \geq 25$

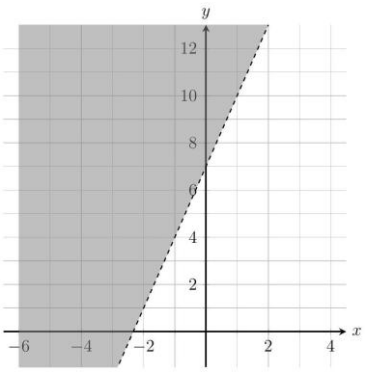
B) $x \geq 50$

C) $25 \leq x \leq 50$

D) $x \leq 75$

33. In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x. The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

34.



The shaded region shown represents the solutions to which quality?

A) $y > 3x + 7$

B) $y > 3x - 7$

C) $y < 3x + 7$

D) $y < 3x - 7$

35. [May 2025]

$$y > 5x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
4	11
6	21
9	36

B)

x	y
4	16
6	26
9	41

C)

x	y
4	21
6	31
9	46

D)

x	y
4	21
6	21
9	41

36. A school is ordering tablet computers for its students. The tablets cost $170 each, and 7% tax is added to the total cost of the order. If the school can spend no more than $53,400 on the tablets, including tax, what is the maximum number of tablets that the school can order?

37. A school is ordering tablet computers for its students. The tablets cost $164 each, and 6% tax is added to the total cost of the order. If the school can spend no more than 44,800 dollars on the tablets, including tax, what is the maximum number of tablets that the school can order?

38. A shopper goes to a pet store to buy dog food and treats. The shopper has $20. Each can of food costs $1.87, and each bag of treats costs $2.37. Which inequality represents the relationship between the number of cans of food, c, and the number of bags of treats, t, the shopper can buy?

A) $1.87c + 2.37t \leq 20$

B) $1.87c + 2.37t \geq 20$

C) $2.37c + 1.87t \leq 20$

D) $2.37c + 1.87t \geq 20$

39. The sum of 2 numbers is 85 . If the smaller number is 45 less than the larger number, what is the larger number?

A) 40
B) 55
C) 65
D) 70

40.

$$y > 3x + 1$$
$$2y + x < 13$$

In the xy-plane, point A is contained in the graph of the solution set of the system of inequalities above. Which of the following could be the coordinates of point A ?

A) $(-2, -8)$
B) $(1,5)$
C) $(1,7)$
D) $(3,3)$

41.

$$2x > 15$$
$$3y - 2x > 21$$

Which of the following consists of the y-coordinates of all the points that satisfy the system of inequalities below?

A) $y > 36$
B) $y > 12$
C) $y > 7$
D) $y > 3$

42. The maximum value of a is 13 more than 6 times another number b. Which inequality shows the possible values of a ?

A. $a \geq 13 + 6b$
B. $a \geq 13b + 6$
C. $a \leq 13 + 6b$
D. $a \leq 13b + 6$

43. In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to the $6y - 4x > 16$?

A) The region where $x > 0$ and $y > 0$
B) The region where $x < 0$ and $y > 0$
C) The region where $x > 0$ and $y < 0$
D) The x-axis.

44. Which of the following inequalities includes the y-coordinates of all points that satisfy the system of inequalities below?

$$y > 3x - 3$$
$$3x > 12$$

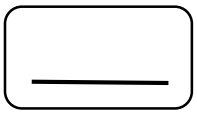
A) $y > 1$
B) $y > 4$
C) $y > 9$
D) $y > 15$

Quatratic function → kvadratik funksiya

1. [September 2025]

$$\frac{1}{cx} = \frac{x}{96} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?



2. [August 2025]

$$x^2 - 9x + 1 = 0$$

What is the one solution to the given equation?

A) $\frac{9+\sqrt{77}}{2}$

B) $\frac{9+\sqrt{85}}{2}$

C) $\frac{-9+\sqrt{77}}{2}$

D) $\frac{-9+\sqrt{85}}{2}$

3. [March US 2023]

$$h(t) = -4.9t^2 + 10t$$

The function h models the height $h(t)$, in meters, of a football t seconds after it is kicked. What is the interpretation of $h(2) = 0.40$ in this context?

A) The football has a maximum height of 0.40 meter

B) The football has a maximum height of 2 meters

C) The football has a height of 0.40 meter 2 seconds after it is kicked

D) The football has a height of 2 meters 0.40 second after it is kicked

4. [August 2025]

$$(x + k)^2 + 31 = 31$$

In the given equation, k is a constant. How many distinct real solutions does this equation have?

A) Zero

B) Exactly one

C) Exactly two

D) Infinitely many

5. [August 2025]

$$f(x) = 4x^2 + 56x + 197$$

The function g is defined by $g(x) = f(x + 6)$. What is the minimum value of $g(x)$?

A) -13
B) -7
C) 1
D) 7

6. [August 2025]

$$(x - k)^2 = (k - 4a)(x - k)$$

In the given equation, a and k are constants, where $k > 4a$ The sum of the solutions to the equation is $3k + 37$ What is the value of a ?

7. [September 2025]

If $\left(\frac{1}{2}x - \frac{21}{2}\right)^2 - 81 = 0$, what is the value of $(x - 21)^2$?

A) 324
B) 162
C) 21
D) 9

8. [September 2025]

$$y = 7x^2 - bx - 6$$

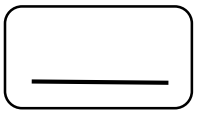
Which of the following equations is equal to the given equation above, where b is a positive constant.

A) $y = 7\left(x - \frac{b}{14}\right)^2 - 6 - \frac{b^2}{28}$
B) $y = 7\left(x - \frac{b}{14}\right)^2 - 6$
C) $y = 7\left(x + \frac{b}{14}\right)^2 - 6 - \frac{b^2}{28}$
D) $y = 7\left(x + \frac{b}{14}\right)^2 - 6$

9. [September 2025]

$$-x^2 + bx - 49 = 0$$

In the given equation, b is an integer. The equation has no real solution. What is least possible value of b ?



10. [August 2025]

$$x^2 - 9x + 1 = 0$$

What is one of the solutions to the given equation?

A) $\frac{9+\sqrt{77}}{2}$

B) $\frac{9+\sqrt{85}}{2}$

C) $\frac{-9+\sqrt{77}}{2}$

D) $\frac{-9+\sqrt{85}}{2}$

11. [August 2025]

$$y - x = 46$$
$$y = x^2 - 44x$$

The graphs of the equations in the given system intersect at the point (x, y) in the xy-plane. What is a possible value of y ?

A) 46

B) 47

C) 91

D) 92

12. [September 2025]

$$-5x^2 - x + 9 = 0$$

What is the greatest solution to the given equation?

A) $-\frac{1}{5} + \frac{\sqrt{181}}{5}$

B) $-\frac{1}{10} + \frac{\sqrt{181}}{10}$

C) $-\frac{1}{5} - \frac{\sqrt{181}}{5}$

D) $-\frac{1}{10} - \frac{\sqrt{181}}{10}$

13. [September 2025]

If $(x+2)^2 = 28$, what is the value of $x^2 + 4x$?
A) 56
B) 28
C) 32
D) 24

14. [September 2025]

$$\frac{1}{78}x^2 + \left(s - \frac{1}{78}t\right)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k ?

15. The function g is a quadratic function. In the xy-plane, the graph of $y = g(x)$ has a vertex at (-1, -4) and passes through the points ($-2, -43$) and ($1, -160$). What is the values of $g(0) - g(2)$?
A) -121
B) 0
C) 117
D) 312

16. [September 2025]

$$kx^2 - 16x = 27x^2 - 8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

17. [September 2025]

The expression $x^2 + kx + 14$, where k is a constant, can be rewritten as $(x + n)(x + 7)$, where n is a constant. What is the value of k ?

A) 2
B) 5
C) 9
D) 7

18. [September 2025]

$$-5x^2 - x + 9 = 0$$

What is the greatest solution to the given equation?

A) $-\frac{1}{5} + \frac{\sqrt{181}}{5}$
B) $-\frac{1}{10} + \frac{\sqrt{181}}{10}$
C) $-\frac{1}{5} - \frac{\sqrt{181}}{5}$
D) $-\frac{1}{10} - \frac{\sqrt{181}}{10}$

19. [September 2025]

In the xy-plane, the graph of the equation $y = -x^2 + 7x - 104$ intersects the line $y = c$ at exactly one point. What is the value of c ?

A) $-\frac{367}{4}$
B) $-\frac{7}{2}$
C) -104
D) $-\frac{465}{4}$

20. [September 2025]

$$\frac{1}{78}x^2 + \left(s - \frac{1}{78}t\right)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k ?

21. [September 2025]

The expression $x^2 + kx + 55$, where k is a constant, can be rewritten as $(x + n)(x + 11)$, where n is a constant. What is the value of k ?

A) 16
B) 6
C) 11
D) 5

22. [September 2025]

$$-3x^2 - 5x + 6 = 0$$

What is the greatest solution to the given equation?

A) $\frac{-5}{6} - \frac{\sqrt{97}}{6}$
B) $\frac{-5}{3} + \frac{\sqrt{97}}{3}$
C) $\frac{-5}{6} + \frac{\sqrt{97}}{6}$
D) $\frac{-5}{3} - \frac{\sqrt{97}}{3}$

23. [September 2025]

$$x^2 + 49x + 1 = 0$$

What is one of the solutions to the given equation?

A) $\frac{-49+\sqrt{(49)^2-4(1)(1)}}{2(1)}$
B) $\frac{-49+\sqrt{(49)^2-4(49)(49)}}{2(49)}$
C) $\frac{-1+\sqrt{(1)^2-4(1)(49)}}{2(1)}$
D) $\frac{-1+\sqrt{(1)^2-4(49)(1)}}{2(49)}$

24. [September 2025]

$$\frac{1}{24}x^2 + \left(s - \frac{1}{24}t\right)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k ?

25. [September 2025]
In the xy -plane, the graph of the equation $y = -x^2 + 7x - 104$ intersects the line $y = c$ at exactly one point. What is the value of c ?

A) $-\frac{367}{4}$
B) $-\frac{7}{2}$
C) -104
D) $-\frac{465}{4}$

26. [September 2025]

$$f(x) = -16x^2 + 76$$

The function f gives the estimated height, in feet, of an acorn x seconds after the acorn fell from a tree. Based on the function, what is the estimated height, in feet, of the acorn before it fell from the tree?

27. [September 2025]

$$\frac{(x-7)(x-8)}{x-3} = 0$$

What is the sum of the solutions to the given equation?

A) 8
B) 18
C) 15
D) 3

28. [September 2025]

$$-6x + 42px = 84$$

In the given equation, p is a constant. The equation has no solution. What is the value of p ?

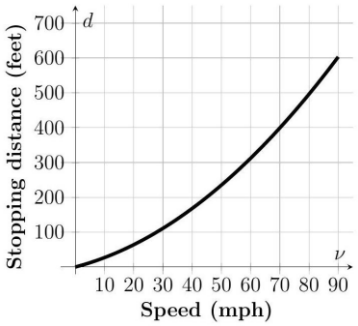
A) $\frac{1}{3}$ B) 0
C) 2 D) $\frac{1}{7}$

29. [September 2025]

If $(x+2)^2 = 18$, what is the value of $x^2 + 4x$?

A) 14
B) 22
C) 36
D) 18

30. [September 2025]



The graph quadratic function shown relates a car's stopping distance d, in feet, to its initial speed v, in miles per hour (mph). Which of the following could be an equation for d ?

A) $d = 2.2v + 0.05v^2$
B) $d = 2.2v - 0.05v^2$
C) $d = 100 + 2.2v + 0.05v^2$
D) $d = 100 + 2.2v - 0.05v^2$

31. [September 2025]

$$nx^2 - 16x = 26x^2 - 8$$

In the given equation, n is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of n ?

32. [October 2025]

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. The model estimates that the object was launched from an initial height of 0.00 feet above the ground and was at a maximum height of 207.36 feet above the ground 3.6 seconds after it was launched. The model also estimates that the object was at a height of 184.32 feet above the ground x seconds after it was launched. What is a possible value of x ?

33. [October 2025]

$$x^2 + 49x + 1 = 0$$

A) $\frac{-49+\sqrt{(49)^2-4(1)(1)}}{2(1)}$

B) $\frac{-49+\sqrt{(49)^2-4(49)(49)}}{2(49)}$

C) $\frac{-1+\sqrt{(1)^2-4(1)(49)}}{2(1)}$

D) $\frac{-1+\sqrt{(1)^2-4(49)(1)}}{2(49)}$

34. [October 2025]

$15x^2 = 15(9)$

What is the negative solution to the given equation?

35. [October 2025]

The y-intercept of the graph of $y = 5x^2 + 72$ is $(0, y)$.

What is the value of y ?

36. [October 2025]

$$80x^2 + 80x = -10$$

One solution to the given equation can be written as $x = \frac{-8-k\sqrt{2}}{16}$, where k is a constant. What is the value of k ?

A) 4
B) 80
C) 32
D) 128

37. [October 2025]

$$k^2 - 47 = 97$$

What is the positive solution to the given equation?

A) 144
B) 12
C) 72
D) 50

38. [October 2025]

What is the y-intercept of the graph of $y = x^2 + 27$ in the xy-plane?

A) $(0, -27)$
B) $(0,28)$
C) $(0,27)$
D) $(0,0)$

39. [October 2025]

$$\frac{x+1}{5x^2} = \frac{k}{x}$$

In the given equation, k is a constant. The solution to the given equation is $\frac{1}{174}$. What is the value of k ?

40. [October 2025]

$$x^2 - \frac{81}{16} = 0$$

How many distinct real solutions does the given equation have?
A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

41. [October 2025]

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 19.36 feet above the ground when it is dropped at $t = 0$. How many seconds after being dropped does the function estimate the object will hit the ground?

42. [October 2025]

$$x^2 + (\sqrt{k-3})x + 42 = 0$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k ?
A) 171
B) 168
C) 165
D) 45

43. [October 2025]

$$9x^2 + 8 = nx$$

In the given equation, n is a constant. The equation has exactly one solution. What is the value of $\frac{n^2}{8}$

44.

$$\frac{1}{x}+\frac{1}{x-3}=\frac{6}{x^2-3x}$$

What are all the solutions to the given equation?

A) -1 and 6

B) 0 and 3

C) 3

D) $\frac{9}{2}$

Quadratics 2 – vazifa

1. [December 2023]

$$x^2 - 12x + c = 0$$

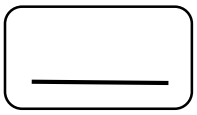
In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

A) -36
B) -12
C) 12
D) 36

2.

$$22x^2 + (22s + r)x + rs = 0$$

In the given equation, r and s are positive constants. The product of the solutions to the given equation is krs , where k is a constant. What is the value of k ?



3.

$$y = -4x^2 - 23$$
$$y = qx - 19$$

In the given system of equations, q is a positive constant. The graphs of the equations in the given system intersect at exactly one point (x, y) , in the $xy-$plane. What is the value of x?

A) -42
C) -8
B) -27
D) -1

4. $\sqrt{k-x} = 52 - x$

In the given equation, k is a constant.
The equation has exactly one real solution.
What is the minimum possible value of $4k$?

5. The function f is a quadratic function. In the xy-plane, the graph of $y = f(x)$ has a vertex at $(1,9)$ and passes through the points $(2,33)$ and $(-1,105)$. What is the value of $f(-2) - f(0)$?

A) 81
B) 129
C) 192
D) 225

6.

$$y = x - c$$

$$y = -8(x - 10)^2$$

In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c ?

A) 5
B) 9
C) $\frac{319}{32}$
D) 15

7.

$$y = -0.5$$
$$y = x^2 + 10x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

8.

$$y = 2x^2 - 19x + 40$$
$$y = x + a$$

In the given system of equations, a is a constant. The graph of the equations in the given system intersect at exactly one point (x, y), in the xy-plane. What is the value of x ?

A) -10
B) -5
C) 5
D) 10

9. The function $f(x) = \frac{1}{9}(x-4)^2 + 5$ gives a toy car's height above the ground $f(x)$ in inches, x seconds after it started moving on an elevated track, where $0 \le x \le 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy-plane?

A) The toy car's minimum height was 5 inches above the ground.
B) The toy car's minimum height was 4 inches above the ground.
C) The toy car's height was 5 inches above the ground when it started moving.
D) The toy car's height was 4 inches above the ground when it started moving

10. An object is kicked from a platform. The equation $h = -4.9t^2 + 5t + 9$ represents this situation, where h is the height of the object above the gound, in meters, t seconds after it is kicked. Which number represent the height, in meters, from which the object was kicked?

A) 0
B) 4.9
C) 5
D) 9

11. A machine launches a baseball from ground level. The baseball reaches a maximum height of 92.16 meters above the ground at 2.4 seconds and hits the ground at 4.8 seconds. Which equation represents the height above ground h, in meters, of the baseball t seconds after it is launched?

(A) $h = -t^2 + 4.8$
(B) $h = -t^2 + 92.16$
(C) $h = -16(t - 2.4)^2 - 4.8$
(D) $h = -16(t - 2.4)^2 + 92.16$

12. An object is launched into the air from ground level. According to a quadratic model, 2.1 seconds after the object is launched, it reaches its maximum height of 70.56 feet above ground level. Which equation represents this model, where $f(t)$ is the object's height, in feet, above ground level t seconds after it was launched?

A) $f(t) = -16(t - 2.1)^2 + 70.56$
B) $f(t) = 16(t + 2.1)^2 + 70.56$
C) $f(t) = -16(t + 2.1) + 70.56$
D) $f(t) = (t - 2.1)^2 + 70.56$

13.

$$f(x) = -16x^2 + 99$$

The function f gives the estimated height, in feet, of an acorn x seconds after the acorn fell from a tree. Based on the function, what is the estimated height, in feet, of an acorn before it fell from the tree?

14. The function $f(x) = \frac{1}{9}(x-6)^2 + 3$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \le x \le 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy plane?

A) The toy car's minimum height was 3 inches above the ground.
B) The toy car's minimum height was 6 inches above the ground.
C) The toy car's height was 3 inches above the ground when it started moving.
D) The toy car's height was 6 inches above the ground when it started moving.

15. A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,600 feet 10 seconds after it was launched. Based on the model, what was the height, in feet, of the object 7 seconds after it was launched?

16.

$$x^2 + 11x + 19 = 0$$

One solution to the given equation can be written as $x = \frac{-11-\sqrt{k}}{2}$, where k is a constant. What is the value of k ?

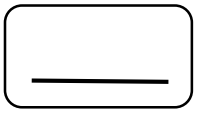
17.

$$f(x) = \sqrt{5x+6}$$

The function f is defined by the given equation. If $f(a) = -5a$, where a is a constant, what is the value of a ?

A) $\frac{3}{5}$
B) $\frac{2}{5}$
C) $-\frac{2}{5}$
D) $-\frac{3}{5}$

18. The function f is defined by $f(x) = ax^2 + bx + c$, where a, b, and c are constants and $1 < a < 4$. The graph of $y = f(x)$ in the xy-plane passes through the points $(11,0)$, and $(-5,0)$. If a is an integer, what could be the value of $a + b$?



19.

$$f(x) = 14x^2 - 35x + 7$$

The given equation defines the function f. For what values of x does $f(x)$ reach its minimum?
A) 1.25
B) 1.75
C) 7
D) 35

20.

$$y = 6x^2 - 48x + 98$$
$$y + 4 = 0$$

How many solutions are there to the given system of equations have?
A) There is exactly 1 solution.
B) There are exactly 2 solutions.
C) There are exactly 3 solutions.
D) There are no solutions.

21.

$$\sqrt{k-x} = 54 - x$$

In the given equation, k is a constant. The equation has exactly one real solution.
What is the minimum possible value of $4k$?

22. $-16(5x-2)^2 + 6(5x-3)^2$

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$

where a, b, and c are constants. Wat is the value of $a+b+c$?

A) -720
B) -360
C) -120
D) -20

23. The tunction f is defined by $f(x) = ax^2 + bx + c$. where a, b , and c are constants and $1 < a < 4$. The graph of $y = f(x)$ in the $xy-$ plane passes through the points $(12,0)$ and $(-4,0)$. If a is an integer, what could be the value of $a+b$?

24.

$$x^2 - 4x - 1 = 0$$

A solution to the given equation is $x = \sqrt{k} + 2$.
What is the value of k ?

25. $-16(5x-3)^2+4(5x-4)^2$

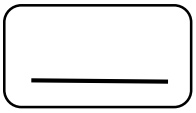
The given expression can be rewritten as

$$\frac{a}{6}x^2+\frac{b}{6}x+\frac{c}{6}$$

where a, b, and c are constants. What is the value of $a+b+c$?
A) -360
B) -180
C) -60
D) -10

26. An object is kicked from a platform. The equation $h=-4.9t^2+13t+15$ represents this situation, where the h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?
A) 0
B) 4.9
C) 13
D) 15

27. The expression $(9x-28)(15x+4)$ is equivalent to the expression ax^2+bx+c, where a, b, and c are constants. What is the value of b ?



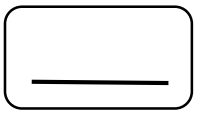
28. Which of the following expressions is (are) a factor of $4x^2+31x-45$?

I. $x-9$
II. $4x-5$

A) I only B) II only
C) I and II
D) Neither I nor II

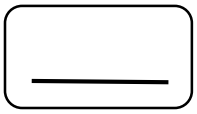
29. $f(x) = x + 4$
$g(x) = 3x^2 - rx + 48$

The functions f and g are given. In function g, r is a constant. If $f(x) \cdot g(x) = 3x^3 + 192$, what is the value of r ?



30. $x(kx - 68) = -4$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

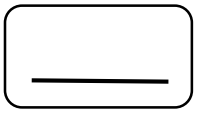


31. The function f is defined by $f(x) = ax^2 + bx + c$ where a, b and c are constants. The graph of $y = f(x)$ in the xy-plane passes through the points $(9,0)$ and $(-2,0)$. If a is an integer greater than 1 , which of the following could be the value of $a + b$?

A) 8
B) 7
C) -6
D) -12

32. $\frac{1}{cx} = \frac{x}{152} + \frac{1}{c}$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?



33. $y = 2x^2 - 23x + 64$
$y = x + a$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x ?

A) -8
B) -6
C) 6
D) 8

34. $-4x^2 + bx - 25 = 0$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b ?

35. The function g is defined by $g(x) = (x + 17)(t - x)$ where t is a constant. In the xy-plane, the graph of $y = g(x)$ passes through the point $(23,0)$. What is the value of $g(0)$?

36. $x^2 + 5x = 47$

The solution to the given equation are $\frac{m+\sqrt{q}}{2}$ and $\frac{m-\sqrt{q}}{2}$, where m and q are integers. what is the value of $m + q$?

37.

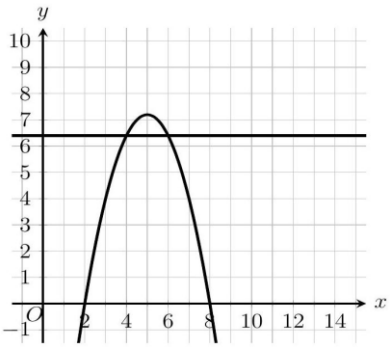
$$y = 2x^2 - 15x + 23$$

$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x ?

A) -9
B) -4
C) 4
D) 9

38.



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y). What is a possible value of x ?

A) 4
B) 5
C) 6.4
C) 7

39.

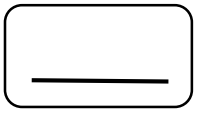
$$(x - k)^2 = (k - 4a)(x - k)$$

In the given equation, a and k are constants, where $k > 4a$. The sum of the solutions to the equation is $3k + 35$. What is the value of a ?

40. An object is launched from a height of 121 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 2 seconds after the object is launched, it reaches a maximum height of 185 feet above the ground. Which equation defines f ?

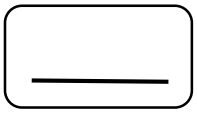
A) $f(t) = -16(t + 185)^2 - 2$
B) $f(t) = -16(t - 185)^2 + 2$
C) $f(t) = -16(t + 2)^2 - 185$
D) $f(t) = -16(t - 2)^2 + 185$

41. One of the factors of $3x^3 + 60x^2 + 288x$ is $x + k$, where k is a positive constant. What is the largest possible value of k ?



42. $x^2 - 60x + c = 0$

In the given equation above, c is a constant. The equation has no real solutions if $c > k$. What is the least possible value of the integer k ?



43. In the xy-plane, a parabola has a vertex $(5,1)$ and intersects the x-axis at two points. If the equation of the parabola is written in the form $y = -ax^2 + bx + c$, where a, b, and c are constants and $a > 0$, which of the following could be a value of c ?

A) -5 B) 2
C) 3 D) 9

44. Which of the following equations has $x + 3a$ as a factor where a is a positive integer?

A) $3x^2 - 24x + 18a = 0$

B) $3x^2 + 8x + 18a = 0$

C) $3x^2 + 16x + 18a = 0$

D) $3x^2 + 24x + 18a = 0$

Polynomials, Factor and Remainder theorem
→Ko'phadlar ko'paytuvchi va qoldiqli bo'lish

1. The graph of the polynomial function f in the xy-plane, where $y = f(x)$, has x intercepts of $(-7,0)$ and $(9,0)$. Which of the following must be true?

A) $f(0) = -7$
B) $f(9) = -7$
C) $f(-7) = 0$
D) $f(-7) = 9$

2. $\frac{4}{7}(4x + 7)(x + \sqrt{4k + 7})(x - \sqrt{4k + 7}) = 0$

In the given equation, k is a positive constant. The product of the solutions to the equation is 77. What is the value of k ?

3. $7x^2(4x - 13)(4x - u) = 0$

In the given equation, u is a positive constant. The sum of the the solutions to the equations is 9 . What is the value of u ?

4. Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A) $3x^2 + 9x + 18b$
B) $3x^2 + 24x + 18b$
C) $3x^2 + 30x + 18b$
D) $3x^2 + 39x + 18b$

5.

$$34z^{14} + bz^7 + 70$$

In the given expression b is a positive integer. If $qz^7 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

6. The graph of the function f in the xy-plane, where $y = f(x)$, has x-intercepts of $(-5,0), (3,0)$, and $(6,0)$. Which of the following must be true?

A) $f(0) = -5$
B) $f(0) = 3$
C) $f(5) = 0$
D) $f(6) = 0$

7. $f(x) = 3(x-a)(x-b)(x-c)$

The function f is defined by the given equation, where a, b, and c are distinct constants. When $a < x < b$, the value of $f(x)$ is positive. The graph of $y = f(x)$ in the xy-plane contains the point (r, s), where r and s are constants. If $s = 8$, which of the following could be true?

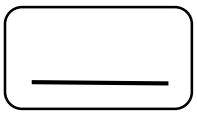
I. $r < a$
II. $b < r < c$
III. $r > c$

A) I only
B) III only
C) I and III only
D) II and III only

8. The function p is defined by

$$p(x) = ((x+6)^2 - b)((x+6)^2 - c),$$

where a, b, and c are constants. In the xy-plane, the graph of $y = p(x)$ passes through the points $(-7,24)$ and $(0,899)$. What is the value of $p(-12) + p(-5)$?



9. The function f is defined by

$$f(x) = (x-8)(x-5)(x-3).$$

In the xy-plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 3 units. What is the value of $g(0)$?

10. The given equation is:

$$0 = (x - k)(7x + t) - (x - k)$$

In the given equation, k and t are positive constants. A solution to the equation is $x = -\frac{22}{7}$. What is the value of t ?

11. The function h is defined by

$$h(x) = (x + p)(x - 2)(2x - 12),$$

where p is a constant. In the xy-plane the graph of $y = h(x)$ passes through the point $(-3,0)$. What is the value of $h(0)$?

A) -36
B) -3
C) 5
D) 72

12. The function p is defined by

$$p(x) = a((x + 5)^2 - b)((x + 5)^2 - c)$$

where a, b, and c are constants. In the xy plane, the graph of $y = p(x)$ passes through the points $(-6,30)$ and $(0,342)$. What is the value of $p(-10) + p(-4)$?

13. One of the factors of $2x^3 + 44x^2 + 234x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

14. Which expression is a factor of $49p^{19} - 121p^{17}$?

A) $7p + 11$
B) $49p^2 + 121$
C) $11p - 7$
D) $-72p^{40}$

15. $26z^{18} + bz^9 + 70$

In the given expression, b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

16. In the xy-plane, the graph of function f, where $y = f(x)$, has exactly $8x$-intercepts. One of these x-intercepts is $(17,0)$. The rational function g is defined by

$$g(x) = \frac{f(x)}{x - 17}$$

In the xy-plane, how many x-intercepts does the graph of $y = g(x)$ have?

17. One of the factors of $4x^3 + 88x^2 + 468x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

18. For the polynomial function f, the grapf of $y = f(x)$ in the xy-plane passes through the points $(-3,0), (1,0)$ and $(4,0)$. Which of the following must be a factor of $f(x)$?

A) $x + 1$
B) $x + 4$
C) $x - 1$
D) $x - 3$

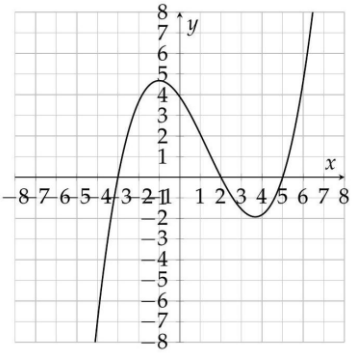
19. For the polynomial function f, the graph of $y = f(x)$ in the xy -plane passes through the points $(-4,0), (3,0)$, and $(5,0)$. Which of the following must be a factor of $f(x)$?

A) $x + 5$
B) $x + 3$
C) $x - 3$
D) $x - 4$

20. For the polynomial function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(-4,0), (1,0)$, and $(9,0)$. Which of the following must be a factor of $f(x)$?

A) $x - 1$
B) $x + 9$
C) $x - 4$
D) $x + 1$

21.



The graph of the polynomial function f is shown, where $y = f(x)$. What is the value of $f(5)$?

A) 2 B) 5
C) -3 D) 0

22.

$$f(x) = 4(g(x)) - 2$$
$$g(x) = |13x - 8|$$

The functions f and g are defined by the given equations. What is the value of $f(-10)$?

A) -554
B) -42
C) 550
D) 138

23. The function f is defined by $f(x) = 20x^3$. The graph of $y = f(-x) + c$ in the xy-plane, where c is a positive integer constant, has an x-intercept at ($r, 0$) and a y-intercept at $(0, t)$, where r and t are constants. Which of the following must be true about r and t ?

A) $r > 0$ and $t > 0$
B) $r > 0$ and $t < 0$
C) $r < 0$ and $t > 0$
D) $r < 0$ and $t < 0$

24.

$$f(x) = 3d(26x + 27) + 18$$

Which of the following represents the x intercept of the graph of $y = f(x) + 6$ in the xy -plane, where d is a constant?

A) $(81d + 24,0)$
B) $\left(\frac{-81d-24}{78d}, 0\right)$
C) $\left(6 - \frac{81d+18}{78d}, 0\right)$
D) $\left(\frac{-45}{78d+6}, 0\right)$

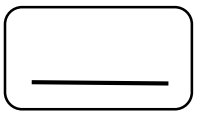
25. Function f is defined as $f(x) = (2x - 5)^2$, and function g is defined as $g(x) = 2(2x + 7)$. Which of the following is equivalent to $f(x) + g(x)$?

A) $4x^2 - 16x + 11$
B) $4x^2 - 16x + 39$
C) $4x^2 - 6x + 39$
D) $4x^2 + 24x + 11$

Percent; Ratio&Proportion → Foiz nisbat va proporsiya

1. [August 2025]

A certain bonefish can swim at a maximum speed of 17 meters per second. At this maximum speed, how many meters would this bonefish swim in 10 seconds?



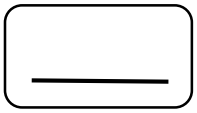
2. [September 2025]

The ratio 6 to 3 is equivalent to the ratio 24 to $3k$, where k is a constant. What is the value of k ?

A) 0.75
B) 4
C) 12
D) 72

3. [September 2025]

For the positive quantities h, j, and $\mathrm{k}, 11\%$ of h is equivalent to 33% of j, and j is equivalent to 14% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)



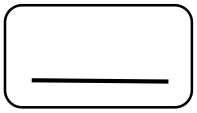
4. An object travels x inches in y seconds. At this rate, which expression represents the time, in seconds, it takes the object to travel a distance of $22x$ inches?

A) $\frac{22}{y}$
B) $\frac{y}{22}$
C) $22y$
D) $22 + y$

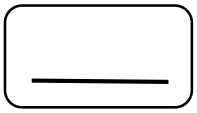
5. A certain goose species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this goose species fly in 6 seconds?
A) 96
B) 16
C) 10
D) 22

6. [August 2025]

For the positive quantities h, j, and k, 12% of h is equivalent to 95% of j, and j is equivalent to 24% of k. What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)



7. For the positive quantities h, j, and k, 13% of h is equivalent to 19% of j, and j is equivalent to of 91% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)



8. The ratio of k to p is constant. If $k = 7$, then $p = 6$.
What is the value of k when $p = 336$?
A) 288
B) 56
C) 48
D) 392

9. [October 2025]

What is 10% of 450 ?
A) 35
B) 440
C) 405
D) 45

10. The mass of object A is 444% of the mass of object B , and the mass of object A is 0.740% of the mass of object C . If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

11. Charles saves $\frac{3}{5}$ of the $\$225$ he earns each week from his summer job. If Charles continues to save at this rate, how much money, in dollars, will Charles save in 3 weeks?

12. The speeds of particles A , B , and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of particle A is $7{,}900\%$ of the speed of particle C and the speed of particle C is 0.008% of the speed of particle B , which expression represents the value of $a + b$ in terms of c ?

A) $12{,}579c$
B) $8{,}025c$
C) $7{,}908c$
D) $2{,}040c$

13. If $\frac{2a}{b} = 5.8$ and $\frac{a}{bn} = 23.2$, what is the value of $\frac{1}{n}$?

14. There is a proportional relationship between p and t. If $p = 2{,}340$ when $t = 3{,}120$, then what is the value of p when $t = 2{,}496$?

A) 1,716
B) 1,872
C) 3,276
D) 3,328

15. For the positive quantities h, j, and k. 94% of h is equivalent to 47% of j, and j is equivalent to 30% of k. What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if you answer is 39%, enter 39)

16. The mass of object A is 483% of the mass of object B , and the mass of object A is 0.069% of the mass of object C . If the mass of object C is $p\%$ of the mass of object B , what is the value of $\frac{p}{1{,}000}$?

17. The positive number a is $2{,}352\%$ of the sum of the positive numbers b and c, and b is 84% of c. What percent of b is a ?
A) $5{,}152\%$
B) $2{,}436\%$
C) 51.52%
D) 24.36%

18. The ratio of 12 to x is equal to the ratio of 156 to y. Which expression represents y in terms of x ?
A) $\frac{1}{13}x$
B) $13x$
C) $\frac{1}{12}x$
D) $12x$

19. Julia purchased 800 feet of fencing. She used 60% of this fencing to surround a vegetable garden. How many feet of fencing did Julia use to surround a vegetable garden?
A) 12
B) 30
C) 480
D) 740

20. The speeds of particles A , B , and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of particle A is $6{,}000\%$ of the speed of particle C and the speed of particle C is 0.008% of the speed of particle B , which expression represents the value of $a + b$ in terms of c ?

A) $12{,}560c$

B) $6{,}125c$

C) $6{,}008c$

D) $1{,}850c$

21. [May 2025]

A number f is 120% greater than a positive number g. A number h is 90% less than the number f. The number h is how many times the number g ?

A) 0.12

B) 0.22

C) 1.08

D) 1.98

22. A scientist analyzed a soil sample with a mass of 900 grams and determined that it contained 189 grams of water. What is the percentage of water, by mass, in this soil sample?

1. An object's speed is increasing at a rate of 6.80 meters per second squared. What is this rate in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

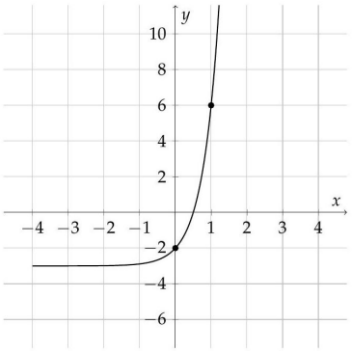
2. A certain town has an area of 13,753,344 square yards. What is the area, in square miles, of this town? (1 mile = 1,760 yards)

A) 7,814.4
B) 2.11
C) 3,708.55
D) 4.44

3. A dog's mass is 24 kilograms. What is the dog's mass, in grams? (1 kilogram = 1,000 grams)

Exponential function → Darjali funksiya

1.



The graph of the function f is shown, where $y = f(x)$. What is the value of $f(0)$?

A) 6
B) 2
C) -2
D) -6

2. $P(t) = 1{,}200(1.06)^t$

The function P gives the estimated number of sea lions in a certain area, where t is the number of years since a study began. What is the best interpretation of $P(0) = 1{,}200$ in this context?

A) The estimated number of sea lions in the area increased by 1,200 each year during the study.
B) The estimated number of sea lions in the area increased by 106 each year during the study.
C) The estimated number of sea lions in the area was 1,200 when the study began.
D) The estimated number of sea lions in the area was 106 when the study began.

3. The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xyplane, the graph of $y = f(x)$ passes through the points $(c, 4)$ and $(2c, 114)$, where c is a constant. Which of the following could be the value of b ?

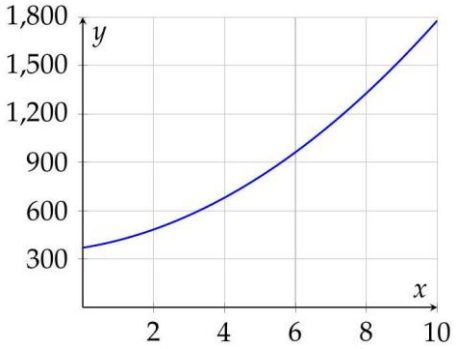
A) 7
B) 14
C) 110
D) 118

4. The number of bacteria in a growth medium is expected to increase by 160% every 2 hours during a period of observation. The number of bacteria in the growth medium was estimated to be 8,000 when the period of observation began. Which function P gives the expected number of bacteria in this growth medium t hours after the period of observation began?

A) $P(t) = 8000(1.60)^{\frac{t}{2}}$
B) $P(t) = 8000(1.60)^{2t}$
C) $P(t) = 8000(2.60)^{\frac{t}{2}}$
D) $P(t) = 8000(2.60)^{2t}$

Expected = kutilgan, tahmin qilindi
increase by 160% = 160 foizga ortishi
every 2 hours = har ikki soatda
during= davomida
a period = bir davr, muddat
observation = kuzatuv
estimated = tahmin qilindi
to be= bo'lishi
when = qachonki
observation began = kuzatuv boshlanganda
t hours after = t soatdan keyin

5.



The graph shows the estimated number of cell phone subscribers, y , in a town x years since 1990. Which equation represents the exponential relationship between the number of cell phone subscribers in the town and the number of years since 1990, where $0 \leq x \leq 10$?

A) $y = 370(0.17)^x$
B) $y = 370(1.17)^x$
C) $y = 1779(0.83)^x$
D) $y = 1779(1.83)^x$

6. The function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. In the xy-plane, the graph of $y = f(x)$ has a y-intercept at $(0, -22)$ and passes through the point $(2,41)$. What is the value of $a + b$?

7. The population of a certain city doubled every 75 years from 1651 to 1951. The population of this city was 160,000 in 1951. What was the population of this city in 1651?

Doubled = ikki karra bo'lmoq (x2)

Triple = uch karra bo'lmoq (x3)

Bu yerda natijaviy miqdor dastlabkisini 2 baravariga teng bo'ladi degan manoda gap ketayapdi

8. A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 27% every 14.07 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

Researcher = tadqiqotchi
Observes = kuzatadi
a sample = bir namuna
decreases by 27% = 27 foizga kamayadi (-27%)
began = boshlagan

A) $M = 100(0.27)^{\frac{t}{14.07}}$
B) $M = 100(0.27)^{t+14.07}$
C) $M = 100(0.73)^{\frac{t}{14.07}}$
D) $M = 100(0.73)^{t+14.07}$

9. The function f is defined by $f(x) = 2x^3 + \frac{1}{3}x^2 - x$. What is the value of $f(1)$?

A) $\frac{4}{3}$
B) 2
C) $\frac{10}{3}$
D) 6

10. The function f is defined by $f(x) = a(4)^x$, where a is a positive constant. In the xy-plane, the graph of $y = f(x)$ passes through the point $(0,13)$. What is the value of $f(0)$?

A) 4
B) 1
C) 13
D) 9

11. For $x = 0, y$ has a value of 200. The value of y increases by 40% for every increase of 1 in the value of x. Which equation represents this situation?
A) $y = 200(x)^{40}$
B) $y = 200(1.40)^x$
C) $y = 200(40)^x$
D) $y = 200(x)^{1.40}$

12. The relationship between the variables x and y is defined by an exponential equation. When $x = 0$, the value of y is 40 , and for every increase in the value of x by 1 , the corresponding value of y increases by 50% of its previous value. Which equation represents this relationship?
A) $y = 40(1.50)^x$
B) $y = 40(1.05)^x$
C) $y = 50(1.40)^x$
D) $y = 50(1.04)^x$

13. The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0,10)$ and $(2,13)$. What is the value of ab ?
A) 13
B) 18
C) 20
D) 26

14. $f(x) = 4000(0.90)^x$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 4000 in this context?
A) The estimated percent decrease in the insect population for this species and area every 8 years after 2008
B) The estimated initial insect population for this species and area in 2008

C) The estimated percent decrease in the insect population for this species and area each year after 2008
D) The estimated insect population for this species and area 8 years after 2008

15. The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p ?
A) 19
B) 44
C) 56
D) 81

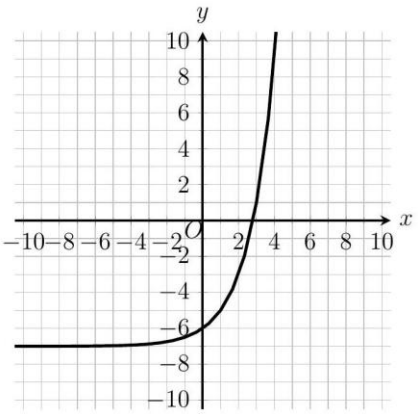
16. A certain neighborhood had a population of 1,340 in 2009. Each year for the next 5 years, the population of the neighborhood increased by approximately 3% of the population the previous year. Which of the following equations represents the population, N, of the neighborhood t years after 2009, where $t \leq 5$?
A) $N = 0.03(1{,}340)^t$
B) $N = 1{,}340(0.03)^t$
C) $N = 1{,}340(1.03)^t$
D) $N = 1.03(1{,}340)^t$

17.

$$f(x) = 7.3(0.98)^x$$

The function shown models the annual soil erosion rate, in tons per acre year, on US croplands x years since 1982 . According to the model, which is the best interpretation of $7.3(0.98)^{10}$ in this context?
A) The number of acres of US cropland in 1982
B) The number of acres of US cropland in 1992
C) The annual soil erosion rate, in tons per acre per year, in 1982
D) The annual soil erosion rate, in tons per acre per year, in 1992

18.



The graph of the equation $y = 2^x + k$ is shown, where k is a constant. What is the value of k ?

19. What is the y-intercept of $y = -(7)^x - 22$ in the xy-plane?

A) $(0, -23)$

B) $(0, -29)$

C) $(0, -22)$

D) $(0, -21)$

20. The y -intercept of the graph of $y = 6(1.85)^x$ in the xy-plane is $(0, y)$. What is the value of y ?

21. The function g is defined by $g(x) = 10^{3x-1}$. What is the value of x when $g(x)$ is equal to 10,000 ?

A) $\frac{4}{3}$

B) $\frac{5}{3}$

C) 3

D) 4

22. The function $f(t) = 10{,}000(2)^{\frac{t}{50}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

23. $f(x) = 7020(0.28)^{\frac{x}{12}}$

The function f gives the value of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what si the value of p ?

A) 2
B) 10
C) 28
D) 72

24. For the function h, the table gives three values of x and their corresponding values of $h(x)$. Which equation could define h ?

x	$h(x)$
0	30
1	31
2	33

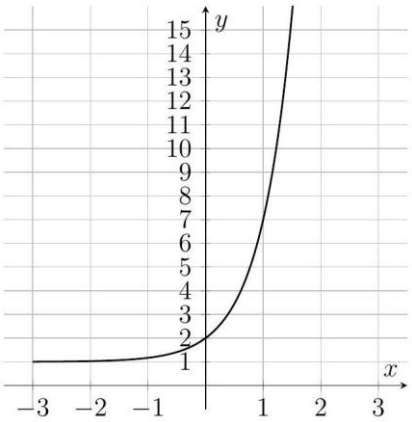
A) $y = 2^x$
B) $y = 2^x + 2$
C) $y = 2^x + 28$
D) $y = 2^x + 29$

25. $f(x) = 240(2)^x$

The function f is defined by the given equation. If $g(x) = f(x - 3)$, which of the following equations defines the function g ?

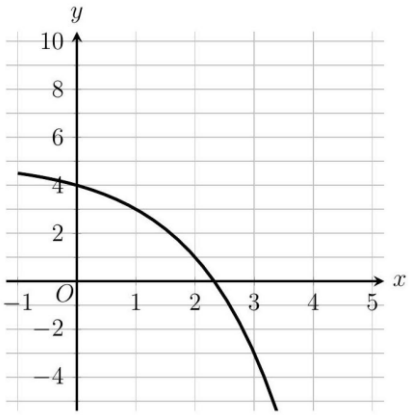
A) $g(x) = 30(2)^x$
B) $g(x) = 80(2)^x$
C) $g(x) = 240(8)^x$
D) $g(x) = 240(6)^x$

26.



The graph of $y = f(x) + 1$ is shown. Which equation defines the function f ?
A) $f(x) = 6^x$
B) $f(x) = 2^x$
C) $f(x) = 6^x + 2$
D) $f(x) = 2^x + 1$

27.



The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?
A) $f(x) = -2^x + 1$
B) $f(x) = -2^x + 3$
C) $f(x) = -2^x + 4$
D) $f(x) = -2^x + 5$

28. For $x > 0$, the function f is defined as follows:

$$f(x) \text{ equals } 111\% \text{ of } x$$

Which of the following could describe this function?
A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

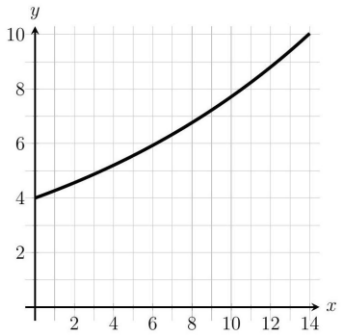
29. The function f is defined by $f(x) = |x - 6x|$. What value of a satisfies $f(7) - f(a) = -15$?
A) -22
B) 3
C) 10
D) 75

30. The function f is defined by

$$f(x) = (x - 9)(x - 6)(x - 5)$$

In the xy-plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units.
What is the value of $g(0)$?

31.



The graph models the population, in thousands, of a town x years since 2003, where $0 \leq x \leq 15$.
Which statement is the best interpretation of the point $(1,4.24)$?
A) In 2003, the estimated population of the town was 4,240 .
B) In 2004, the estimated population of the town was 4,240 .
C) In 2017, the estimated population of the town was 4,240 .
D) In 2018, the estimated population of the town was 4,240 .

32.

x	-20	-8	10
$g(x)$	4	0	6

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+5}$ and f is a linear function. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A) $(0, -8)$
B) $(0,5)$
C) $(0,8)$
D) $(0,40)$

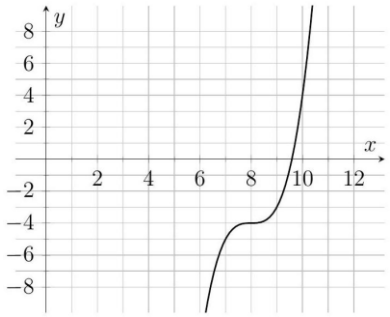
33. For $x > 0$, the function f is defined as follows:

$f(x)$ equals 300% of x

Which of the following could describe this function?

A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

34.



The complete graph of $y = f(x)$ is shown. For how many values of x does $f(x) = 0$?

A) Zero
B) One
C) Two
D) Three

35. The table shows selected values from the function f.

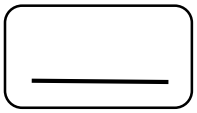
x	$f(x)$
-1	$\frac{1}{69}$
0	1
1	69
2	4,761

Which of the following is the best description of function f ?
A) Increasing linear
B) Decreasing linear
C) Increasing exponential
D) Decreasing exponential

36. The function k is defined by

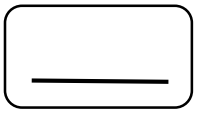
$$k(s) = \sqrt{s + 110}$$

If $k(53p) = p$, where p is a constant, what is the value of p ?



37. A certain neighborhood had a population of 1,130 in 2006. Each year for the next 5 years the population of the neighborhood increased by approximately 4% of the population of the previous year. Which of the following equations represents the population N, of the neighborhood t years after 2006, where $t \leq 5$?
A) $N = 0.04(1{,}130)^t$
B) $N = 1{,}130(0.04)^t$
C) $N = 1{,}130(1.04)^t$
D) $N = 1.04(1{,}130)^t$

38. The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy plane, the graph of $y = f(x)$ has an x intercept at $(2,0)$ and y-intercept at $(0,-255)$. What is the value of a ?



39. At the start of an experiment, the pressure inside a container was 19 pounds per square inch (psi). The pressure inside the container doubled every 28 minutes after the start of the experiment. Which equation represents this situation, where y is the pressure, in psi, inside the container x minutes after the start of the experiment?

A) $y = 19(2)^{\frac{x}{28}}$
B) $y = 19(28)^{2x}$
C) $y = 19(x + 28)^2$
D) $y = 19x^2 + 28$

40.

x	$h(x)$
0	1.46
2	2.32
4	3.68

The table shows exponential relationship between the number of years, x, since Isela started training in pole vault, and the estimated height $h(x)$,in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

A) $h(x) = 1.26(0.46)^x$
B) $h(x) = 1.26(1.46)^x$
C) $h(x) = 1.46(0.26)^x$
D) $h(x) = 1.46(1.26)^x$

41. The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0,8)$ and $(2,32)$. What is the value of ab ?

A) 32
B) 35
C) 40
D) 64

42. The number of bacteria in a liquid medium doubles every day. There are 72,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y, in the liquid medium t days after the start of the observation?

A) $y = \frac{1}{2}(72{,}000)^t$
B) $y = 2(72{,}000)^t$
C) $y = 72{,}000\left(\frac{1}{2}\right)^t$
D) $y = 72{,}000(2)^t$

43. The function $f(t) = 10{,}000(2)^{\frac{t}{330}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

44. The number of bacteria in a liquid medium doubles every day. There are 75,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y, in the liquid medium t days after the start of the observation?

A) $y = \frac{1}{2}(75{,}000)^t$
B) $y = 2(75{,}000)^t$
C) $y = 75{,}000\left(\frac{1}{2}\right)^t$
D) $y = 75{,}000(2)^t$

2-vazifa

1. For the exponential function f, the value of $f(1)$ is k, where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A) $f(x) = 50(1.4)^{x+1}$
B) $f(x) = 70(1.4)^{x}$
C) $f(x) = 98(1.4)^{x-1}$
D) $f(x) = 137.2(1.4)^{x-2}$

2. The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

1. $f(x) = 16(1.24)^{x} + 43$
2. $g(x) = 8(0.72)^{x}$

A) 1 only
B) 2 only
C) 1 and 2
D) Neither 1 nor 2

3.

Time(years)	Total amount (dollars)
0	867.00
1	870.47
2	873.95

Sam opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Sam opened the account and the total amount n, in dollars, in the account. If Sam made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n ?

A) $n = 0.004(1 + 867)^t$
B) $n = 867(1 + 0.004)^t$
C) $n = (1 + 0.004)^t$
D) $n = (1 + 867)^t$

4. The number of bacteria in a liquid medium doubles every hour. There are 44000 bacteria in the liquid medium at the start of the observation. Which of the following represents number of bacteria, y, in the liquid medium t hours after the the start of the observation?
A) $y = \frac{1}{2}(44000)^t$
B) $y = 2(44000)^t$
C) $y = 44000\left(\frac{1}{2}\right)^t$
D) $y = 44000(2)^t$

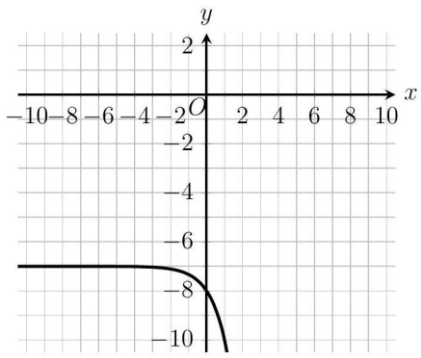
5. The function f and g are defined by the equations, where $x > 0$. Which of the following equations displays, as a constant or a coefficient, the maximum value of he function it defines, where $x \geq 0$?
I. $f(x) = 16(1.25)^x + 43$
II. $g(x) = 6(0.75)^x$
A) I only
B) II only
C) I and II
D) Neither I not II

6.

$$f(t) = 80{,}000(2)^{\frac{t}{790}}$$

gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?
A) 2
B) 790
C) 1,580
D) 80,000

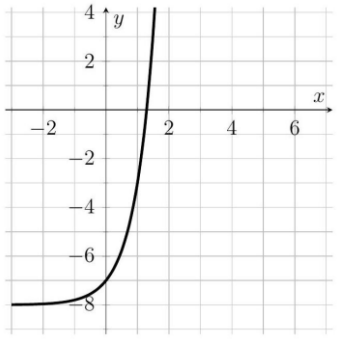
7.



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a, b, and c are constants. For how many values of x does $f(x) = 0$?

A) Three
B) Two
C) One
D) Zero

8.



The graph of the equation $y = 5^x + k$ is shown, where k is a constant. What is the value of k ?

A) -8
C) 7
B) -7
D) 8

9. What is the y-intercept of the graph of $y = 12(5)^{x+2}$ in the xy-plane?

A) (0,12)
B) (0,120)
C) (0,300)
D) (0,3,600)

10. In the xy-plane, the graph of the equation $y = 9^x$ is translated 8 units down. What is an equation of the resulting graph?

A) $y = 9^x + 8$

B) $y = 9^x - 8$

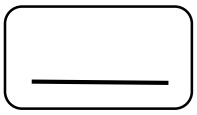
C) $y = 9^{(x+8)}$

D) $y = 9^{(x-8)}$

11. The function f is defined by

$$f(x) = a^x + b$$

where a and b are constants and $a > 0$. In the xy-plane, the graph of $y = f(x)$ has a y-intercept at $(0, -20)$ and passes through the point $(2,43)$. What is the value of $a - b$?



12. The population of a certain city doubled every 25 years from 1660 to 1760 . The population of this city was 208,000 in 1760 . What was the population this city in 1660 ?

A) 13,000

B) 26,000

C) 104,000

D) 416,000

13.

$$f(x) = 25(1.40)^{\frac{x}{3}}$$

For the given function f, the value of $f(x)$ increases by $p\%$ for every increase of x by 6 . What is the value of p ?

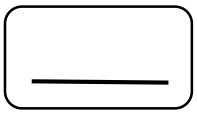
A) 40

B) 49

C) 80

D) 96

14. The population of a certain city doubled every 75 years from 1652 to 1952. The population of this city was 160,000 in 1952 . What was the population of this city in 1652?



15. The relationship between the variables x and y is defined by an exponential equation. When $x = 0$, the value of y is 90 , and for every increase in the value of x by 1 , the corresponding value of y increases by 20% of its previous value. Which equation represents this relationship?

A) $y = 90(1.20)^x$
B) $y = 90(1.02)^x$
C) $y = 20(1.90)^x$
D) $y = 20(1.09)^x$

16. $z(w) = (0.829)^{2w}$

The function z is defined by the equation shown. The value of $z(w)$ decreases by $p\%$ for each increase by 1 in the value of w. Which of the following is closest to the value of p ?

A) 0.171
B) 0.829
C) 17.1
D) 31.3

17. $f(x) = 66(b)^x$

In the given function, b is a positive constant. For every increase in the value of x by 1 , the value of $f(x)$ increases by $c\%$, where $0 < c < 100$. Which expression gives the value of c in terms of b ?

A) $1 + \frac{b}{100}$
B) $b + 100$
C) $100(b + 1)$
D) $100(b - 1)$

18. The equation $N(m) = 65(Q)^{\frac{m}{4}}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 120 seconds. What is the value of p in terms of Q ?

A) $100\left(Q^{\frac{1}{2}} + 1\right)$

B) $100(Q^{30} + 1)$

C) $100\left(Q^{\frac{1}{2}} - 1\right)$

D) $100(Q^{30} - 1)$

19. The function is defined by

$$f(x) = -10(3)^x + \frac{1}{k}$$

where k is a constant. If $y = f(x)$ is graphed in the xy-plane, what is the y-coordinate of the y-intercept of the graph?

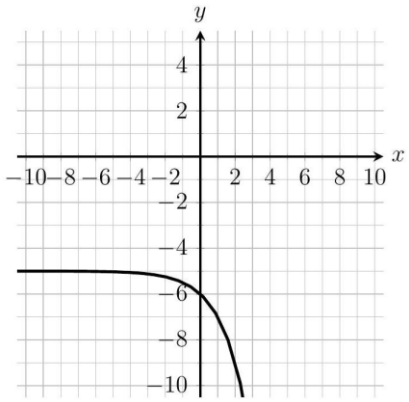
A) -10

B) $\frac{1-10k}{k}$

C) $\frac{1}{k}$

D) $\frac{1-k}{k}$

20.



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a, b, and c are constants. For how many values of x does $f(x) = 0$?

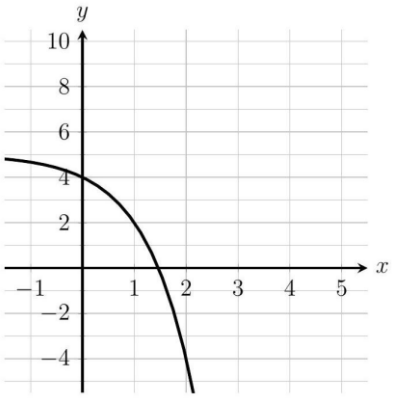
A) Three

B) Two

C) One

D) Zero

21.



The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?

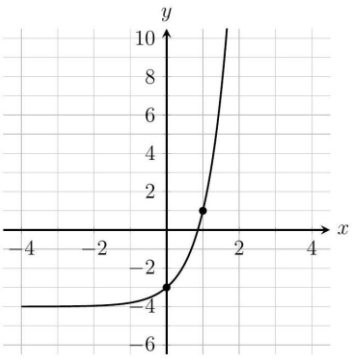
A) $f(x) = -3^x + 1$

B) $f(x) = -3^x + 3$

C) $f(x) = -3^x + 4$

D) $f(x) = -3^x + 5$

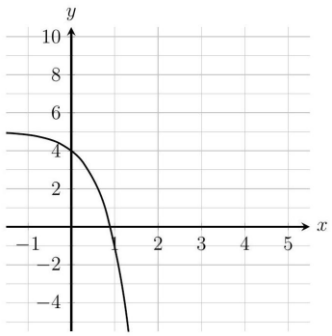
22.



The equation of the graph shown is $y = a^x + b$, where a and b are constants. What is the value of $a - b$?

23. The function f is defined by $f(x) = ab^{\frac{x}{n}}$, where a, b, and n are constants, and b and n are integers. If $f(5) = 7$ and $f(8) = 189$, what is the value of $f(10)$?

24.



The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?

A) $f(x) = -6x^2 + 1$

B) $f(x) = -6x^2 + 3$

C) $f(x) = -6x^2 + 4$

D) $f(x) = -6x^2 + 5$

25. The value of a share of stock was $\$240$ on the day it was purchased. Every year, the value of the share of stock increases by 1% of its value the previous year. Which function V gives the value, in dollars, of the share of stock t years after the day it was purchased?

A) $V(t) = 240(1.01t)$

B) $V(t) = 240(1.1t)$

C) $V(t) = 240(1.01)^t$

D) $V(t) = 240(1.1)^t$

26. A company opens an account with an initial balance of $\$35{,}600.00$. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function. A, where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 15 years is $\$129{,}672.38$. Which equation could define A ?

A) $A(t) = 35{,}600.00(1.09)^t$

B) $A(t) = 35{,}600.00(0.09)^t$

C) $A(t) = 94{,}072.38(1.09)^t$

D) $A(t) = 94{,}072.38(0.09)^t$

27. The function f is defined by $f(x) = -39^x$. The function g is a decreasing linear function. In the xy-plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m), where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

A) $x > k$
B) $x < h$
C) $x > k$ or $x < h$
D) $h < x < k$

28. $f(x) = 3{,}950(0.38)^x$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p ?

A) 3
B) 8
C) 38
D) 62

29. $P(t) = 260(1.03)^{\left(\frac{3}{2}\right)t}$ The function P models the population, in thousands, of a certain city t years after 2009. According to the model, the population is predicted to increase by $n\%$ every 8 months. What is the value of n ?

A) 0.22
B) 1.03
C) 2
D) 3

According to the model = modelga ko'ra
in thousands = mingliklarda
population = aholi
predicted = kutildi, tahmin qilindi

30. The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants. If $f(s) = t$ and $f(s + 1) = t - 0.87t$, where s and t are constants, what is the value of b ?

A) 0.13
B) 0.87
C) 1.13
D) 1.87

31. The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy-plane, which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function?

I. $f(x) = a(4.2)^{x+b}$

II. $g(x) = a(4.2)^x + b$

A) I only

B) II only

C) I and II

D) Neither I nor II

32. A model predicts that the population of a certain town was 3,000 in 2007. The model also predicts that each year for the next 5 years, the population p increased by 8% of the previous year's population. Which equation best represents this model, where t is the number of years after 2007, for $0 \leq t \leq 5$?

previous year's population = o'tgan yilgi aholi

(A) $p = 0.92(3{,}000)$

(B) $p = 1.08(3{,}000)$

(C) $p = 3{,}000(0.92)^t$

(D) $p = 3{,}000(1.08)^t$

33. The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy-plane, which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function?

I. $f(x) = a(2.1)^{x+b}$

II. $g(x) = a(2.1)^x + b$

A) I only

B) II only

C) I and II

D) Neither I nor II

34. $y = 7900(0.86)^x$

The given equation gives the value, in dollars, of a certain car x years after it was purchased. According to the given equation, what is the estimated value, in dollars, of the car at the time it was purchased?

A) 790
B) 6794
C) 7900
D) 8600

35. The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 18(1.4)^{x+5}$
II. $g(x) = 18(1.16)(1.4)^{x+3}$

A) I only
B) II only
C) I and II
D) Neither I nor II

36. The function f is defined by

$$f(x) = a(3.1^x + 3.1^b)$$

where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f, where k and m are constants. Which of the following equations displays the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane as a constant or coefficient?

I. $g(x) = a(3.1^x + k)$
II. $h(x) = a(3.1)^x + m$

A) I only
B) II only
C) I and II
D) Neither I nor II

37. The y-intercept of the graph of $y = 81 + 65^x$ in the xy-plane is $(0, y)$. What is the value of y ?

38. $f(x) = 24(1.30)^{\frac{x}{3}}$

For the given function f, the value of $f(x)$ increases by $p\%$ for every increase of x by 6 . What is the value of p ?

A) 30
B) 41
C) 60
D) 69

39. The function $P(x) = 533(1.011)^x$ models the population of a certain town each year from 1951 through 1966, where x is the number of years after 1951. Which of the following is the best interpretation of " $P(5)$ is approximately equal to 563" in this context?

A) The population of the town is estimated to be approximately 5 greater in 1956 than in 1951.
B) The population of the town is estimated to be approximately 563 in 1956.
C) The population of the town is estimated to be approximately 5 times greater in 1956 than in 1951.
D) The population of the town is estimated to increase by approximately 563 every 5 years between 1951 and 1956.

population of … = …ning aholisi
certain town= malum bir qishloq
each year = har yili
from 1951 through 1966 = 1951 dan 1966 gacha
Approximately = tahminan
Estimated = tahmin qilindi
to be = bo'lishi

40. At the time of posting a video, a social media channel has 88 subscribers. Each day after the video was posted, the number of subscribers doubled from the number of subscribers on the previous day. Which equation gives the total number of subscribers, n, to the channel d days after the videos were posted?

A) $n = (88)^d$
B) $n = 88(2)^d$
C) $n = 88\left(\frac{1}{2}\right)^d$
D) $n = 88^2 + d$

At the time = vaqtda
posting a video = vide joylashtirish
a social media = bir ijtimoiy tarmoq
subscribers = obunachilar
doubled = 2 karra bo'ldi
previous day = o'tgan kun

41.

x	$h(x)$
0	1.25
2	1.80
4	2.59

The table shows the exponential relationship between the number of years, x, since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

A) $h(x) = 1.20(0.25)^x$
B) $h(x) = 1.20(1.25)^x$
C) $h(x) = 1.25(0.20)^x$
D) $h(x) = 1.25(1.20)^x$

42. The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy-plane, which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function?

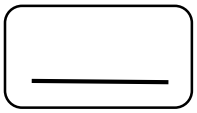
I. $f(x) = a(3.3)^{x+b}$
II. $g(x) = a(3.3)^x + b$

A) I only
B) II only
C) I and II
D) Neither I nor II

43. A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

A) $M = 100(0.24)^{1+22.96t}$
B) $M = 100(0.24)^{\frac{t}{22.96}}$
C) $M = 100(0.76)^{t+22.96}$
D) $M = 100(0.76)^{\frac{t}{22.96}}$

44. The function f is defined by $f(x) = ab^x$, where a, b, and n are constants, and b and n are integers, If $f(4) = 5$ and $f(7) = 135$, What is the value of $f(9)$?



45. $f(x) = 7(4)^x$

The function f is defined by the given equation. If $g(x) = f(x + 3)$, which of the following equations defines the function g ?

A) $g(x) = 21(4)^x$

B) $g(x) = 448(4)^x$

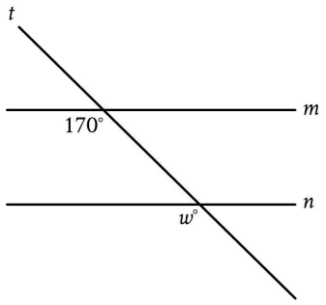
C) $g(x) = 21(12)^x$

D) $g(x) = 343(64)^x$

Lines and angles → Chiziqlar va burchaklar

Parallel lines with transversal

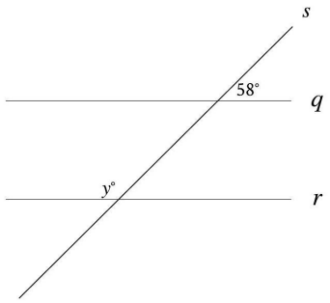
1.



In the figure, line m is parallel to line n. What is the value of w ?

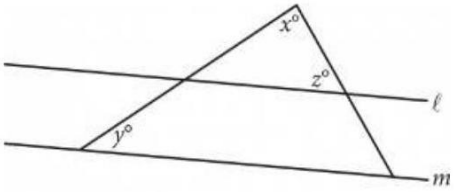
A. 17
B. 30
C. 70
D. 170

2.



In the figure, line q is parallel to line r, and both lines are intersected by line s. If $y = 2x + 8$, what is the value of x ?

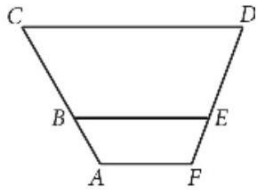
3.



In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

A. 120
B. 100
C. 90
D. 80

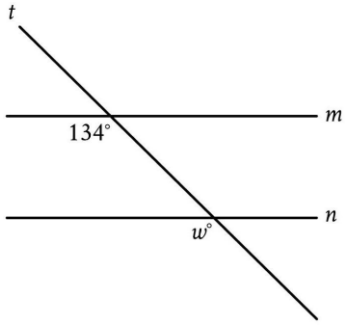
4.



In the figure above, $\overline{AF}, \overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9, BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

A. 16.8
B. 17.5
C. 18.4
D. 19.6

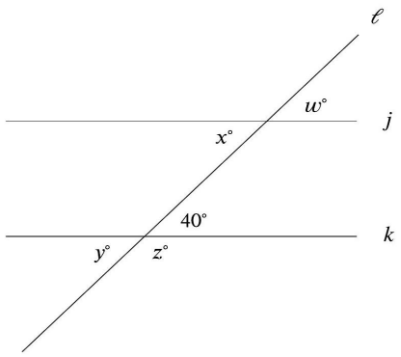
5.



In the figure, line m is parallel to line n. What is the value of w ?

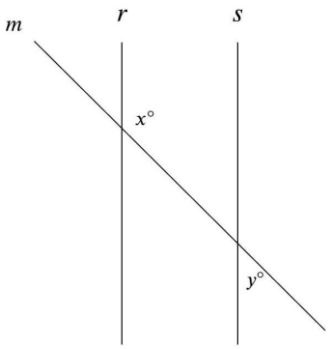
A. 13
B. 34
C. 66
D. 134

6. In the figure shown, line ℓ intersects lines j and k. Which additional piece of information is sufficient to prove that lines j and k are parallel?



A. $w = 40$
B. $x = 140$
C. $y = 40$
D. $z = 140$

7.

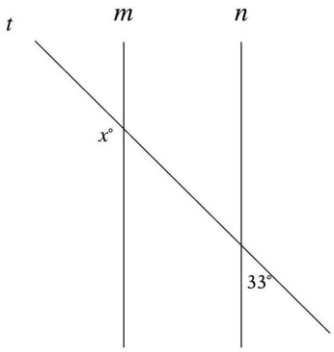


Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

A. $x < 115$
B. $x > 115$
C. $x + y < 180$
D. $x + y > 180$

8.

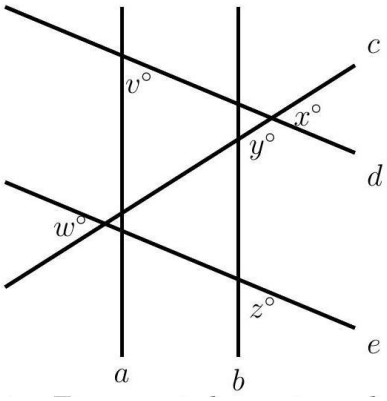


Note: Figure not drawn to scale.

In the figure, line m is parallel to line n, and line t intersects both lines. What is the value of x ?

A. 33
B. 57
C. 123
D. 147

9. [June 2023]

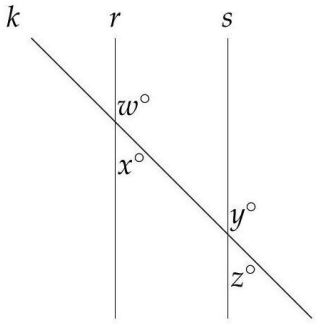


Note: Figure not drawn to scale

In the figure, parallel lines a and b are intersected by lines c, d, and e. If $x = 49, y = 136$, and $v < z$, which statement about x and w must be true?

A) $x < w$
B) $x > w$
C) $x = w$
D) $x + w = 90$

10. [October 2025]

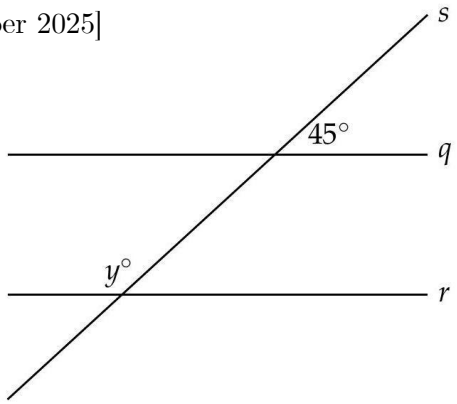


Note: Figure not drawn to scale

In the figure shown, line k intersects lines r and s. If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

A) $x = 33$
B) $y = 147$
C) $w + y = 180$
D) $y + z = 180$

11. [October 2025]



Note: Figure not drawn to scale.;

In the figure, line q is parallel to line r, and both lines are intersected by line s. If $y = 2x + 9$, what is the value of x ?

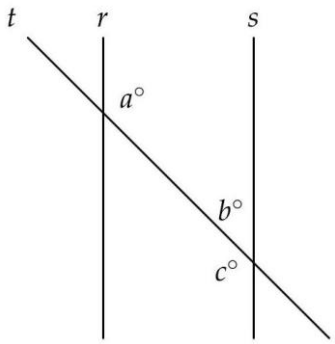
A) 27

B) 73

C) 63

D) 18

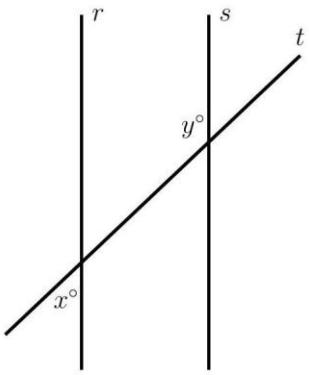
12. [October 2025]



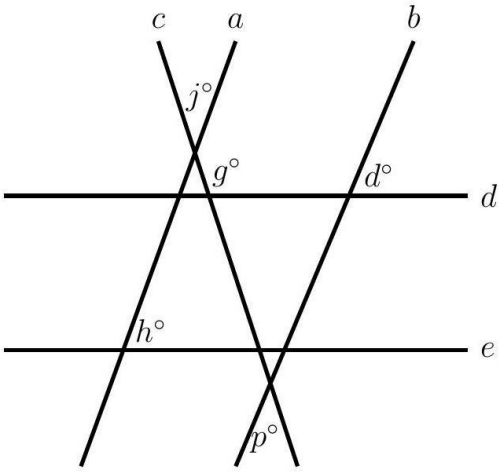
In the figure, lines r and s are parallel and line t intersects both lines. If $a = 4k + 57$ and $b = \frac{k}{2} + 55$, what is the value of c ?

13. [May 2024]

In the figure shown, line r is parallel to line s, and both lines are intersected by line t. If $x = 2w + 24, y = 5w + 29$, and $7w = a$, where a is a constant, what is the value of a ?



14. [August 2024]



Note: Figure not drawn to scale

In the figure, parallel lines r and s are intersected by lines t, u, and w. If $d = 42$, $g = 143$, and $h > d$, which statement about j and p must be true?

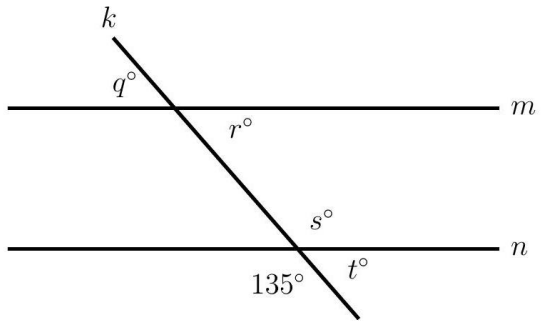
A) $j = p$

B) $j + p = 90$

C) $j > p$

D) $j < p$

15. [August US 2024]



In the figure shown, line m is parallel to line n, and line k intersects both lines. Which of the following statements is true?

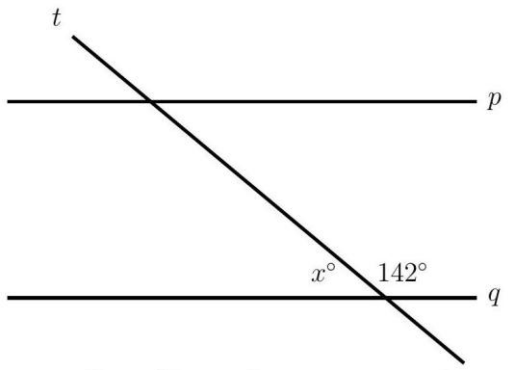
A) $q = 135$

B) $r = 135$

C) $s = 135$

D) $t = 135$

16. [August US 2024]

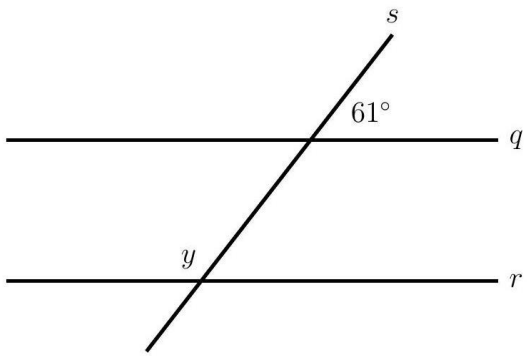


Note: Figure drawn not to scale

In the figure, line p is parallel to line q, and t intersects both of them. What is the value of $x + 142$?

A) 52
B) 90
C) 142
D) 180

17. [October US 2024]



Note: Figure not drawn to scale

In the figure, line q is parallel to line r, and both lines are intersected by line s. If $y = 2x + 9$, what is the value of x ?

A) 26
B) 35
C) 55
D) 65

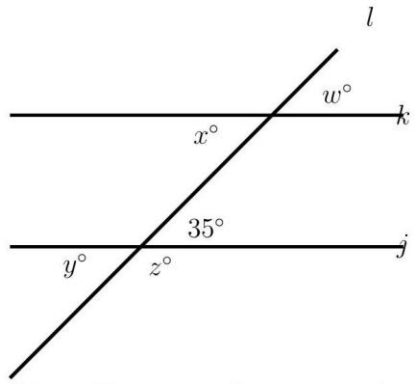
18. [October 2024]

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)°$. The sum of the measures of four of the eight angles is k. Which of the following could NOT be equivalent to k, for all values of x ?

A) $-14x + 1{,}540$
B) $14x - 320$
C) $-28x + 1{,}720$
D) 360

19. A line intersects two parallel lines, forming four acute and four obtuse angles. The measure of one of the acute angles is $(7x - 619)°$. The sum of the measure of one of the acute and three of the obtuse angles is $(-14x + w)°$. What is the value of w ?

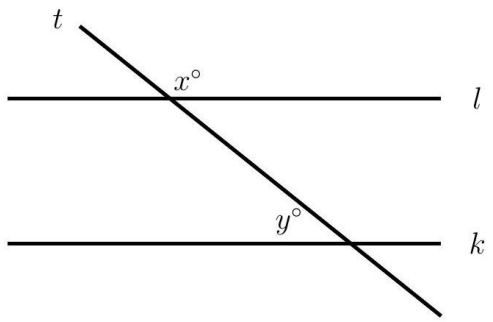
20. [October US 2024]



In the figure shown, line l intersects lines j and k. Which additional piece of information is sufficient to prove that lines j and k are parallel?

A) $w = 35°$
B) $x = 145°$
C) $y = 35°$
D) $z = 145°$

21. [November 2024]



In the figure shown, the lines l and k are parallel and line t intersects both lines. If $x > 154$, which of the following must be true?

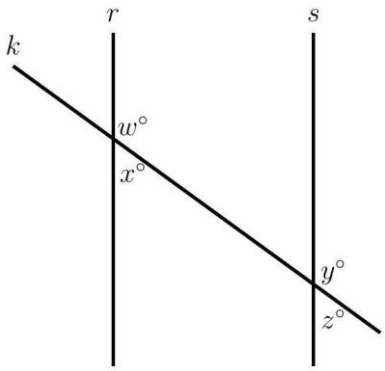
A) $y < 26$

B) $y > 26$

C) $x + y < 180$

D) $x + y > 180$

22. . [March 2025]



In the figure shown, line k intersects lines r and s. If $w = 146°$, which additional piece of information is sufficient to prove that lines r and s are parallel?

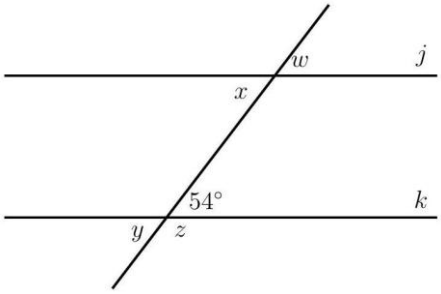
A) $x = 34$

B) $y = 146$

C) $w + y = 180$

D) $y + z = 180$

23. [May 2025]



In the figure shown, line t intersects lines j and k. Which additional piece of information is sufficient to prove that lines j and k are parallel?

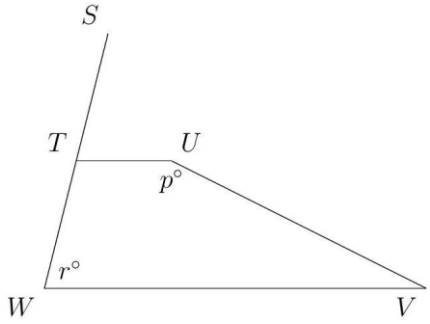
A) $w = 54$

B) $x = 126$

C) $y = 54$

D) $z = 126$

24. [June US 2025]



In the figure shown, point T lies on segment SW, and segment TU is parallel to segment VW. The measure of angle V is $26°$, and the measure of angle STU is $74°$. What is the value of $p - r$?

25. Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(7x - 80)°$. Which of the following could NOT be the sum of the measures of any two of these angles?

A) (14x - 160)°

B) (-14x + 520)°

C) (14x - 520)°

D) $180°$

26. 18 Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is (9x – 110)°. Which of the following could NOT be the sum of the measures of any two of these angles?

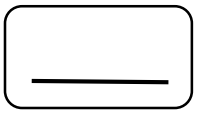
A) (-18x + 220)°

B) (-18x + 580)°

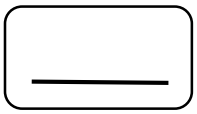
C) (18x – 220)°

D) 180°

27. A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is (7x - 610) deg The sum of the measures of one of the acute angles and three of the obtuse angles is (- 14x + w) deg What is the value of w?



28. A line intersects 2 parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is (9x-440). The sum of the mesures of one of the acute angles of three of the obtuse angles is (-18x+w)° What's the value of w?



29. A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(3x - 370)°$ .The sum of the measures of four of the eight angles is k Which of the following could NOT be equivalent to k, for all values of x?

A $-6x + 2{,}020°$
B) $6x - 560°$
C $-12x + 2{,}200°$
D $360°$

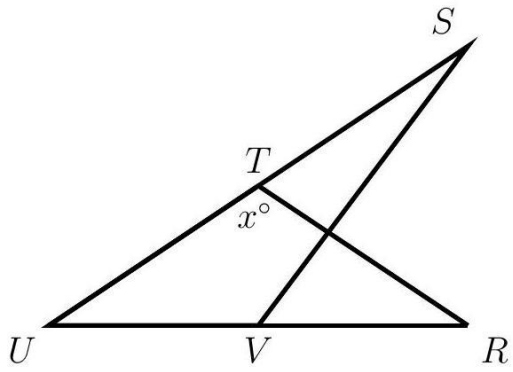
30. Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)°$. Which of the following could NOT be the sum of the measures of any two of these angles?

A) $(-18x + 280)°$
B) $(-18x + 640)°$
C) $(18x - 280)°$
D) $180°$

31. Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 190)°$. Which of the following could NOT be the sum of the measures of any two of these angles?
A) $(-18x + 380)°$
B) $(-18x + 740)°$
C) $(18x - 380)°$
D) $180°$

Angles

1. [November 2023]



In the figure, $RT = TU$, the measure of angle VST is $23°$ and the measure of angle RVS is $44°$. What is the value of x ?

2. [March 2024]

A polygon has exactly 73 sides. If the measure of each of the 73 interior angles of this polygon is $(180p)°$, what is the value of p ?

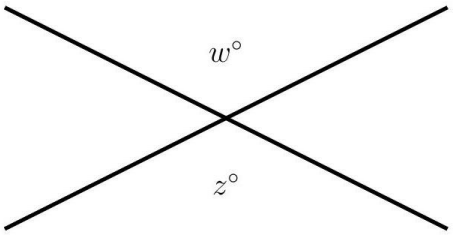
3. [March 2024]

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{13\pi}{18}$ radians greater than the measure of angle R. What is the measure of angle T, in degrees?

4. [March 2024]

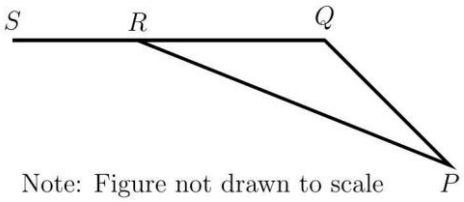
A polygon has exactly 83 sides. If the measures of each of the 83 interior angles of this polygon is $(180p)°$, what value of p ?

5. [March 2024]



Note: Figure not drawn to scale.
In the figure, two lines intersect at a point. If $w = 150$, what is the value of z ?
A) 30
B) 50
C) 75
D) 150

6.



In triangle PQR, $\overline{QR}$ is extended to point S. The measure of $\angle PQR$ is $142°$, and the measure of $\angle PRS$ is $166°$. What is the measure of $\angle QPR$?
A) $14°$
B) $19°$
C) $24°$
D) $38°$

7. [October US 2024]

In $\triangle JKL$, the measures of both $\angle J$ and $\angle K$ are equal, and the measure of $\angle L$ is $134°$. What is the measure of $\angle J$?
A) $23°$
B) $46°$
C) $67°$
D) $90°$

8. [August 2024]

The difference between the measure of angle A and the measure of angle B is $-\frac{3}{4}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B, in degrees?

A) $-\frac{3}{4}\cdot\frac{180^\circ}{\pi}$

B) $-\frac{3}{4}\pi\cdot\frac{\pi}{180^\circ}$

C) $-\frac{3}{4}\pi\cdot\frac{180^\circ}{\pi}$

D) $-\frac{3}{4}\pi\cdot\frac{360^\circ}{\pi}$

9. In triangle ABC, the measure of angle A is 47°, the measure of angle B is 90°, and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

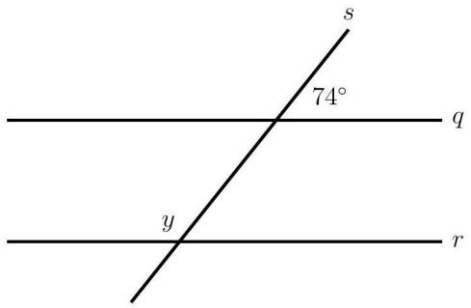
A) 43

B) 45

C) 86

D) 94

10. [December 2024]



Note: Figure not drawn to scale

In the figure, line q is parallel to line r, and both lines are intersected by line s. If $y = 2x + 8$, what is the value of x ?

A) 33

B) 41

C) 49

D) 59

11. [March 2025]

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{12}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B, in degrees?

A) $-\frac{5}{12}\pi \cdot \frac{180^\circ}{\pi}$
B) $-\frac{5}{12}\pi \cdot \frac{\pi}{180^\circ}$
C) $-\frac{5}{12}\pi \cdot \frac{\pi}{360^\circ}$
D) $-\frac{5}{12}\pi \cdot \frac{360^\circ}{\pi}$

12. [May 2025]

In triangle ABC, the measure of angle A is 34° and the measure of angle B is 90°. What is the measure, in degrees, of angle C ?

A) 34
B) 56
C) 90
D) 180

13. In $\triangle DEF$, the measure of $\angle D$ is 114°. Which of the following could be the measure, in degrees, of $\angle E$?

A) 115
B) 90
C) 67
D) 65

14. [August 2025]

The measure of angle G is $\frac{\pi}{10}$ radians. If the measure of angle G is $9n$ in degrees, where n is a constant, what is the value of n ?

A) 2
B) 9
C) 10
D) 18

15. [August 2025]

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(7x - 410)°$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-14x + w)°$. What is the value of w ?

16. [August 2025]

The measure of angle S is $\frac{7\pi}{17}$ radians. The measure of angle T is 2 times the measure of angle S . Which expression represents the measure, in degrees, of angle T ?

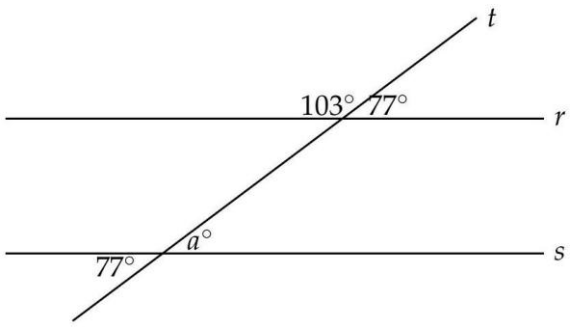
A) $\frac{7}{17}(900)(2)$

B) $\frac{7}{17}(180)(2)$

C) $\frac{7}{17\pi}(90)(2)$

D) $\frac{7\pi}{17}180 \cdot (2)$

17. [September 2025]



Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line t intersects both lines. What is the value of a ?

A) 180

B) 77

C) 167

D) 90

18. [September 2025]

The measure of angle K is $\frac{\pi}{2(12)}$ radians. The measure of angle L is 12 times the measure of angle K. What is the measure, in degrees, of angle L ?

A) 102
B) 12
C) 78
D) 90

19. [October 2025]

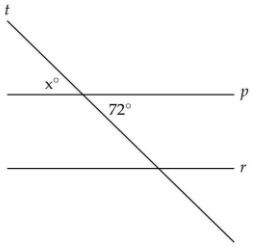
In a right triangle, the measure of one of the acute angles is $14°$. What is the measure of the other acute angle, in degrees? (Disregard the degree when entering your answer.)

20. [October 2025]

The measure of angle Z is $45°$. What is the measure, in radians, of angle Z ?

A) $\frac{1}{2}\pi$
B) $\frac{3}{4}\pi$
C) $\frac{1}{8}\pi$
D) $\frac{1}{4}\pi$

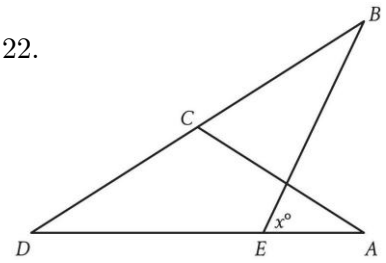
21. [October 2025]



In the figure, line p is parallel to line r, and line t intersects both lines. What is the value of x ?

A) 36
B) 72
C) 180
D) 252

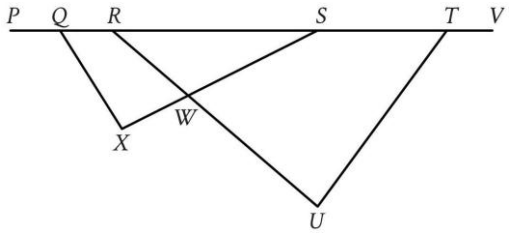
22.



Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is $45°$, and the measure of angle ACD is $104°$. What is the value of x ?

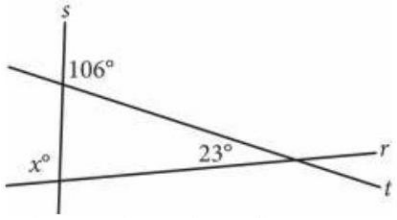
23.



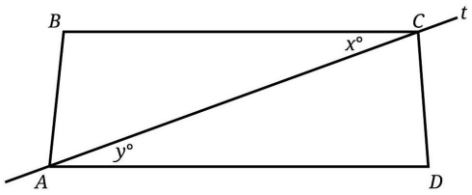
Note: Figure not drawn to scale.

In the figure shown, points Q, R, and T lie on line segment PV, and line segment RU intersects line segment SX at point W. The measure of $\angle SQX$ is $48°$, the measure of $\angle SXQ$ is $86°$, the measure of $\angle SWU$ is $85°$, and the measure of $\angle VTU$ is $162°$. What is the measure, in degrees, of $\angle TUR$?

24. Intersecting lines r, s, and t are shown below.



What is the value of x ?

25. 

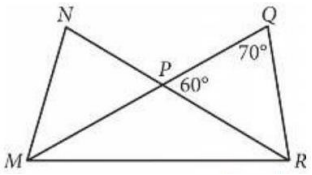
Note: Figure not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C. What is the value of y ?

A. 22
B. 44
C. 68
D. 158

26. In triangle PQR, the measures of angle P is $(3x + 5)°$, the measure of angle Q is $(2x + 9)°$, and the measure of angle R is $(4y + 5)°$. If side QR is extended through point R to point S, and the measure of angle PRS is $(x + y)°$, what is the value of $x + y$?

A. 39 B. 46 C. 102 D. 51

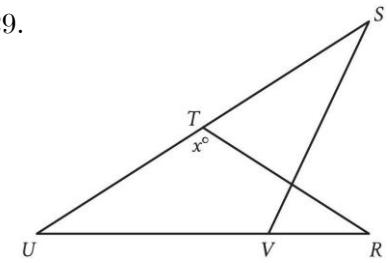
27. 

In the figure above, MQ and NR intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

28. In $\triangle XYZ$, the measure of $\angle X$ is $24°$ and the measure of $\angle Y$ is $98°$. What is the measure of $\angle Z$?

A. $58°$
B. $74°$
C. $122°$
D. $212°$

29.



Note: Figure not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is $29°$, and the measure of angle RVS is $41°$. What is the value of x ?

30. In triangle JKL, the measures of $\angle K$ and $\angle L$ are each $48°$. What is the measure of $\angle J$, in degrees? (Disregard the degree symbol when entering your answer.)

Triangles → Uchburchaklar

Triangles → Additional information questions

1. [September 2025]

In triangle ABC, the measure of angle A is 52 and $\mathrm{AC} = 40$. In triangle PQR , the measure of angle P is 52 and $\mathrm{PR} = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

A) $\mathrm{AB} = 45$ and $\mathrm{PQ} = 45$.
B) $\mathrm{AB} = 45$ and $\mathrm{QR} = 135$.
C) The measures of angle B and angle R are $40°$ and $88°$, respectively.
D) The measures of angle B and angle Q are $52°$ and $40°$, respectively

2. [August 2025]

The length of each side of triangle A is 270 millimeters. The length of each side of triangle B is double the length of each side of triangle A. What is the length, in millimeters, of each side of triangle B ?
A) 270
B) 272
C) 540
D) 1,080

3. [August 2025]

In the figure shown, the length of AB is 22. Which statement provides sufficient information to prove that triangle ABC is isosceles?
A) The length of AC is 11 .
B) The length of BC is 22 .
C) The measure of angle A is $70°$.
D) The sum of the measures of angles A, B, and C is $180°$.

4. In triangle ABC, the measure of angle A is $56°$ and $AC = 35$. In triangle PQR, the measure of angle P is $56°$ and $PR = 105$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

A) $AB = 50$ and $PQ = 50$

B) The measures of angle B and angle R are $32°$ and $92°$, respectively.

C) The measures of angle B and angle Q are $56°$ and $32°$, respectively.

D) $AB = 50$ and $QR = 50$

5. [May 2024]

In triangle ABC and triangle DEF, sides AB and DE each have a side length of 6 inches, and angles A and D each have an angle measure of $40°$. Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

I. The measures of angles B and C are equal.

II. The lengths of sides AC and DF are equal.

III. The lengths of sides BC and EF are equal.

A) I is sufficient, but II and III are not.

B) II is sufficient, but I and III are not.

C) III is sufficient, but I and II are not.

D) II is sufficient and III is sufficient, but I is not.

6. For triangle PQR and triangle STU, angles P and S each measure $30°$. $QR = 16$ and $TU = 12$. Which additional piece(s) of information is(are) sufficient to prove that triangle PQR is similar to triangle STU ?

I. The length of PQ is $\frac{1}{3}$ the length of ST.

II. The length of PR is $\frac{1}{3}$ the length of SU.

A) I is sufficient, but II is not.

B) II is sufficient, but I is not.

C) I is sufficient, and II is sufficient.

D) Neither I nor II is sufficient.

7. In triangle FGH and triangle KLM, the measures of angles G and L are each $60°$. The lengths of GH and LM are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle FGH is congruent to triangle KLM?
A) The lengths of FG and KM are each 17 centimeters.
B) The measures of angles H amd M are each $30°$.
C) The measures of angles F amd K are each $90°$.
D) No additional information is necessary.

8. [August US 2024]

Triangles ABC and DEF are similar, where A corresponds to D, and B corresponds to E. The measure of angle A is $26°$, and $AB = 4$. Which of the following statements must be true?
I. The measure of angle D is $26°$
I. $DE = 4$
A) I only
B) II only
C) I and II
D) Neither I nor II

9. [November US 2024]

In triangle ABC and triangle DEF, AB and DE are each equal to 5 centimeters, and angles A and D each have measure $35°$. Which additional piece of information is sufficient to prove that triangle ABC is congruent to triangle DEF ?
A) The lengths of sides BC and EF are equal.
B) The lengths of sides AC and DF are equal.
C) The measures of angles B and C are equal.
D) No additional information is necessary to prove that the two triangles are congruent.

10. [March 2025]

In triangle $ABC, BC = 36\sqrt{3}$. What is the value of $\cos A$?

In triangle ABC, the measure of angle A is $52°$ and $AC = 35$.
In triangle PQR, the measure of angle P is $52°$ and $PR = 105$.
Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?
A) $AB = 30$ and $PQ = 30$
B) $AB = 30$ and $QR = 90$
C) The measures of angle B and angle R are $34°$ and $94°$, respectively.
D) The measure of angle B and angle Q are $52°$ and $34°$, respectively.

11. [March 2025]

In triangle ABC, the measure of angle A is $58°$ and $AC = 30$.
In triangle PQR, the measure of angle P is $58°$ and $PR = 120$.
Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?
A) $AB = 35$ and $PQ = 35$
B) $AB = 35$ and $QR = 140$
C) The measures of angle B and angle R are $34°$ and $88°$, respectively
D) The measures of angle B and angle Q are $58°$ and $34°$, respectively

12. [June 2025]

In triangle FGH, the measure of angle F is $63°$. Which additional piece of information is sufficient to prove that triangle FGH is a right triangle?
A) The length of $\overline{FH}$ is 63 .
B) The length of $\overline{GH}$ is 126 .
C) The measure of angle H is $27°$.
D) The sum of the measures of angles F, G, and H is $180°$.

13. [June US 2025]

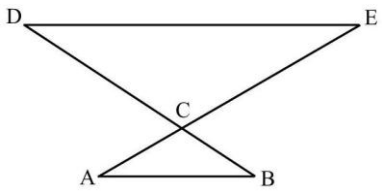
In triangle DEF, the measure of angle D is $47°$ and the measure of angle E is $97°$. In triangle RST, the measure of angle R is $47°$ and the measure of angle S is $97°$. Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?
A) The measure of angle F
B) The measure of angle T
C) The measure of angle F and the measure of angle T
D) No additional information is needed.

14. [June US 2025]

In triangle ABC and triangle DEF, sides AB and DE each have a side length of 10 inches, and angles A and D each have an angle measure of $40°$. Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?
I. The measures of angles B and C are equal. II. The lengths of sides AC and DF are equal.
III. The lengths of sides BC and EF are equal.
A) I is sufficient, but II and III are not.
B) II is sufficient, but I and III are not.
C) III is sufficient, but I and II are not.
D) II is sufficient and III is sufficient, but I is not.

15. In triangles XYZ and DEF, angles X and D each have measures of $37°$ and angles Z and F each have measures of $52°$. Which additional piece of information is sufficient to determine whether triangle XYZ is congruent to triangle DEF?
A) The measure of angle Y
B) The length of side EF
C) The lengths of sides XY and DE
D) No additional information is necessary.

16. In the figure below, $\overline{AB}$ and $\overline{DE}$ intersect at point C. Which of the following additional statements is (are) sufficient to prove that triangle ABC is similar to triangle DCE?



I. The length of $\overline{DE}$ is 3 times the length of $\overline{AB}$.
II. AB is parallel to DE .
A) I is sufficient, but II is not.
B) II is sufficient, but I is not.
C) I is sufficient, and II is sufficient.
D) Neither I nor II is sufficient.

17. In triangles ABC and LMN, angles B and M both have measure of $65°$, angles A and L correspond, angles C and N correspond, and $AB = \frac{9}{2}$ and $AC = \frac{27}{4}$. Which additional piece of information is sufficient to prove that triangle ABC and LMN are congruent?
A) $LM = \frac{9}{2}$
B) $LM = \frac{9}{2}$ and $MN = \frac{27}{4}$
C) $LM = \frac{9}{2}$ and angles A and L are congruent.
D) The measures of angles A and L are congruent, and the measure of angles C and N are congruent.

18. In triangles ABC and JKL, angles C and L each have measures of $57°, BC = 7$, and $KL = 28$. Which additional piece of information is sufficient to prove that triangles ABC and JKL are similar?
A) $AB = 11$ and $JK = 44$
B) $AC = 12$ and $JL = 12$
C) The measures of angles A and K are $58°$ and $65°$, respectively.
D) The measure of A and J are $58°$ and $65°$, respectively.

19. In triangles ABC and DEF, AB and DE each are 7 inches. Angle A and Angle D are both 37°. Which of the following is sufficient to determine that triangles ABC and DEF are congruent?

I. The measures of angles B and C are equal.

II. $AC = DF$

III. $BC = EF$

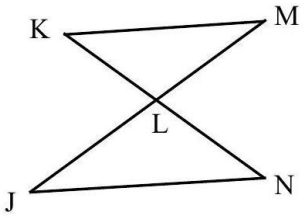
A) I only

B) II only

C) III only

D) II or III only

20. In the figure shown, $\overline{JM}$ and $\overline{KN}$ intersect at point L. Which of the following additional pieces of information is NOT sufficient to prove that $\triangle KLM$ is similar to $\triangle JLN$? (Note: Figure not drawn to scale.)



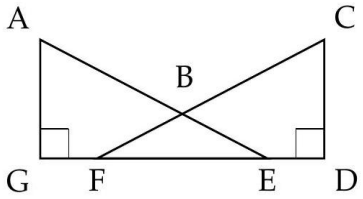
A) The measure of $\angle M$ is equal to the measure of $\angle K$, and $\angle N$ is equal to the measure of

B) $\frac{\angle J}{KM}$ is parallel to $\overline{JN}$.

C) The measure of $\angle M$ is equal to the measure of $\angle J$

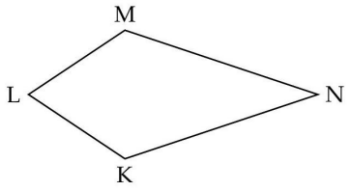
D) The length of $\overline{LM}$ is $\frac{5}{4}$ the length of $\overline{JL}$.

1. [August 2025]



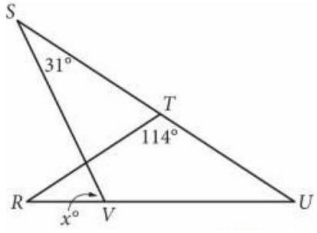
In the figure, triangle AEG is congurent to triangle CFD, where E corresponds to F. The measure of angle CBE is $56°$. What is the measure in degrees, of angle EAG ?

2.



In quadrilateral $KLMN$ shown, $KL = 3$, $LM = 3, KN = 27$, and $MN = 27$. Diagonals KM and LN (not shown) intersect at point G (not shown), where $GK = 1$ and $GM = 1$ if the length of diagonal LN is $\sqrt{p} + \sqrt{w}$, where p and w are integers, what is the value of $p + w$?

3.



In the figure above, $RT = TU$. What is the value of x ?

A. 72
B. 66
C. 64
D. 58

4. [October 2025]

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles C and F are right angles. If $\tan A = \frac{612}{35}$, what is the value of $\sin D$?

A) $\frac{612}{613}$
B) $\frac{35}{613}$
C) $\frac{613}{612}$
D) $\frac{613}{35}$

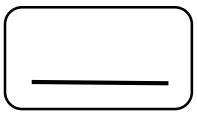
5. [June 2023]

Triangles ABC and DEF are congruent, where A corresponds to D, and B and E are right angles. The measure of angle A is $62°$. What is the value of angle F ?

A) $28°$
B) $62°$
C) $90°$
D) $118°$

6. [June 2023]

In triangle RST, the measure of angle R is 10 degrees and the measure of angle T is 50 degrees. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. What is the measure, in degrees, of angle SKL ? (Disregard the degree symbol when entering your answer.)

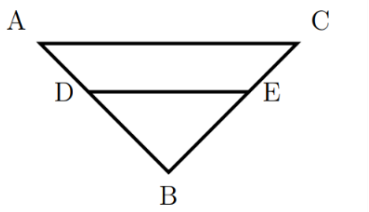


7. [November 2023]

A triangle has a base length of 82 centimeters and a height of 164 centimeters. What is the area, in square centimeters, of the triangle?

A) 246 B) 3,362
C) 6,724 D) 13,448

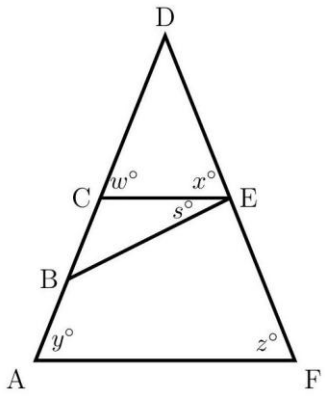
8 [August US 2023]



In the figure shown, line segments AC and DE are parallel. The measure of angle ACB is $58°$, and $AB = 1.5DB$.

What is the measure, in degrees, of angle DEB ?
(Disregard the degree symbol when entering your answer.)

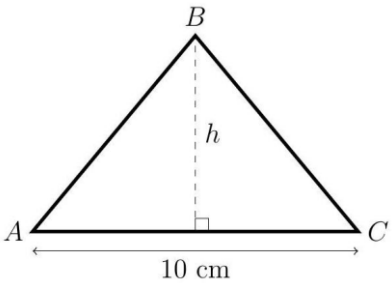
9.



In figure shown, B and C lie on $\overline{AD}$, E lies on $\overline{DF}$, and $\overline{CE}$ is parallel to $\overline{AF}$. Which of the following is enough to prove that $\triangle BDE$ is similar to $\triangle EDC$?

A) $w + y = x + z$
B) $w + x = y + z$
C) $s + x = z$
D) $s + x = y$

10.

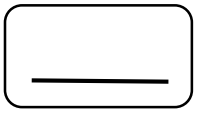


The area of triangle ABC is 30 square centimeters. What is the height h, in centimeters, of this triangle?

A) 3
B) 6
C) 10
D) 12

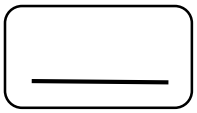
11. [August US 2023]
The area of a triangle is equal to x^2 square inches. The base of triangle is $5 + 2x$ inches, and the height of the triangle is $x - 2$ inches. What is the value of x ?
A) 2.5
B) 2.7
C) 5
D) 10

12. [December 2023]
In triangle RST, the measure of angle R is 11 degrees and the measure of angle T is 77 degrees. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. What is the measure, in degrees, of angle SKL ?(Disregard the degree symbol when entering the answer.)



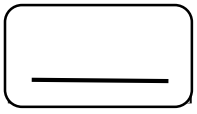
13. [December 2023]

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 16$ centimeters, and the height of the triangle $x - 6$ centimeters. What is the value of x ?



14. [March 2024]

Triangles ABC and DEF are congruent, where A corresponds to D, and B and E are right angles. The measure of angle A is $27°$. What is the measure, in degrees, of angle F ?

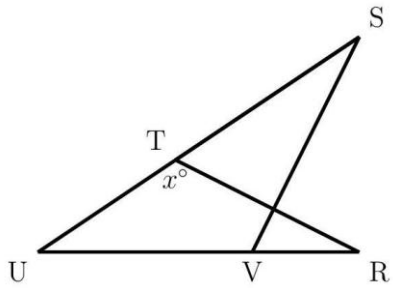


15. Triangles ABC and DEF are congruent, where A corresponds to D, and D, and B and E are right angles. The measure of angle A is $71°$. What is the measure, in degrees, of angle F ?

16. [May US 2024]

Triangles ABC and DEF are similar. Each side length of triangle ABC is 2 times the corresponding side length of triangle DEF. The area of triangle ABC is 300 square centimeters. What is the area, in square centimeters, of triangle DEF ?

17. [May 2024]



Note: Figure not drawn to scale
In the figure, $RT = TU$, the measure of angle VST is $23°$, and the measure of angle RVS is $34°$. What is the value of x ?

18. [May 2024]

In isosceles triangle ABC, sides AB and AC are congruent. Point D divides BC such that the length of $\overline{BD}$ is $\frac{5}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD. If the length of $\overline{BE}$ is 10 , what is the length of $\overline{CF}$?

A) 4 B) 14 C) 50 D) 70

19. A perimeter of an isosceles right triangle is $38 + 38\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

A) 19

B) $19\sqrt{2}$

C) 38

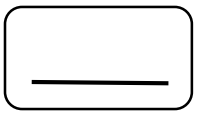
D) $38\sqrt{2}$

20. In a triangle ABC, the measure of angle A is $48°$, the measure of angle B is $90°$, and the measure of angle C is $\left(\frac{k}{2}\right)°$. What is the value of k ?

A) 42 B) 45 C) 84 D) 96

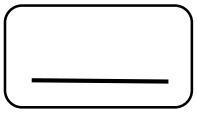
21. [August 2024]

Triangle QRS is similar to triangle FGH such that Q, R, and S correspond to F, G, and H, respectively. Each side of triangle FGH has $\frac{1}{4}$ the length of its corresponding side in triangle QRS. In triangle FGH, the measure of angle F is $35°$, the measure of angle G is $8°$, and $FG = a$. In triangle QRS, the measure of angle Q is $b°$ and $QR = 20$. What is the value of $a + b$?



22. [August 2024]

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 12.5$ centimeters, and the height of the triangle is $x - 5$ centimeters. What is the value of x ?



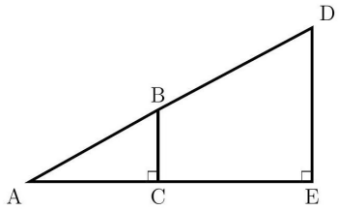
23. If the perimeter of an equilateral triangle with a height of 46 units is $a\sqrt{3}$ units, what is the value of a ?

A) 46 B) 92 C) $\frac{2,116}{3}$ D) $\frac{4,232}{3}$

24. [October US 2024]

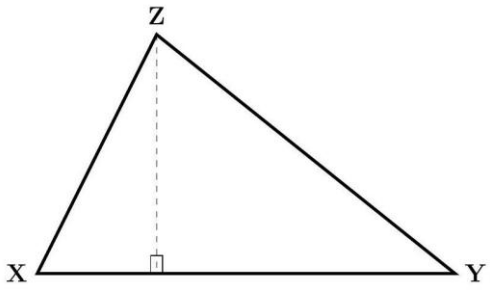
In $\triangle FGH$, the measure of $\angle F$ and $\angle G$ are each $39°$.
What is the measure of $\angle H$?
A) $78°$
B) $102°$
C) $129°$
D) $141°$

25. [October US 2024]



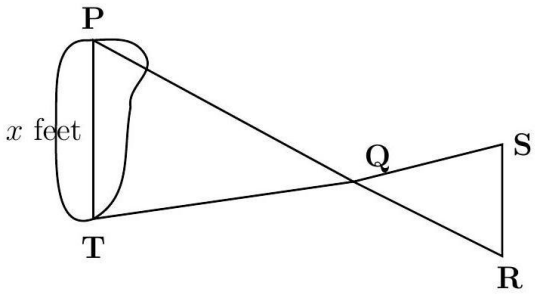
In the figure shown $AB = \sqrt{89}$ units, $AC = 8$ units, and $CE = 24$ units. What is the area, in square units, of triangle ADE ?

26. [December 2024]



Note: Figure not drawn to scale.
In the figure shown, the measure of angle X is $52°$. The length of $\overline{XY}$ is 24 units and the length of $\overline{XZ}$ is 17 units.
What is the area, in square units, of triangle XYZ ?
A) 204
B) 408
C) $204\sin 52°$
D) $408\sin 52°$

27. [December 2024]



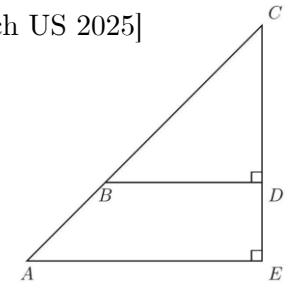
Note: Figure not drawn to scale.

A property owner wants to find the length x, in feet, across a pond as represented in the figure. PR intersects ST at point Q, and $\triangle PTQ$ is similar to $\triangle RSQ$. The length represented by PQ, TQ, QS, and RS were determined to be 4,200 feet, 8,400 feet, 1,400 feet, and 1,600 feet, respectively. What is the value of x ?

28. [March 2025]

In triangle RST, the measure of angle R is $39°$; the measure of angle S is $x°$, and the measure of angle T is $(5x - 3)°$. Point L lies on RS, point K lies on ST, and LK is parallel to RT. What is the measure, in degrees, of angle SKL ? (Disregard the degree symbol when entering your answer.)

29. [March US 2025]



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD. The measure of angle CBD is $57°$, and $AE = 21(BD)$. What is the measure of angle CAE?

A) $(21 \cdot 57)°$

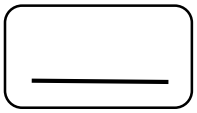
B) $(21 + 57)°$

C) $57°$

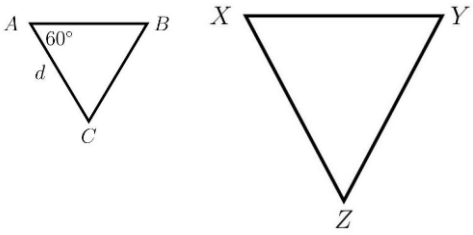
D) $21°$

30. [September 2025]

In triangle RST, angle T is a right angle, point L lies on RS, point K lies on ST, and LK is parallel to RT. If the length of RT is 72 units, the length of LK is 24 units, and the area of RST triangle is 396 square units, what is the length of KT, in units?



31. [December US 2024]



Note: Figure not drawn to scale

For the triangles shown, triangle ABC dilated by a factor of 5 to obtain triangle XYZ, where $d = 19$. What is the measure, in degrees, of angle X ?

A) 12 B) 55 C) 60 D) 65

32. [November US 2024]

The area of a triangle is 270 square centimeters. The length of the base of the triangle is 12 centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

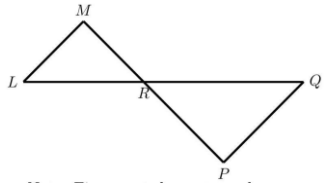
A) 15
B) 18
C) 30
D) 36

33. [October 2024]

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

A) 9
B) $9\sqrt{2}$
C) 18
D) $18\sqrt{2}$

34.



Note: Figure not drawn to scale

In the figure, LQ intersects MP at point R, and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of MR, LR, and RP are 9,10 , and 19 , respectively. What is the length of LQ ?

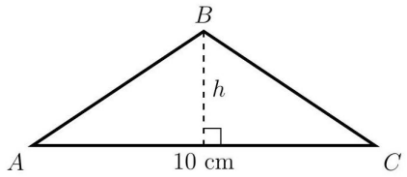
A) $\frac{280}{19}$
B) $\frac{190}{9}$
C) $\frac{271}{9}$
D) $\frac{280}{9}$

35. The table gives the perimeters of similar triangles TUV and XYZ, where TU corresponds to XY. The length of TU is 16 .

	Perimeter
Triangle TUV	44
Triangle XYZ	352

What is the length of XY ?
A) 2
B) 16
C) 60
D) 128

36.



Note: Figure not drawn to scale
The area of triangle ABC is 220 square centimeters.
What is the height h, in centimeters, of this triangle?
A) 10
B) 22
C) 44
D) 88

37. [March US 2025]

In triangle ABC, point D is the midpoint of segment AB and point E is the midpoint of segment BC. If $BD = 8$ and $DE = 14$, what is the length of segment AC ?
A) 28
B) 26
C) 22
D) 16

Triangle inequality

31. A triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side length of 8 and 13 , which of inequality represents the possible length, x, of the third side of the triangle?
A) $x < 21$
B) $x > 21$
C) $5 < x < 21$
D) $x < 5$ or $x > 21$

32. [June US 2025]

The perimeter of an obtuse triangle is 27 meters. The lengths of two sides of this triangle are 5 meters and 9 meters. What is the length, in meters, of the third side of this triangle?
A) 13
B) 14
C) 18
D) 22

33.

D

Margin of Error → Nisbiy Xatolik

- **Survey** – So'rovnoma
- **Sample** – Tanlanma (bosh to'plamdan olingan kichik namuna guruh)
- **Population** – Bosh to'plam (tadqiqot qilinayotgan butun aholi yoki obektlar guruhi)
- **At random / Randomly selected** – Tasodifiy ravishda tanlangan
- **Estimate / Estimated** – Taxmin, baholash / Taxmin qilingan, baholangan
- **Margin of error** – Xatolik chegarasi (marjasi)
- **Plausible values** – Ehtimoliy qiymatlar (haqiqatga yaqin, bo'lishi mumkin bo'lgan qiymatlar)
- **In favor of** – Qo'llab-quvvatlovchi, tarafdor
- **Average / Mean** – O'rtacha qiymat
- **Conclusion** – Xulosa
- **Defective** – Yaroqsiz, nuqsonli (brak)
- **Proportion** – Ulush, nisbat
- **Characteristic** – Xususiyat, belgi
- **Increase / Decrease** – Oshish / Kamayish

- **Estimate / Estimated** – Taxmin, baholash / Taxmin qilingan, baholangan PDF
- **Describe / Describes** – Tasvirlamoq, ifodalamoq / Tasvirlaydi, ifodalaydi PDF
- **Cannot be...** – ...bo'lishi mumkin emas / ...bo'la olmaydi PDF
- **Lower than...** – ...dan pastroq / ...dan kichikroq PDF
- **Higher than...** – ...dan yuqoriroq / ...dan kattaroq PDF
- **Margin of error** – Xatolik chegarasi (marjasi) PDF
- **Plausible values** – Bo'lishi mumkin bo'lgan (haqiqatga yaqin) qiymatlar PDF
- **Conclusion** – Xulosa PDF
- **Interpretation** – Talqin, ma'no PDF
- **Best** – Eng yaxshi, eng to'g'ri PDF
- **Most appropriate** – Eng mos keladigan, eng o'rinli PDF
- **Most likely** – Eng ehtimoliy, eng kutiladigan PDF

1. [October 2025]

For a survey, a sample of 996 people was selected at random from all registered voters in a county. Based on the survey, it is estimated that 42% of all registered voters in the county are in favor of a new law, with an associated margin of error of 3.07%. Which of the following best describes the plausible values for the percent of all registered voters in the county who are in favor of the new law?

A) Any value greater than 38.93% and less than 45.07%
B) Any value greater than 38.93%
C) Any value less than 38.92% or greater than 45.07%
D) Any value less than 45.07%

- **Greater than...** – ...dan katta / ...dan ko'p
- **Less than...** – ...dan kichik / ...dan kam
- **More than...** – ...dan ko'proq
- **Fewer than...** – ...dan kamroq
- **Between ... and ...** – ... va ... orasida
- **Exactly...** – Aynan / Naqd
- **Approximately...** – Taxminan / Taxminiy ravishda
- **Lower / Higher** – Pastroq / Yuqoriroq

2. [October 2025]

A snack company advertises that bags of pretzels produced by the company weigh, on average, 1 pound each. To test this, Sam selected at random 40 bags of pretzels produced by the company and weighed each bag. Based on his measurements, Sam estimated that for all bags of pretzels produced by the company, the average weight per bag is 1.05 pounds, with a margin of error of 0.06 pounds. For all bags of pretzels produced by the company, which of the following is the most appropriate conclusion about?

A) $0.99 \leq w \leq 1.05$
B) $w = 1.05$
C) $1.05 \leq w \leq 1.11$
D) $0.99 \leq w \leq 1.11$

- **Produces / Produced** – Ishlab chiqaradi / Ishlab chiqarilgan
- **Fills / Filled** – To'ldiradi / To'ldirilgan
- **Weighs / Weighed** – Vazni ... keladi / Tortib ko'rdi
- **Advertises** – Reklama qiladi
- **Selected / Chosen** – Tanlab olingan
- **Tested** – Sinovdan o'tkazilgan / Tekshirilgan
- **Repeated** – Takrorlangan
- **Spends** – Sarflaydi
- **Calculated** – Hisoblangan

3. [March 2024]

A machine fills bags with approximately 10 ounces of cranberries. To test the accuracy of the filling process, 148 bags of cranberries were selected at random and weighed. Based on the sample, it is estimated that the average weight of all bags of cranberries filled by the machine in an 8 -hour period is 9.89 ounces, with an associated margin of error of 0.14 ounce. Which of the following is the best interpretation of this estimate?

- **Best describes...** – Eng yaxshi tasvirlaydi / ifodalaydi
- **Best interpretation** – Eng to'g'ri talqin / ma'no
- **Appropriate conclusion** – Mos / To'g'ri xulosa
- **Plausible values** – Bo'lishi mumkin bo'lgan qiymatlar
- **Likely effect** – Kutilayotgan ta'sir / Ehtimoliy oqibat
- **Guarantees** – Kafolatlaydi

A) Plausible values for the average weight of all bags of cranberries filled by the machine are between 9.75 ounces and 10.03 ounces.
B) Plausible values for the average weight of all bags of cranberries filled by the machine are less than 9.75 ounces or greater than 10.03 ounces.
C) The average weight of all bags of cranberries filled by the machine is greater than 9.99 ounces.
D) The average weight of all bags of cranberries filled by the machine is less than 9.99 ounces.

4. [August US 2024]

A company that produces only one type of item wants to estimate the percent on the items produced in a typical week that are defective. A random sample of 400 of the items produced in a certain week were tested. Based on the sample, it is estimated that 8.5% of all items produced by the company in this week were defective, with an associated margin of error of 2.73%. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all items produced by the company during this week.
A) A 2.73% of all items are defective.
B) It is plausible that between 5.77% and 11.23% of the items are defective.
C) 8.5% of all items are defective.
D) It is plausible that more than 11.23% of the items are defective.

5. [June 2024]

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.2 , with an associated margin of error of 0.08 . Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

A) It is plausible that the proportion is between 0.12 and 0.28 .
B) It is plausible that the proportion is less than 0.12 .
C) The proportion is exactly 0.2 .
D) It is plausible that the proportion is greater than 0.28 .

6. [August US 2024]

The manager of a gym selected a sample of 139 members at random to estimate the percentage of the gym's members that would continue to pay for a membership if the price increased. From the survey, the manager estimates that 82% of the gym's members would continue to pay for membership if the price increased, with associated margin of error of 6.39%. If the survey is repeated with a random sample of 278 members and the results are calculated in the same way, which of the following will be the most likely effect of using the larger random sample compared to the smaller random sample?

A) The margin of error will be lower
B) The margin of error will be higher
C) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be lower
D) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be higher

7. [October US 2024]

A machine fills bags with approximately 29 ounces of walnuts. To test the accuracy of the filling process, 169 bags of walnuts were selected at random and weighed. Based on the sample, it is estimated that the average weight of all bags of walnuts filled by the machine in an 8 -hour period is 28.88 ounces, with an associated margin of error of 0.13 ounce. Which of the following is the best interpretation of this estimate?

A) Plausible values for the average weight of all bags of walnuts filled by the machine are between 28.75 ounces and 29.01 ounces.
B) Plausible values for the average weight of all bags of walnuts filled by the machine are less than 28.75 ounces or greater than 29.01 ounces.
C) The average weight of all bags of walnuts filled by the machine is less than 28.99 ounces.
D) The average weight of all bags of walnuts filled by the machine is greater than 28.99 ounces.

8. [October US 2024]

A researcher is designing a study to investigate the average number of hours a student at a high school spends reading per day. The researcher will report the average number of hours a student at the high school spends reading with an associated margin of error. The researcher is considering using a random sample of either 105 or 210 students from the high school. Which of the following could be the most likely effect of using the larger random sample compared to the smaller random sample?

A) The reported margin of error would be lower.
B) The reported margin of error would be higher.
C) The reported average number of hours would be lower.
D) The reported average number of hours would be higher.

9. [October US 2024]

A survey was conducted in Franklin to estimate number of adult residents who supported the addition of a parking garage in their town. Based on the survey, it is estimated that 77% of all adult residents of Franklin support the addition of a parking garage in their town, with an associated margin of error of 5%. Which of the following is the most appropriate conclusion about all adult residents of Franklin?

A) Plausible values for the percent of all adult residents of Franklin who support the addition of a parking garage in their town are between 72% and 82%
B) The addition of a parking garage in the town is supported by exactly 72% of all adult residents of Franklin
C) Plausible values for the percent of all adult residents of Franklin who support the addition of a parking garage in their town are either less than 72% or more than 82%
D) The addition of a parking garage in the town is supported by more than 82% of all adult residents of Franklin

10. [December 2024]

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325 degrees Fahrenheit ($^{\circ}F$). The research manager recorded the amount of time, in minutes, each oven takes to preheat to $325(^{\circ}F)$. Based on the first sample, the research manager estimated that this type of oven takes an average of 15.2 minutes to preheat to $325(^{\circ}F)$, with an associated margin of error of 1 minute. Based on the second sample, the research manager estimated that this type of oven takes an average of 15.4 minutes to preheat to $325(^{\circ}F)$, with an associated margin of error of 2.2 minutes. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

A) The first sample contained more ovens than the second sample
B) The first sample contained fewer ovens than the second sample.
C) The first sample took more time on average to preheat to $325(^{\circ}F)$ than the second sample.
D) The first sample took less time on average to preheat to $325(^{\circ}\mathrm{F})$ than the second sample.

11. [June 2025]

A survey was conducted in Madison to estimate the percent of adult residents who support the addition of streetlights at every corner in their town. Based on the survey, it is estimated that 40% of all adult residents of Madison support the addition of streetlights at every corner in their town, with an associated margin of error of 6%. Which of the following is the most appropriate conclusion about all adult residents of Madison?

A) Plausible values for the percent of all adult residents of Madison who support the addition of streetlights at every corner in their town are between 34% and 46%.
B) The addition of streetlights at every corner in the town is supported by exactly 34% of all adult residents of Madison.
C) Plausible values for the percent of all adult residents of Madison who support the addition of streetlights at every corner in their town are either less than 34% or more than 46%.
D) The addition of streetlights at every corner in the town is supported by more than 46% of all adult residents of Madison.

12. A group of 40 employees selected at random from all the employees at a company were surveyed about the number of books they read last year. From the data collected, it was estimated that the mean number of books employees at the company read last year is 6.8, with an associated margin of error of 1.1. Which of the following is the most appropriate conclusion?

A) It is plausible that the actual mean number of books employees at the company read last year is less than 5.7.
B) It is plausible that the actual mean number of books employees at the company read last year is between 5.7 and 7.9.
C) It is plausible that every employee at the company read between 6.8 and 10.1 books last year.
D) It is plausible that the actual mean number of books employees at the company read last year is greater than 7.9.

13. A sample was selected at random from a population. Based on the sample, it is estimated that the mean value of a variable for this population is 217, with an associated margin of error of 1. Based on the estimate and associated margin of error, which of the following best describes the range of plausible values for the population mean?

A. Any value between 215 and 217
B. Any value between 216 and 218
C. Any value less than 216
D. Any value greater than 218

14. A survey was conducted in Madison to estimate the percent of adult residents who support the addition of a gas station at a popular intersection in their town. Based on the survey, it is estimated that 27% of all adult residents of Madison support the addition of a gas station at a popular intersection in their town, with an associated margin of error of 6%. Which of the following is the most appropriate conclusion about all adult residents of Madison?

A. Plausible values for the percent of all adult residents of Madison who support the addition of a gas station at a popular intersection in their town are between 21% and 33%.
B. Plausible values for the percent of all adult residents of Madison who support the addition of a gas station at a popular intersection in their town are either less than 21% or more than 33%.
C. The addition of a gas station at a popular intersection in the town is supported by exactly 21% of all adult residents of Madison.
D. The addition of a gas station at a popular intersection in the town is supported by more than 33% of all adult residents of Madison.

15. A researcher conducted a study to estimate the average number of hours high school students spend on homework per week. Based on a random sample of students, the mean was estimated to be 14 hours, with an associated margin of error of 1.5 hours. Which of the following is the most appropriate conclusion about the population mean?

A) The average number of hours for all high school students is exactly 14 hours.

B) It is plausible that the true average number of hours for all high school students is between 12.5 and 15.5 hours.

C) Most high school students spend between 12.5 and 15.5 hours on homework per week.

D) The margin of error guarantees that the population mean cannot be less than 12.5 hours.

16. A polling organization wants to estimate the proportion of voters in a city who support a new education policy. In a preliminary poll of 400 randomly selected voters, the estimate had a margin of error of 4%. If the organization repeats the poll with a new random sample of 1,600 voters under identical conditions, which of the following is the most likely margin of error for the new poll?

A) Around 1%

B) Around 2%

C) Around 8%

D) Around 16%

Boxplots

Probability → Ehtimollik

Ehtimollik

1. [March US 2025]

The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	34%
$18-40$ years old	23%
$41-65$ years old	22%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?
A) 0.21
B) 0.32
C) 0.45
D) 0.62

2. [May 2025]

At a convention center, there are a total of 375 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48 , and the probability of selecting a visitor who is located in room B is 0.24 . How many visitors are located in room C ?
A) 28
B) 43
C) 90
D) 105

3.

Voice type	Number of singers
Countertenor	4
Tenor	6
Baritone	10
Bass	5

A total of 25 men registered for singing lessons. The frequency table shows how many of these singers have certain voice types. If one of these singers is selected at random, what is the probability he is a baritone?

A. 0.10
B. 0.40
C. 0.60
D. 0.67

4. In a bag, there are 7 red, 4 white, 33 blue, and 33 yellow cubes. If one of these cubes is selected at random, what is the probability of selecting a cube that is neither blue nor yellow?

A. $\frac{6}{7}$
B. $\frac{7}{11}$
C. $\frac{1}{3}$
D. $\frac{1}{7}$

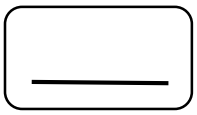
5. At a convention center, there are a total of 375 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.32 . How many visitors are located in room C ?

A) 20
B) 58

C) 60
D) 75

6. [May 2025]

Hakim has a bag of 50 tiles, where 45 are labeled with a letter and 5 are black. If Hakim selects a tile from the bag at random, what is the probability of selecting a black tie? (Express your answer as a decimal or fraction, not as a percent.)



7. Each rock in a collection of 70 rocks was classified as either igneous, metamorphic, or sedimentary, as shown in the frequency table.

Classification	Frequency
igneous	10
metamorphic	33
sedimentary	27

If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

A. $\frac{10}{27}$
B. $\frac{10}{33}$
C. $\frac{10}{60}$
D. $\frac{10}{70}$

8. A band with 45 members has 11 members who play saxophone. If one band member is selected at random, what is the probability of selecting a band member who plays saxophone?

A. $\frac{1}{45}$
B. $\frac{11}{45}$

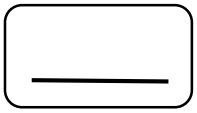
C. $\frac{34}{45}$
D. $\frac{45}{45}$

9. [August US 2023]

The table summarizes the distribution of age and assigned group for participants in a study.

	0 – 9 years	10 – 19 years	20 + years	Total
Group A	14	19	27	60
Group B	27	7	26	60
Group C	19	34	7	60
Total	60	60	60	180

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age? (Express your answer as a decimal or fraction, not as a percent.)



10. [August 2025]

Survey Results	
Annuals	159
Perennials	295

A total of 454 gardeners selected at random were asked if they preferred perennials or annuals. The results of the survey are shown in the table above. According to these results, if 5,448 gardeners were choosing plants, how many more would prefer perennials than would prefer annuals?

A) 136
B) 1632
C) 1908
D) 3540

11. There are n nonfiction books and 12 fiction books on a bookshelf. If one of these books is selected at random, what is the probability of selecting a nonfiction book, in terms of n ?

A. $\frac{n}{12}$
B. $\frac{n}{n+12}$
C. $\frac{12}{n}$
D. $\frac{12}{n+12}$

12. The table summarizes the distribution of age and assigned group for 105 participants in a study.

	$0-9$ years	$10-19$ years	$20+$ years	Total
Group A	5	19	11	35
Group B	4	8	23	35
Group C	26	8	1	35
Total	35	35	35	105

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age

A) $\frac{2}{7}$
B) $\frac{3}{7}$
C) $\frac{19}{35}$
D) $\frac{6}{7}$

13. A factory produced a batch of 200 light bulbs, 8 of which were found to be defective. If one light bulb is selected at random from this batch, what is the probability that it is defective?

A. 0.004
B. 0.04
C. 0.4
D. 4

14. A researcher calculated the mitochondrial genome sizes g, in kilobase pairs (kbp), and the number of mitochondrial genes for 34 different species. The table summarizes these data.

Number of mitochondrial genes	Mitochondrial Genome Size (kbp)				Total
	$g < 16$	$16 \leq g < 18$	$18 \leq g < 20$	$g \geq 20$	
36 or less	3	0	0	0	3
37	1	19	0	0	20
38 or more	2	0	4	0	6
Total	6	19	4	0	29

One of these species is selected at random. What is the probability of selecting a species with a mitochondrial genome size greater than or equal to $18kbp$, given that it has 37 or more mitochondrial genes?
(Express your answer as a decimal or fraction, not as a percent.)

15. For a science project, Anka recorded whether it rained each weekday and weekend day for 12 weeks. Her results are summarized in the table below.
Weekday and Weekend Day Rain for 12 Weeks

	Rain	No rain	Total
Number of weekdays	12	48	60
Number of weekend days	8	16	24
Total	20	64	84

If one of the days on which there was no rain is selected at random, what is the probability the day was a weekend day?

A. $\frac{4}{21}$ B. $\frac{1}{4}$
C. $\frac{2}{3}$ D. $\frac{3}{4}$

16. A box contains 13 red pens and 37 blue pens. If one of these pens is selected at random, what is the probability of selecting a red pen? (Express your answer as a decimal or fraction, not as a percent.)

A) $\frac{13}{50}$
B) $\frac{37}{50}$
C) $\frac{26}{50}$
D) $\frac{3}{5}$

17.

	Live north of Center St.	Live south of Center St.	Total
Less than 45 years old	13	12	25
At least 45 years old	22	88	110
Total	35	100	135

The table summarizes members of a local organization by age and whether they live north or south of Center St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

A) $\frac{25}{135}$
B) $\frac{35}{135}$
C) $\frac{100}{135}$
D) $\frac{110}{135}$

18.

	Even integers	Odd integers	Total
List A	12	44	56
List B	40	4	44
Total	52	48	100

The given table shows the number of even and odd integers contained in list A and list B . No integer is

contained in both lists. What fraction of even integers represented in the table are in list A ?

A) $\frac{12}{52}$ B) $\frac{12}{56}$ C) $\frac{52}{100}$ D) $\frac{52}{56}$

19. For 100 neurons, the table summarizes the distribution of classification and cell body diameter:

Classification	Cell body diameter (micrometers)		
	Less than 20	20 to 30	Greater than 30
Sensory neuron	13	7	3
Motor neuron	0	15	18
Interneuron	10	34	0

One of these neurons will be selected at random. What is the probability of selecting a neuron with a cell body diameter that is less than or equal to 30 micrometers, given that it is not classified as a motor neuron?

20. [March 2025]

At a convention center, there are a total of 375 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.68 , and the probability of selecting a visitor who is located in room B is 0.24 . How many visitors are located in room C ?

A) 8
B) 30
C) 61
D) 165

21. [May 2024]

A choir of 48 singers has 12 members who sing bass. If one singer from the choir is selected at random, what is the probability of selecting a singer who sings bass?

A) $\frac{1}{48}$
B) $\frac{12}{48}$
C) $\frac{36}{48}$
D) $\frac{48}{48}$

22. The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	27%
18 – 40 years old	24%
41 – 65 years old	28%
Greater than 65 years old	21%

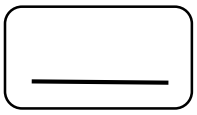
If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?
A) 0.21
B) 0.29
C) 0.35
D) 0.41

23. [December US 2024]
A sleep study consisted of 59 participants, of which 53 participants each had an average of more than 120 minutes of rapid eye movement (REM) sleep per night. If a participant from this study is selected at random, what is the probability of selecting a participant that had an average of more than 120 minutes of REM sleep per night?
A) 53/100
B) 59/100
C) 53/59
D) 59/53

24. [March 2024]
There are 20 marbles in the bag: 8 red marbles, 10 green marbles, and 2 blue marbles. if one of these marbles is selected at random, what is the probability of selecting a

red marble?
A) 2/20
B) 8/20
C) 10/20
D) 12/20

25. [June 2023]
A certain book has 250 pages, and 21 of these pages have an illustration. If one of the book's pages is selected at random, what is the probability of selecting a page with an illustration? (Express your answer as a decimal or fraction, not as a percent.)



26. Andy has a bag of marbles. If the bag has 30 blue marbles, 25 red marbles, and 15 green marbles, what is the probability that he will randomly select a red marble?
A) $\frac{5}{14}$
B) $\frac{6}{15}$
C) $\frac{3}{7}$
D) $\frac{5}{7}$

27. [June US 2025]

Each rock in a collection of 70 rocks was classified as either igneous, metamorphic, or sedimentary, as shown in the frequency table.

Classification	Frequency
igneous	10
metamorphic	38
sedimentary	22

If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

A) $\frac{10}{70}$ B) $\frac{10}{60}$
C) $\frac{10}{38}$ D) $\frac{10}{22}$

28. A bin contains a mixture of T -shirts for two sports teams. The table shows the number of T-shirts in the bin, classified by size and sports team.

	Flames	Panthers	Total
Small	6	9	15
Medium	24	33	57
Large	18	24	42
Total	48	66	114

One T-shirt from the bin will be selected at random. What is the probability of selecting a T -shirt that is medium, given that the T -shirt is a Flames T -shirt? (Express your answer as a decimal or fraction, not as a percent.)

29. [March 2025]

At a convention center, there are a total of 275 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.68 , and the probability of selecting a visitor who is located in room B is 0.24 . How many visitors

are located in room C ?

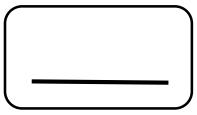
A) 8

B) 22

C) 45

D) 121

30. Each face of a fair 12-sided die is labeled with a number 1 through 12, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 7 ?



31. A pencil will be randomly selected from a bag. The probability of selecting a white pencil is $\frac{3}{13}$. The probability of selecting a blue pencil is $\frac{2}{13}$. What is the probability of selecting a white or blue pencil?

A) $\frac{1}{13}$

B) $\frac{5}{13}$

C) $\frac{5}{26}$

D) $\frac{6}{26}$

32. There are 20 buttons in a bag: 8 white buttons, 2 orange buttons, and 10 brown buttons. If one of these buttons is selected at random, what is the probability of selecting a white button?

A. $\frac{2}{20}$

B. $\frac{8}{20}$

C. $\frac{10}{20}$

D. $\frac{12}{20}$

33. $-6,14,18$

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a negative number?

A. 0
B. $\frac{1}{3}$
C. $\frac{2}{3}$
D. 1

34.

Diet	Type of Weight Change		Total
	Gained	Lost	
A	50	100	150
B	75	75	150

If one of the dogs in the table is selected at random, what is the probability of selecting a dog who lost weight and received diet B ?

A) $\frac{75}{300}$
B) $\frac{75}{150}$
C) $\frac{100}{175}$
D) $\frac{100}{300}$

35. For a school competition, each student in the sixth, seventh, and eighth grades is assigned to either the red team or the green team. The table shows the distribution of grade and team for the students in the school.

	Red team	Green team	Total
Sixth grade	47	42	89
Seventh grade	49	36	85
Eighth grade	38	52	90
Total	134	130	264

A student from the competition will be selected at random. What is the probability of selecting a

student who is in the sixth or seventh grade, given that the student is on the green team? (Express your answer as a decimal or fraction, not as a percent.)

Right triangles and Trigonometry

1. Triangle FGH is similar to triangle JKL, where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

A. $\frac{75}{317}$ B. $\frac{308}{317}$
C. $\frac{317}{308}$ D. $\frac{317}{75}$

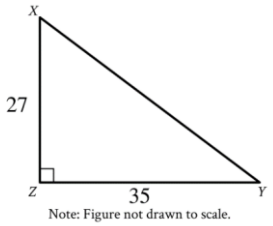
2. In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

A. $\frac{\sqrt{15}}{15}$
B. $\frac{\sqrt{15}}{4}$
C. $\frac{4\sqrt{15}}{15}$
D. $\sqrt{15}$

3. In triangle XYZ, angle Z is a right angle and the length of $\overline{YZ}$ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

A. 188 B. 168 C. 84 D. 71

4.

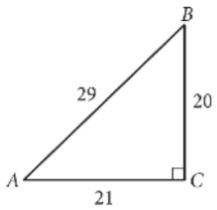


Note: Figure not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

A. $\tan X$
B. $\tan Y$
C. $\cos X$
D. $\cos Y$

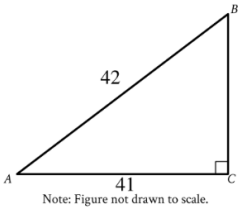
5.



In the figure above, what is the value of $\tan(A)$?

A. $\frac{20}{29}$ B. $\frac{21}{29}$
C. $\frac{20}{21}$ D. $\frac{21}{20}$

6.



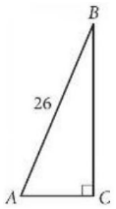
Note: Figure not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

A. $\frac{42}{41}$ B. $\frac{41}{42}$
C. $\frac{1}{42}$ D. $\frac{1}{41}$

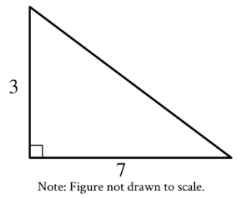
7. In triangle XYZ, angle Y is a right angle, the measure of angle Z is $33°$, and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k\tan 33°$, where k is a constant, what is the value of k ?

8.



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

9.

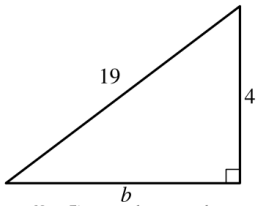


Note: Figure not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

A. 3.2 B. 5
C. 7.6 D. 20

10.



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

A. $4b = 19$
B. $4 + b = 19$
C. $4^2 + b^2 = 19^2$
D. $4^2 - b^2 = 19^2$

11. The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

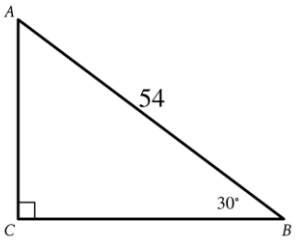
A. 9
B. $9\sqrt{2}$

C. 18
D. $18\sqrt{2}$

12. In $\triangle ABC, \angle B$ is a right angle and the length of $\overline{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

A. 34 B. 102 C. 136 D. 170

13.

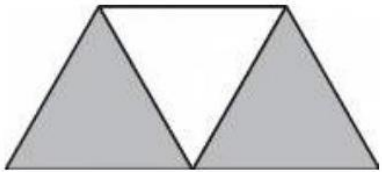


Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

A. $\frac{\sqrt{3}}{54}$ B. $\frac{1}{\sqrt{3}}$
C. $\sqrt{3}$ D. $27\sqrt{3}$

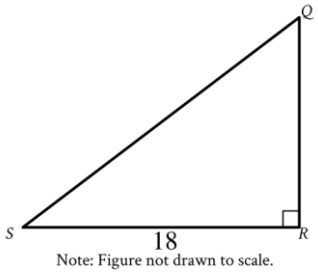
14.



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

A. $2\sqrt{3}$
B. $4\sqrt{3}$
c. $8\sqrt{3}$
D. 16

15. In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?



Note: Figure not drawn to scale.

A. $18\cos Q$
B. $18\sin Q$
C. $\frac{18}{\cos Q}$
D. $\frac{18}{\sin Q}$

16. The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

17. In triangle ABC, angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

A. $14\sqrt{37}$
B. $26\sqrt{37}$
C. $34\sqrt{37}$
D. $\sqrt{34 \cdot 37}$

18. Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D, and angles C and F are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

A. $\frac{1}{22}$ B. $\frac{1}{23}$
C. $\frac{21}{22}$ D. $\frac{22}{23}$

19. An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

A. $29\sqrt{2}$
B. $58\sqrt{2}$
C. $58 + 58\sqrt{2}$
D. $58 + 116\sqrt{2}$

20. An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

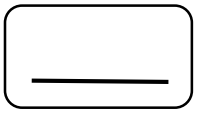
A. 47
B. 94
C. $47\sqrt{2}$
D. $94\sqrt{2}$

21.
$RS = 440$
$ST = 384$
$TR = 584$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A. $\frac{48}{73}$ B. $\frac{55}{73}$
C. $\frac{48}{55}$ D. $\frac{55}{48}$

22. A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

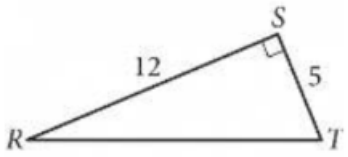


23.
$RS = 20$
$ST = 48$
$TR = 52$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A. $\frac{5}{13}$ B. $\frac{5}{12}$
C. $\frac{12}{13}$ D. $\frac{12}{5}$

24.



In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

25. A right triangle has legs with lengths of 28 centimeters and 20 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

A. $4\sqrt{74}$ B. $8\sqrt{6}$

C. 48 D. 1,184

26. Triangle ABC is similar to triangle DEF, where A corresponds to D and C corresponds to F. Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 125$, what is the length of $\overline{DE}$?

A. $125\frac{\sqrt{3}}{3}$ B. $125\frac{\sqrt{3}}{2}$

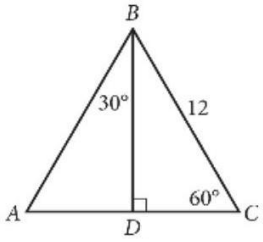
C. $125\sqrt{3}$ D. 250

27. A right triangle has legs with lengths of 11 centimeters and 9 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

A. $\sqrt{40}$ B. $\sqrt{202}$

C. 20 D. 202

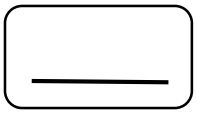
28.



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

A. 4 B. 6
C. $6\sqrt{2}$ D. $6\sqrt{3}$

29. The length of a rectangle's diagonal is $3\sqrt{17}$, and the length of the rectangle's shorter side is 3 . What is the length of the rectangle's longer side?



Mixture → Aralashma

1. How many gallons of water must be added to 40 gallons of a 10% alcohol solution to produce an 8% alcohol solution?

A) 5 B) 8 C)10 D) 12

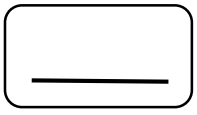
2. How many gallons of a 20% salt solution must be added to 10 gallons of a 50% salt solution to produce 30% salt solution?

A) 5 gallons
B) 10 gallons
C) 15 gallons
D) 20 gallons

3. How many quarts of alcohol must be added to 10 quarts of a 25% alcohol solution to produce a 40% alcohol solution?

A) 2.5 quarts
B) 8 quarts
C) 10 quarts
D) 15 quarts

4. How many liters of a 40% saline solution must be mixed with 10 liters of a 10% saline solution to produce a 25% saline solution?



5. How many gallons of acid must be added to G gallons of a k% acid solution to bring it up to an m% solution?

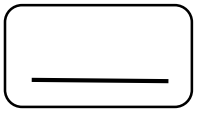
A) $\frac{G}{100-m}$

B) $\frac{G*m}{100-m}$

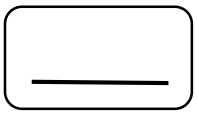
C) $\frac{G*(m-k)}{100-m}$

D) $\frac{100-m}{G*(m-k)}$

6. A 30% alcohol solution is mixed with a 70% alcohol solution to create 50 liters of a 58% alcohol solution. How many liters of the 70% solution were used?



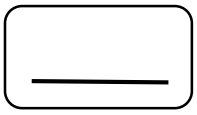
7. A store sells two types of coffee: Type A is 20% Colombian beans, and Type B is 60% Colombian beans. A blend of 5 kg is created using both types, containing 44% Colombian beans. How many kilograms of Type B were used?



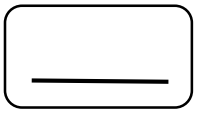
8. A chemist mixed x liters of a 2% saline solution with y liters of a 5% saline solution to produce a 3% saline solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)

A) $0.02x + 0.05y = 3(x + y)$
B) $0.02x + 0.05y = 0.03(x + y)$
C) $0.2x + 0.5y = 3(x + y)$
D) $0.2x + 0.5y = 0.3(x + y)$

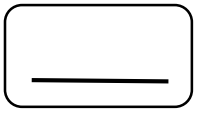
9. 7.5 gallons of a $p\%$ chlorine solution are added to 10 gallons of a 5% chlorine solution to make a 20% chlorine solution. What is the value of p?



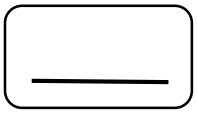
10. A scientist is creating a sugar water solution for his lab. If there are currently 10 liters of a 32% sugar solution in a bucket, how many liters of a 12% sugar solution need to be added to the bucket to create a 20% sugar solution?
A scientist is creating a sugar water solution for his lab. If there are currently 10 liters of a 32% sugar solution in a bucket, how many liters of a 12% sugar solution need to be added to the bucket to create a 20% sugar solution??



11. How many liters of a 40% saline solution must be added to 8 liters of a 10% saline solution to obtain a 30% saline solution?



12. How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?



13. A sample of a certain alloy has a total mass of 50.0 grams and is 50.0% silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. What was the mass, in grams, of the silicon in the second piece?

A. 9.0 B. 16.0
C. 20.0 D. 30.0

14. [October 2025]

A chemist combines 25 grams of compound A, 35 grams of compound B, and x grams of compound C . The resulting mixture has a mass of 125 grams. Find the value of x.
A) 25
B) 65
C) 35
D) 10

15. [September 2025]

$$34x + 79y = 260$$

The given equation represents the volume of mulch x, in cubic feet, and the volume of soil y, in cubic feet, in a mixture of mulch and soil that weighs 260 pounds. If the volume of soil in the mixture is 2 cubic feet, what is the volume of mulch in the mixture, in cubic feet?

16. [September 2025]

$$38x + 85y = 360$$

The given equation represents the volume of mulch x, in cubic feet, and the volume of soil y, in cubic feet, in a mixture of mulch and soil that weighs 360 pounds. If the volume of soil in

the mixture is 2 cubic feet, what is the volume of mulch in the mixture, in cubic feet?

17. A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where x is the number of pounds of Fertilizer A and y is the number of pounds of Fertilizer B?

A. 0.4x + 0.6y = 240
B. 0.6x + 0.4y = 240
C. 40x + 60y = 240
D. 60x + 40y = 240

18. A chemist mixed x liters of a 2% saline solution with y liters of a 5% saline solution to produce a 3% saline solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)
A) $0.02x + 0.05y = 3(x + y)$
B) $0.02x + 0.05y = 0.03(x + y)$
C) $0.2x + 0.5y = 3(x + y)$
D) $0.2x + 0.5y = 0.3(x + y)$

19. [March 2024]
At a constant temperature, a scuba diver combines two different concentrations of nitrox mix (oxygen-nitrogen gas mixture) to fill a tank to a total pressure of 130 atmospheres (atm), of which 34% is contributed by oxygen. Nitrox mix A contributes x atm of pressure to the tank, of which 22% is contributed by oxygen. Nitrox mix B contributes y atm of pressure to the tank, of which 48% is contributed by oxygen. Which

system of equations represents this situation?

A) $x + y = 130$
$0.48x + 0.22y = 0.34(130)$

B) $x + y = 130$
$0.22x + 0.48y = 0.34(130)$

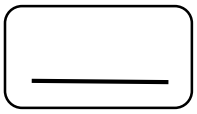
C) $x + y = 0.34(130)$
$0.48x + 0.22y = 130$

D) $x + y = 0.34(130)$
$0.22x + 0.48y = 130$

20. [October US 2024]

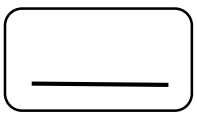
$$0.10x + 0.20y = 0.19(x + y)$$

The equation gives a volume x, in gallons, of a 10% salt water solution that could be mixed with a volume y, in gallons of a 20% saltwater solution to produce a 19% saltwater solution. According to this equation, what volume, in gallons, of the 20% saltwater solution could be mixed with 50.0 gallons of 10% saltwater solution to produce a 19% saltwater solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

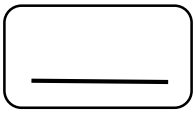


21. [June 2024]

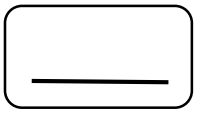
A chemist combines water and isopropanol to make a mixture with a volume of 52 milliliters (mL). The volume of isopropanol in the mixture is 10 mL . What is the volume of water, in mL, in the mixture?(Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)



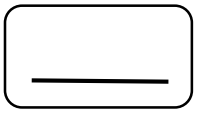
22. A chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL). The volume of acetic acid in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)



23. A chemist has 200 mL of a solution that is 15% acid. How many milliliters of pure acid are in the solution?

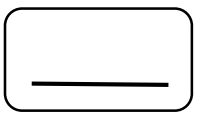


24. How many liters of a 30% chlorine solution must be added to 12 liters of a 10% chlorine solution to obtain a 15% chlorine solution?



25. [June 2024]

A chemist combines water and isopropanol to make a mixture with a volume of 45 milliliters (mL). The volume of isopropanol in the mixture is 10 mL . What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)



26. A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 5% salt by weight, with y kilograms of a high-salt mixture, which is 85% salt by weight, to create 30 kilograms of a mixture that is 12% salt by weight. Which equation represents this situation?

A. $0.85x + 0.05y = (0.12)(30)$

B. $0.05x + 0.85y = (0.12)(30)$

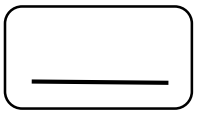
C. $0.85x + 0.05y = 30$

D. $0.05x + 0.85y = 30$

27. A mixture consisting of only vitamin D and calcium has a total mass of 180 grams. The mass of vitamin D in the mixture is 60 grams. What is the mass, in grams, of calcium in the mixture?

A. 240
B. 180
C. 120
D. 60

28. [October US 2024]
The composition of an animal is defined as the muscles, bones, and fat of the animal.A scientist studied the composition of one young swamp buffalo and determined that the buffalo had 139.2 kilograms of muscle, which made up approximately 67.7% of its composition. Of the remaining composition of this buffalo, approximately 51.4% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone.



29. A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 3% salt by weight, with y kilograms of a high-salt mixture, which is 80% salt by weight, to create 40 kilograms of a mixture that is 15% salt by weight. Which equation represents this situation?

A. $0.80x + 0.03y = (0.15)(40)$
B. $0.03x + 0.80y = (0.15)(40)$
C. $0.80x + 0.03y = 40$
D. $0.03x + 0.80y = 40$